

Cyclin-Dependent Kinase (CDK) Inhibitors

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Cyclin-Dependent Kinase (CDK) Inhibitors: Structure–Activity Relationships and Insights into the CDK-2 Selectivity of 6-Substituted 2-Arylamino purines

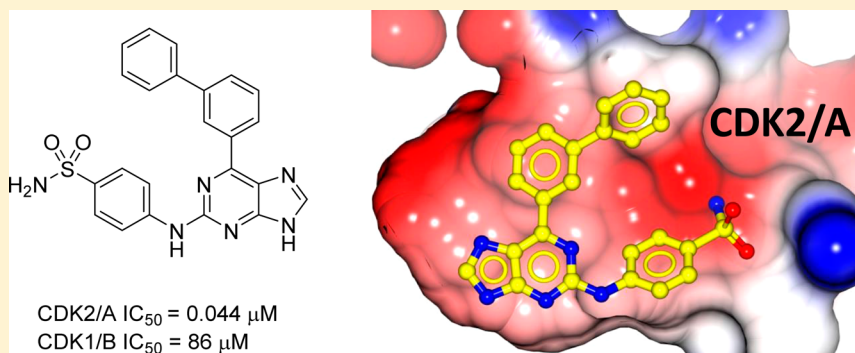
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Supporting Information



ABSTRACT: Purines and related heterocycles substituted at C-2 with 4'-sulfamoylanilino and at C-6 with a variety of groups have been synthesized with the aim of achieving selectivity of binding to CDK2 over CDK1. 6-Substituents that favor competitive inhibition at the ATP binding site of CDK2 were identified and typically exhibited 10–80-fold greater inhibition of CDK2 compared to CDK1. Most impressive was 4-((6-([1,1'-biphenyl]-3-yl)-9H-purin-2-yl)amino) benzenesulfonamide (73) that exhibited high potency toward CDK2 (IC_{50} 0.044 μ M) but was ~2000-fold less active toward CDK1 (IC_{50} 86 μ M). This compound is therefore a useful tool for studies of cell cycle regulation. Crystal structures of inhibitor–kinase complexes showed that the inhibitor stabilizes a glycine-rich loop conformation that shapes the ATP ribose binding pocket and that is preferred in CDK2 but has not been observed in CDK1. This aspect of the active site may be exploited for the design of inhibitors that distinguish between CDK1 and CDK2.

INTRODUCTION

The cyclin-dependent kinase (CDK) family of serine/threonine kinases plays an integral role in the regulation of the eukaryotic cell cycle as well as having key functions in apoptosis, transcription, differentiation, and neuronal function.^{1,2} Deregulation of the cell cycle, a hallmark of human cancer, is frequently associated with aberrant CDK activity through mechanisms that may include mutation/overexpression of CDKs or mutations to genes encoding proteins that directly or indirectly modulate CDK activity.^{3–5} CDK2 activation entails association with partner cyclins A and E, whereas the endogenous proteins p21^{Cip1} and p27^{Kip1} are inhibitory. Cyclin

E overexpression and/or p27^{Kip1} suppression are a common feature of many human tumors.^{6,7}

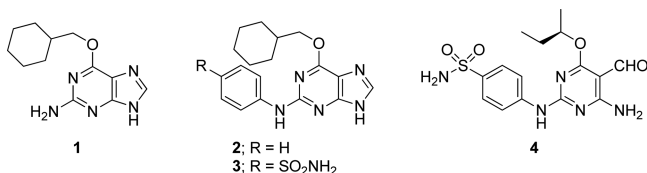
Pharmacological inhibition of CDK family members is a potential therapeutic approach for the treatment of cancer. In this context, a large number of ATP-competitive inhibitors have been evaluated.^{5–9} The clinical development of CDK inhibitors as antitumor agents was originally hampered by poor kinase selectivity, particularly within the CDK family, and uncertainty as to which CDK constitutes the most appropriate therapeutic target.³ However, an appreciation of the roles played by specific

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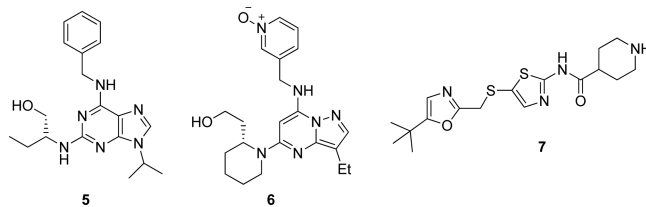
CDKs in sustaining signaling through dysregulated pathways that drive particular cancers is leading to more stringent patient selection and correspondingly improved efficacy. For example, the highly selective CDK4/6 inhibitor palbociclib has been approved for the treatment of ER-positive, HER2-negative breast cancer, while abemaciclib, palbociclib, and ribociclib are in phase III clinical trials to treat advanced breast cancer in which signaling through the CDK4/6-retinoblastoma axis is critical.¹⁰ Similarly, agents that target a subset of CDKs are being evaluated in a range of other clinically defined segments.¹¹

The validity of CDK2 as a cancer therapeutic target was originally called into question both by CDK2 knockdown experiments in which loss of CDK2 failed to induce cell cycle arrest in a number of tumor cell lines¹² and by mouse knockout experiments where the animals were viable.^{13,14} However, applying the rationale outlined above, CDK2 inhibitors are also expected to have utility in settings in which the cancer is addicted to enhanced CDK2 activity or where synthetic lethalties can be identified. Recent studies employing a chemical genetic approach in which CDK2 expression was maintained, but kinase activity was inhibited, provides compelling evidence that CDK2 is a valid cancer target. In a panel of human cancer cells transformed with various oncogenes, highly selective small-molecule CDK2 inhibition resulted in marked growth inhibition.¹⁵ CDK2-selective inhibitors may also have applications in combination therapies in appropriate clinical settings. For example, a recent study demonstrated that a combination of phosphatidylinositol-3-kinase and CDK2 inhibitors induced apoptosis in malignant glioma xenografts via a synthetic-lethal interaction.¹⁶ Accumulating evidence also implicates CDK2 as a prospective therapeutic target in BRCA-deficient cancers,¹⁷ neuroblastoma,¹⁸ and ovarian cancer.¹⁹ These observations, taken together with supportive clinical data,²⁰ have led to a resurgence of interest in CDK2 inhibitors as cancer therapeutic agents.^{21,22} Ideally, such an inhibitor would discriminate between CDK2 and CDK1 to avoid cell toxicity as a result of inhibiting CDK1, which is an essential member of the CDK family.



Our previous studies employing a structure-lead approach resulted in the development of selective ATP-competitive CDK2 inhibitors based on 2-amino-6-alkoxy- (1; NU2058) and 2-arylamino-6-alkoxy- (2) purine derivatives.^{23,24} These efforts were rewarded by the identification of potent purine-based inhibitors, exemplified by 6-cyclohexylmethoxy-2-(4'-sulfamoylanilino)purine (3; NU6102, CDK2 IC₅₀ = 5.0 nM)²⁵ as well as alkoxypyrimidine inhibitors (e.g., 4, CDK2 IC₅₀ = 0.8 nM).²⁶ Importantly, 3 exhibits selectivity for CDK2 over other CDK-family members, proving some 50-fold selective for CDK2 over CDK1 (IC₅₀ = 250 nM), a kinase with close structural homology to CDK2. In addition, 3 is only weakly active against CDK7 and CDK9 (IC₅₀ = 4.4 and 1.1 μ M, respectively). These CDKs play a key role in transcriptional regulation through phosphorylation of RNA polymerase II. Modulation of protein synthesis by off-target inhibition of

CDK7/CDK9 has obfuscated previous studies to elucidate the pharmacological effects of clinically evaluated compounds initially identified as selective CDK2 inhibitors such as seliciclib (5), dinaciclib (6), and SNS-032 (7).^{3,28,29}



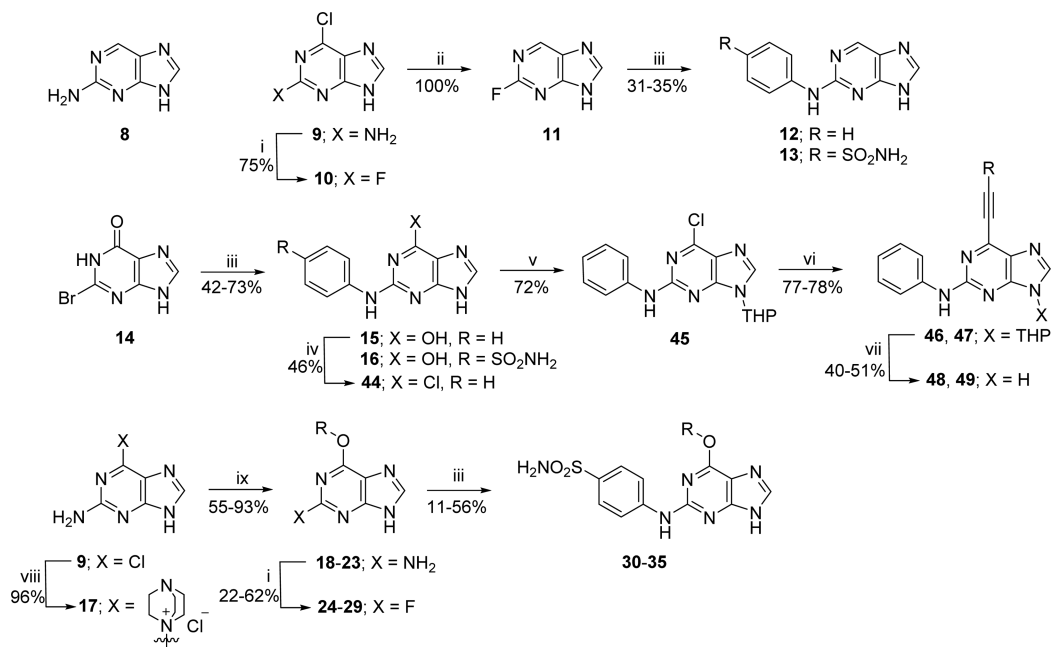
Crystal structures of purines 1–3 in complex with T160-phosphorylated CDK2-cyclin A (CDK2-cyclin A) provided valuable details of inhibitor interactions within the ATP-binding site.^{23,24} The purine heterocycle anchored the inhibitor through a triplet of hydrogen bonds between N-9, N-3, and the 2-amino group, and the backbone carbonyl moiety of Glu81 and amide and carbonyl moieties of Leu83, respectively, located in the hinge region of the kinase (Figure 2). The greater potency and selectivity of 3 compared with 2 and the parent 2-amino-6-cyclohexylmethoxypurine 1, has been attributed, at least in part, to additional interactions of the 2-sulfanilyl substituent of 3 with the specificity surface of CDK2.^{24,30} Notably, two hydrogen bond interactions between the sulfonamide group of 3 and Asp86 facilitate optimal hydrophobic packing of the arylamino ring.

Extensive structure–activity relationship studies (SARs) at the purine 2-position of 3 have further established the importance of these inhibitor–kinase interactions.²⁵ By contrast, although the 6-cyclohexylmethyl group of 3 occupies the ribose-binding pocket, the precise nature of the interactions made by substituents at the purine 6-position with the CDK2 active site remains uncertain, notwithstanding that a large number of derivatives have previously been evaluated empirically.³¹

In this paper, we report the results of studies to characterize interactions between substituents at the purine 6-position and the CDK2 ATP-binding site. The effect upon biological activity of modifying the core purine heterocycle of 3 is also reported. These studies have resulted in the identification of 4-((1,1'-biphenyl)-3-yl)-9H-purin-2-ylamino benzenesulfonamide 73 as a potent and selective CDK2 inhibitor (IC₅₀ = 44 nM), exhibiting some 2000-fold-selectivity over CDK1 (IC₅₀ = 86 μ M).

CHEMISTRY

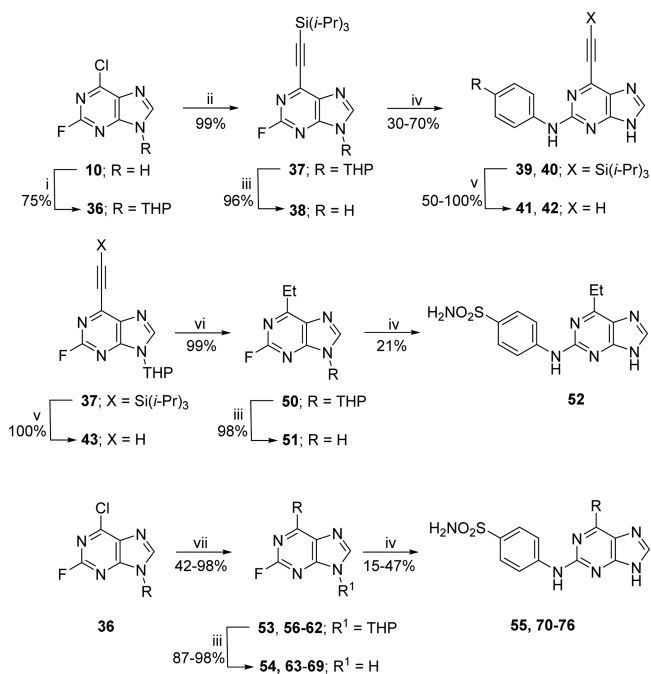
With a view to establishing the overall contribution of the 6-cyclohexylmethoxy substituent of 1 and 3 to inhibitor binding, the simple 6-unsubstituted purines 8, 12, and 13 were required. 2-Aminopurine (8) was commercially available, while the 2-phenylaminopurine (12) and 2-sulfanilylaminopurine (13) derivatives were synthesized as outlined in Scheme 1. Thus, selective reductive dehalogenation of 6-chloro-2-fluoropurine (10), prepared from 2-amino-6-chloropurine (9) following a literature procedure,³² afforded 2-fluoropurine (11). Treatment of 11 with aniline or 4-aminobenzenesulfonamide (sulfanilamide) in 2,2,2-trifluoroethanol (TFE) with catalysis by trifluoroacetic acid (TFA) as described previously³³ gave the respective 2-arylamino purines 12 and 13 in good yield. The required 2-arylamino guanine derivatives 15 and 16 were also readily accessible from commercially available 2-bromohypox-

Scheme 1^a

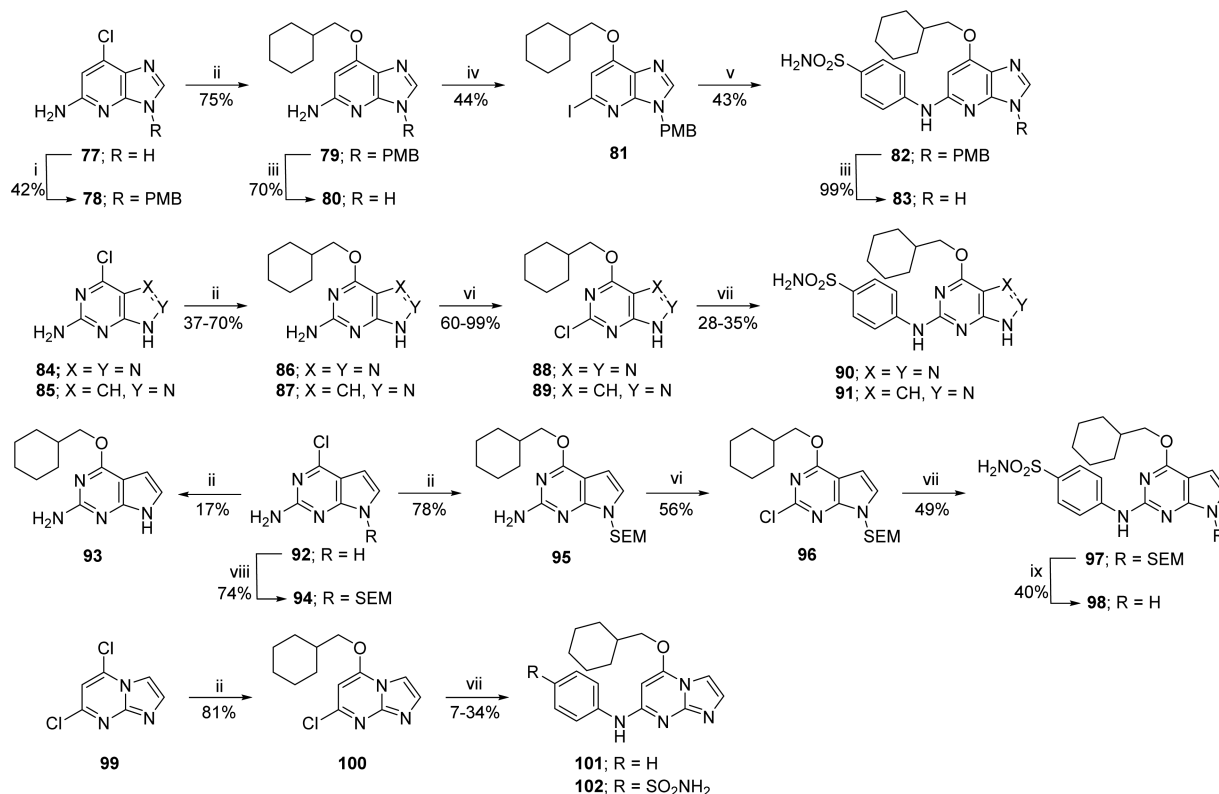
^aReagents and conditions: (i) aq HBF₄, NaNO₂, −15 to 0 °C; (ii) Pd(OH)₂, HCO₂NH₄, MeOH, reflux; (iii) ArNH₂, TFA, TFE, 90 °C; (iv) POCl₃, PhNMe₂, reflux; (v) 3,4-2H-dihydropyran, (*rac*)-camphorsulfonic acid, EtOAc, 75 °C; (vi) R-CCH, PdCl₂(MeCN)₂, dicyclohexyl-(2',6'-triisopropyl-[1,1'-biphenyl]-2-yl)phosphine, Cs₂CO₃, MeCN; (vii) TFA, *i*-PrOH, H₂O, reflux; (viii) DABCO, DMSO, 25 °C. (ix) For 18, 19, 22: ROH, Na, reflux. For 20, 21, 23: ROH, NaH, DMSO, 25 °C. For definition of R groups, see Table 1.

anthine (14) under similar conditions. The 6-alkoxypurine derivatives (30–35) were synthesized by utilizing previously optimized methodology.^{23,25,31} Briefly, introduction of the 6-alkoxy substituent was achieved either by direct reaction of the appropriate sodium alkoxide with 2-amino-6-chloropurine (9) by employing the corresponding alcohol as solvent or, where necessary, following conversion into the “DABCO-purine” intermediate (17).³⁴ Elaboration of the 6-alkoxy-2-aminopurines (18–23) into the corresponding 2-fluoropurines (24–29) under Balz–Schiemann conditions enabled subsequent introduction of the 2-(4-aminobenzenesulfonamide) group using the TFA-TFE procedure to furnish the target purines (30–35) in good overall yields (Scheme 1).

The 6-ethynylpurine derivatives (41 and 42) were synthesized from the THP-protected 6-chloro-2-fluoropurine (36)³⁵ via a Sonogashira alkylation approach (Scheme 2). Coupling of triisopropylacetylene with 36, employing Pd(PPh₃)₂Cl₂/CuI, afforded the protected 6-ethynylpurine (37) in excellent yield, and removal of the N⁹-THP group to give 38 and introduction of the 2-arylamino substituent proceeded in the manner described above to give 39 and 40. Removal of the TIPS group to furnish the target 2-arylamino-6-ethynylpurines 41 and 42 was achieved under standard conditions. The 6-propynyl- and 6-phenylethynyl-purine derivatives 48 and 49 were accessible from the guanine derivative 15 by sequential chlorination at the purine 6-position (44), THP protection (45), introduction of the appropriately substituted ethynyl group under Sonogashira conditions (46, 47), and final N⁹-deprotection to furnish the target purines 48 and 49 (Scheme 1). The 6-ethylpurine derivative (52) was readily obtained by reduction of the 6-ethynyl group of the THP derivative 43 with Lindlar's catalyst-quinoline to afford 50, with N⁹-deprotection (51) and final introduction of the 2-(4-aminobenzenesulfonamide) group, giving 52 in modest overall yield.

Scheme 2^a

^aReagents and conditions: (i) 3,4-2H-dihydropyran, (*rac*)-camphorsulfonic acid, EtOAc, 75 °C; (ii) (*i*-Pr)₃SiCCH, Pd(PPh₃)₂Cl₂, CuI, Et₃N, THF, 25 °C; (iii) TFA, *i*-PrOH, H₂O, reflux; (iv) PhNH₂ or 4-NH₂C₆H₄SO₂NH₂, TFA, TFE, 90 °C; (v) (*n*-Bu)₄NF, THF, 25 °C; (vi) Lindlar's catalyst, quinoline, H₂, EtOAc, RT, 2 h; (vii) RB(OH)₂ or RBF₃K, Pd(OAc)₂, dicyclohexyl-(2',6'-dimethoxybiphenyl-2-yl)-phosphine, K₃PO₄, PhMe, H₂O, 100 °C. For definition of R groups, see Table 1.

Scheme 3^a

^aReagents and conditions: (i) PMB-Cl, K₂CO₃, DMF, 80 °C; (ii) NaH, cyclohexylmethanol, DMSO, 25 °C; (iii) TFA, 80 °C; (iv) CH₂I₂, *iso*-amyl nitrite, CuI, THF, 80 °C; (v) sulfanilamide, Pd₂(dba)₃, XPhos, K₂CO₃, MeCN, 80 °C; (vi) LiCl, *iso*-amyl nitrite, SOCl₂, DMA or THF, 0–25 °C; (vii) ArNH₂, TFA, TFE, 90 °C; (viii) SEM-Cl, NaH, MeCN, 0 °C; (ix) TBAF, THF, 25 °C.

Efforts to synthesize the 6-cyclopropylpurine (**55**) employing Suzuki–Miyaura conditions were unsuccessful, likely due to competing protodeboronation of the cyclopropylboronic acid under the reaction conditions employed. This problem was successfully addressed as reported in the literature by converting the cyclopropylboronic acid into the corresponding potassium trifluoroborate,³⁶ whereby coupling to the THP-protected 6-chloro-2-fluoropurine (**36**) occurred to furnish **53** in acceptable yield. Subsequent transformation into the target 6-cyclopropylpurine (**55**) was achieved readily. The introduction of aryl substituents at the purine 6-position was achieved using Suzuki–Miyaura cross-coupling chemistry with 6-chloro-2-fluoropurine derivative **36** (Scheme 2). Optimization of the reagents and reaction conditions employed for the cross-coupling step enabled the preparation of a defined series of 2-fluoro-6-aryl-purine derivatives (**56**–**62**) in excellent yields, and conversion into the requisite 6-aryl-2-(4-sulfonamidophenyl)-aminopurines (**70**–**76**) proceeded smoothly under the previously described conditions.^{25,33}

The synthesis of prospective CDK2 inhibitors encompassing alternative heterocyclic systems was undertaken as summarized in Scheme 3. In each case, the parent chloroheterocycles (**77**, **84**, **85**, **92**, **99**) were synthesized by adapting literature procedures^{37–41} and the cyclohexylmethyl substituent introduced by treatment with the corresponding alkoxide, with *N*⁹ protected where necessary, as described previously. Iodination (**81**)⁴² or chlorination (**88**, **89**, **96**) of the respective 2-aminoheterocycles under Sandmeyer conditions facilitated introduction of the 2-aryl-amino group under the standard TFA-TFE conditions (**88**, **89**) or employing a Buchwald

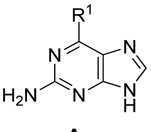
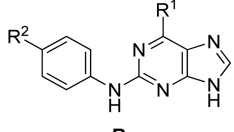
amination procedure (**81**, **96**), followed by deprotection as necessary to afford the target compounds **83**, **90**, **91**, and **98**. The imidazo[1,2-*a*]pyrimidine derivatives (**101**, **102**) were prepared from 5,7-dichloroimidazo[1,2-*a*]pyrimidine (**99**)^{43,44} by regioselective 5-alkoxylation to give **100**, followed by final introduction of the aryl-amino group at the 7-position under TFA-TFE conditions. Given the possibility of alkoxide attack at the 7-position of **99** to furnish the alternative regioisomer, the structure of **102** was unambiguously confirmed by X-ray crystallography (Supporting Information).

RESULTS AND DISCUSSION

The chemical structures, CDK inhibitory activity, and in vitro antitumor activity for the purine derivatives and alternative heterocycles are summarized in Tables 1–3, with the values for compounds 1–3 included for comparative purposes.

Structure–Activity Relationships. The importance of a substituent at the purine 6-position was reaffirmed by the simple 2-amino- and 2-phenylamino-purines (**8** and **12**), where a dramatic reduction in potency was observed compared with the parent 6-cyclohexylmethoxy derivatives **1** and **2**, respectively. A similar effect was also evident for the sulfanilyl-purine **13**, which was some 300-fold less active than **3**. These results are consistent with the putative interaction of this inhibitor class with the ATP-binding site of CDK2, which requires that the 6-substituent occupies a lipophilic pocket close to the ribose binding site, thereby orienting the purine to make the triplet of hydrogen bonds with the hinge region.²⁴ The low-micromolar potency of the 2-sulfanilyl derivative **13** likely reflects the contribution to binding arising through additional interactions

Table 1. Chemical Structures and CDK2-Inhibitory Activity of Purine Derivatives

 A  B									
No.	Structure	R ¹	R ²	CDK2 Inhibition IC ₅₀ (μM) ^a or Inhibition (%) at 100 μM	No.	Structure	R ¹	R ²	CDK2 Inhibition IC ₅₀ (μM) ^a or Inhibition (%) at 100 μM
1	A		–	17 ± 2 ^b	42	B	C≡CH	SO ₂ NH ₂	0.83 ± 0.03
2	B		H	0.97 ± 0.03	48	B	C≡CMe	H	6.4 ± 0.4
3	B		SO ₂ NH ₂	0.005 ± 0.001	49	B	C≡CPh	H	8.9 ± 0.7
8	A	H	–	(4 ± 3%) ^c	52	B	Et	SO ₂ NH ₂	0.22 ± 0.003
12	B	H	H	61 ± 0.1	55	B		SO ₂ NH ₂	0.019 ± 0.001
13	B	H	SO ₂ NH ₂	1.5 ± 0.1	70	B	Ph	SO ₂ NH ₂	0.024 ± 0.002
15	B	OH (C=O) ^d	H	(34 ± 1%)	71	B		SO ₂ NH ₂	0.039 ± 0.013
16	B	OH (C=O) ^d	SO ₂ NH ₂	(52 ± 4%)	72	B		SO ₂ NH ₂	0.052 ± 0.004
30	B	OEt	SO ₂ NH ₂	0.026 ± 0.001	73	B		SO ₂ NH ₂	0.044 ± 0.012
31	B		SO ₂ NH ₂	0.008 ± 0.002	74	B		SO ₂ NH ₂	0.58 ± 0.02
32	B		SO ₂ NH ₂	0.01 ± 0.001	75	B		SO ₂ NH ₂	1.4 ± 0.2
33	B		SO ₂ NH ₂	0.003 ± 0.0003	76	B		SO ₂ NH ₂	6.7 ± 0.5
34	B		SO ₂ NH ₂	0.001 ± 0.0005					
35	B		SO ₂ NH ₂	0.019 ± 0.001					
40	B	C≡CSi(<i>i</i> -Pr) ₃	SO ₂ NH ₂	0.017 ± 0.001					
41	B	C≡CH	H	21 ± 9					

^aIC₅₀ values were determined in accordance with previously described methods.^{20,23} ^bData shown are the mean of at least two independent experiments ± standard deviation. ^cPercent inhibition values are in brackets. ^dThe oxo-tautomers predominate.

between the arylsulfonamide group of **13** with the CDK2 surface adjacent to the ATP binding site on the C-terminal lobe (termed the “specificity surface”) despite the absence of a 6-substituent. Perhaps not surprisingly, the guanine derivatives **15** and **16** were only weakly active, attributable to unfavorable interactions between the 6-oxo functionality and the ribose-binding pocket. These derivatives also exhibited the very poor aqueous solubility characteristic of many guanine derivatives.

Replacement of the 6-cyclohexylmethoxy group of **3** by smaller alkoxy substituents proved informative and was broadly consistent with our earlier investigations with derivatives of **1**.³¹ In that study, 2-amino-6-methoxypurine (*O*⁶-methylguanine) demonstrated negligible activity (CDK2; IC₅₀ > 100 μM), and the relatively poor chemical stability of this purine militated against preparing the corresponding 2-sulfanilyl analogue. The

6-ethoxypurine **30** exhibited high CDK2 inhibitory activity (IC₅₀ = 26 nM), with an approximately 3-fold increase in potency being observed for the *n*-propoxy- (**31**; IC₅₀ = 8 nM) and *iso*-propoxy- (**32**; IC₅₀ = 10 nM) derivatives. Potency was improved further by increasing the size of the 6-alkoxy group, as demonstrated by the activity of the isomeric butoxy analogues **33** and **34** with IC₅₀ values of 3 and 1 nM, respectively. The relationship between the bulk of the 6-alkoxy group and the CDK2-inhibitory activity generally parallels that previously observed for the corresponding 2-amino-6-alkoxy-purines, albeit that the sulfanilylpurines **30**–**34** are several orders of magnitude more potent. For example, **34** (IC₅₀ = 1 nM) is approximately 50000-fold more potent than the corresponding 2-amino-6-*sec*-butoxypurine (CDK2; IC₅₀ = 49 ± 7 μM),³¹ again corroborating the crucial binding

contribution made by the arylsulfonamide function. A propargyloxy group at the purine O⁶-position conferred activity comparable to the ethoxy derivative (compare **35** and **30**), in keeping with the trend reported previously for the 2-amino-6-alkoxypurine series.³¹

The introduction of an ethyl substituent at the purine 6-position (**52**) reduced CDK2-inhibitory activity some 10-fold compared with the 6-ethoxypurine (**30**), and very modest potency was also observed for the 6-ethynyl derivatives **41** and **42**. Although 6-alkoxy and 6-alkyl groups differ with regard to both their electronic and steric character, as well as lipophilicity, the high potency of the 6-triisopropylsilyethynylpurine (**40**; IC₅₀ = 17 nM) shows that steric bulk, as well as electronic factors, is an important feature for binding in the ribose pocket of CDK2. The modest improvement in potency observed for the prop-1-ynyl (**48**) and phenylethynyl (**49**) derivatives compared with the parent 6-ethynylpurine **41** also suggests that the overall shape of the 6-substituent is an important factor for occupancy of the ribose-binding pocket, supported by the activity of the 6-cyclopropylpurine **55** (IC₅₀ = 19 nM). These data also suggest that the oxygen atom of the 6-alkoxypurines, which would act as a weak hydrogen bond acceptor, does not make a significant contribution to binding affinity.

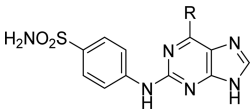
The potency of the 6-phenylpurine (**70**; IC₅₀ = 24 nM) prompted further elaboration, with comparable activity residing in the 3-methoxyphenyl (**71**), 4-methoxyphenyl (**72**), and 6-(3-phenylphenyl)purine (**73**) derivatives. By contrast, the piperonyl derivative (**74**) was some 20-fold less potent than **70**, and the introduction of still larger groups was detrimental, as evident from the very weak CDK2 inhibitory activity of the dibenzofuran-1-yl (**75**) and thianthren-1-yl (**76**) analogues. The bulkier bicyclic and tricyclic heteroaromatic rings of these purines are presumably poorly accommodated within the ribose-binding pocket. It is of interest to note that the CDK2 inhibitory activity of the 6-unsubstituted derivative **13** is equipotent with **75** and superior to that of **76**, implying that the 6-substituents of these purines make a negligible binding contribution.

With a view to assessing selectivity for CDK2, selected purine derivatives were evaluated against a panel of CDKs (Table 2). As expected, removal of the 6-cyclohexylmethyl substituent of **3**, which makes a number of interactions with the conserved ATP ribose binding site to furnish **13** compromised potency against all the CDKs evaluated, although the effect was less pronounced against CDK2.

The 6-alkoxypurines **34** and **35** exhibited good selectivity for CDK2 over the closely related CDK1 (80-fold and 50-fold, respectively), with **34** retaining selectivity over CDKs 4, 7, and 9 comparable with **3**. A similar profile was observed for the 6-phenylpurine **70**, which was some 30-fold selective for CDK2 over CDK1. By contrast, the 6-([1,1'-biphenyl]-3-yl)purine derivative **73** exhibited a very interesting CDK selectivity profile, proving some 2000-fold selective for CDK2 over CDK1 while exhibiting only weak inhibitory activity against CDKs 4, 7, and 9. To the best of our knowledge, this level of selectivity for CDK2 over CDK1 is unprecedented.

Modification of the core purine heterocycle of **1** and **3** afforded further interesting results (Table 3). A comparison of 2-amino-6-cyclohexylmethylpurine **1** with the alternative heterocycles **86**, **87**, and **93** revealed that although these compounds exhibited comparable potency against CDK2, activity against other CDK family members was markedly attenuated. The very weak activity of the imidazopyrimidine

Table 2. CDK Selectivity and Cellular Activity for Selected Compounds



Compd.	R	IC ₅₀ (μM) or % inhibition at 100 μM ^a				
		CDK1/B	CDK2/A	CDK4/D	CDK7/H	CDK9/T
3		0.25 ±	0.005 ±	1.45 ±	4.40 ±	1.07 ±
		0.05 ^b	0.001	0.70	0.90	0.02
13	H	15.8 ± 1.2	1.5 ± 0.09	(12 ± 4%) ^c	75 ± 3	38 ± 11
34		0.097 ±	0.0012 ±	1.30 ±	2.8 ± 0.1	0.57 ±
		0.040	0.0005	0.30		0.05
35		0.94 ± 0.1	0.019 ± 0.001	8.8 ± 1.2	—	—
70	Ph	0.73 ±	0.024 ±	3.8 ± 1.4	6.8 ± 0.4	0.57 ±
		0.003	0.002			0.04
73		86 ± 4	0.044 ± 0.012	(26 ± 2%)	(28 ± 6%)	25 ± 8

^aIC₅₀ values were determined in accordance with previously described methods.^{20,23} ^bData shown are the mean of at least two independent experiments ± standard deviation. ^cPercent inhibition values are in brackets.

derivatives **101** and **102** is not surprising given the absence of the requisite donor–acceptor–donor motif and presumably reflects the combined influence arising from the absence of the N¹ and N⁷ functions found in the corresponding purine derivatives **1** and **3**. The modest CDK inhibition profile of imidazopyridine **83** indicates that removal of the N¹ nitrogen is detrimental to CDK inhibitory activity. However, with the exception of the imidazopyridine (**83**) and imidazopyrimidine (**102**) derivatives, modification of the core purine heterocycle of **3** was generally tolerated without a marked loss of CDK2 inhibitory activity. Thus, triazolopyrimidine (**90**) analogue is 2.5-fold more potent than **3**, while pyrazolopyrimidine (**91**) and pyrrolopyrimidine (**98**) analogues proved only some 4–5-fold less potent than **3** against CDK2 and retained good selectivity over all other CDK family members examined. Notably, derivatives **91** and **98** were approximately 100-fold selective for CDK2 over CDK1.

Cellular Studies. The ability of compounds to inhibit cellular proliferation was examined in a 5-day growth assay using five histologically distinct retinoblastoma protein (Rb)-proficient human tumor lines (A375 melanoma, Calu-6 lung carcinoma, MDA-MB-231 breast adenocarcinoma, SJSA1 osteosarcoma, and SKUT-1B uterine corpus leiomyosarcoma). While compound **73** demonstrated appreciable nanomolar potency versus CDK2 in isolated kinase assays, the maximal concentration tested in cellular assays (30 μM) had no or limited effects on the growth of cells, thereby preventing the concentration that induced a 50% growth inhibitory (GI₅₀) effect to be determined (data not shown). In contrast, compounds **91** and **98**, which were only 2.4- and 1.7-fold more potent respectively in CDK2 kinase assays, dose-dependently inhibited the growth of each tumor cell line with GI₅₀ values ranging from 1–22 μM. The cellular activity of

Table 3. Chemical Structures and CDK Inhibitory Activity of Other Heterocyclic Derivatives

No.	Structure	R	CDK Inhibitory Activity					
			IC ₅₀ (μM) ^a or Inhibition at 10 μM (%)					
			CDK1	CDK2	CDK4	CDK5	CDK7	CDK9
1		H	26 ± 4 ^b	17 ± 2	(33 ± 9%) ^c	97 ± 3	(44 ± 6%)	(42 ± 11%)
3			0.25 ± 0.05	0.005 ± 0.001	1.5 ± 0.7	0.48 ± 0.07	4.4 ± 0.9	1.1 ± 0.2
80		H	-	28.4 ± 0.8	-	-	-	-
83			11 ± 2	0.29 ± 0.01	23 ± 1%	(37 ± 2%)	(37 ± 4%)	(42 ± 5%)
86		H	(21 ± 1%)	18 ± 1	(6.2 ± 3.6%)	(13 ± 1%)	(13 ± 6%)	(16 ± 8%)
90			0.75 ± 0.06	0.002 ± 0.0001	9.6 ± 0.5	3.3 ± 0.2	6.8 ± 0.7	0.75 ± 0.03
87		H	(32 ± 1%)	16 ± 4	(0%)	(14 ± 9%)	(16 ± 7%)	(16 ± 2%)
91			1.6 ± 0.2	0.018 ± 0.001	2.8 ± 0.3	5.9 ± 0.1	7.0 ± 0.3	4.6 ± 0.2
93		H	(11 ± 3%)	(41 ± 10%) ^d	(5.6 ± 7.9%)	(12 ± 1%)	(5.1 ± 5.5%)	(0%)
98			2.7 ± 0.2	0.026 ± 0.001	1.9 ± 0.1	4.4 ± 0.6	4.1 ± 0.2	1.0 ± 0.2
101		H	-	(10 ± 6%)	-	-	-	-
102		SO ₂ NH ₂	-	(36 ± 3%)	-	-	-	-

^aIC₅₀ values were determined in accordance with previously described methods.^{20,23} ^bData shown are the mean of at least two independent experiments ± standard deviation. ^cPercent inhibition values are in brackets. ^dValue determined at 100 μM.

these compounds compared favorably to that of compound 3 (Figure 1).

Structure Determination. Compounds within this series show greater activity toward CDK2 than CDK1, generally varying between 10- and 80-fold (Table 2). However, 73 exhibits exceptional discrimination being a ca. 2000-fold more potent inhibitor of CDK2. To confirm the binding mode and to identify potential interactions that might explain the selectivity of 73 for CDK2 over CDK1, 73 and 3 were cocrystallized with CDK2-cyclin A and CDK1-cyclin B-cyclin-dependent kinases regulatory subunit 2 (CKS2), respectively. The poor potency of 73 toward CDK1 appears to preclude determination of a CDK1-cyclin B-CKS2–73 cocomplex structure, as despite repeated attempts we were unable to visualize the compound in the active site. The statistics for the data sets and for the crystallographic refinements are presented in Table 4. Compound 73 emulates the interactions made by 3 within the CDK2 active site (Figure 2A).²⁴ Indeed, the purine and aniline rings of each inhibitor superimpose very well. The purine rings occupy the CDK2 ATP binding site, and each

makes a triplet of hydrogen bonds with the backbone carbonyl of Glu81 and the amide NH and carbonyl moieties of Leu83 within the hinge sequence (Figure 2A). The respective anilino rings project out of the CDK2 catalytic site so that the sulfonamide NH₂ group and one of the oxygens in each make hydrogen bonds with the side chain and backbone nitrogen of Asp86, respectively.

The structures of 73 and 3 differ in the substitution present at the purine C-6 position. The O⁶-cyclohexylmethyl substituent of 3 occupies the ATP ribose binding pocket and is complementary in shape and forms favorable hydrophobic interactions with an apolar pocket created by the conformation of the CDK2 glycine-rich loop (residues 9–19).²⁴ This loop adopts an identical backbone structure when bound to 73 so that the purine-proximal phenyl ring emulates the position of the cyclohexylmethyl substituent of 3. The distal phenyl ring substituted at the *meta* position can then be comfortably accommodated as it twists toward the aniline moiety (Figure 2A). In contrast, *para* substitutions of the purine-proximal phenyl ring exemplified by 72, 74, 75, and 76 (Table 1) would

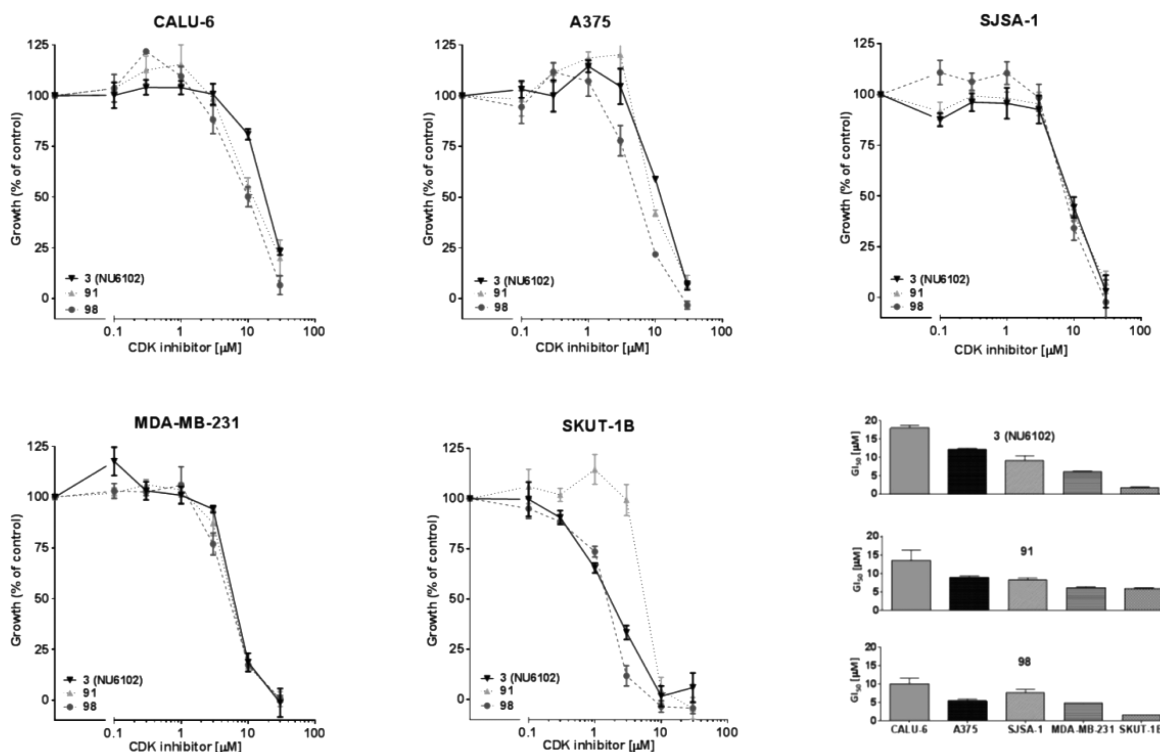


Figure 1. Activity of compounds **3**, **91**, and **98** against tumor cell growth.⁴⁵ Rb-proficient human tumor cell lines were incubated for 120 h with compounds and cellular protein determined by SRB assay. Dose–response curves and GI₅₀ bar charts represent the mean $\pm$ SEM from three or four independent experiments.

Table 4. X-ray Data Collection and Refinement Statistics

	CDK2-cyclin A– 73 ^a	CDK1-cyclin B-CKS2–3 ^a
Data Collection		
space group	<i>P</i> 2 ₁ 2 ₁ 2 ₁	<i>P</i> 1
unit cell (Å)	<i>a</i> = 73.5, <i>b</i> = 132.1, <i>c</i> = 149.2	<i>a</i> = 65.0, <i>b</i> = 67.8, <i>c</i> = 85.1, α = 103.9, β = 90.9, γ = 90.4
resolution (Å)	66.03–2.97	65.76–2.06
(highest resolution shell)	(3.13–2.97)	(2.10–2.06)
total observations	226370 (7244)	169709 (8389)
unique	30773 (1098)	84841 (4161)
<i>R</i> _{merge} ^a	0.086	0.092
mean <i>I</i> / σ (<i>I</i>)	6.7 (1.6)	6.2 (1.2)
completeness %	100 (99.7)	97.3 (95.5)
Refinement		
total number of atoms		
protein	8927	20449
other	64	100
waters	18	291
<i>R</i> (highest resolution shell)	0.213	0.198
<i>R</i> _{free} (highest resolution shell)	0.266	0.254
rmsd bonds (Å)	0.0122	0.0217
rmsd angles (deg)	1.770	2.233

^aThe structures have been deposited in the PDB with accession codes 5LQE (CDK2-cyclin A–73) and 5LQF (CDK1-cyclin B-CKS2–3).

be directed out of the CDK2 active site toward the tip of the glycine-rich loop. Smaller substitutions, for example a methoxy in **72** can be accommodated but larger ring systems as

exemplified by **74** and **76** lead to considerable drops in potency, suggesting they may sterically clash with the CDK2 structure in this region (Table 1).

To probe further the binding mode of the series, CDK1-cyclin B-CKS2 was cocrystallized with **3** (Figure 2B). This structure shows that the purine backbone emulates the interactions made by this inhibitor within the CDK2 binding site and that the *O*⁶-cyclohexylmethyl substituent occupies the ribose binding pocket. As previously reported,⁴⁶ inhibitor binding within the ATP binding site is accompanied by minor remodelling of the CDK1-cyclin B into a conformation compatible with catalysis.

All of the side chains contacted by **73** in the co-complex with CDK2-cyclin A are conserved in CDK1 so that selectivity must derive from indirect readout of remote sequence differences. One mechanism by which such differences can impact inhibitor binding is where they shape the conformational energy landscape, permitting conformations in one kinase that are precluded in another. As described above, the glycine-rich loop makes a significant contribution to the catalytic cleft, shaping the binding site that accommodates the purine C-6 substituents. We note that the glycine-rich loop in the CDK2-cyclin A–73 structure adopts a conformation in which Tyr15 is folded into the active site, contacting residues of the C-helix (Figure 2C). This position for the side chain of Tyr15 is seen in a number of other CDK2-cyclin A–inhibitor complexes, suggesting that it reflects a preferred conformation of CDK2. However, in available structures of apo CDK1-cyclin B, and CDK1-cyclin bound to compound **23**⁴⁶ or **3** (this paper), a comparable location for the side chain of Tyr15 has not been observed (Figure 2C). We speculate, therefore, that **73** binds more tightly to CDK2 than to CDK1 because in doing so, it

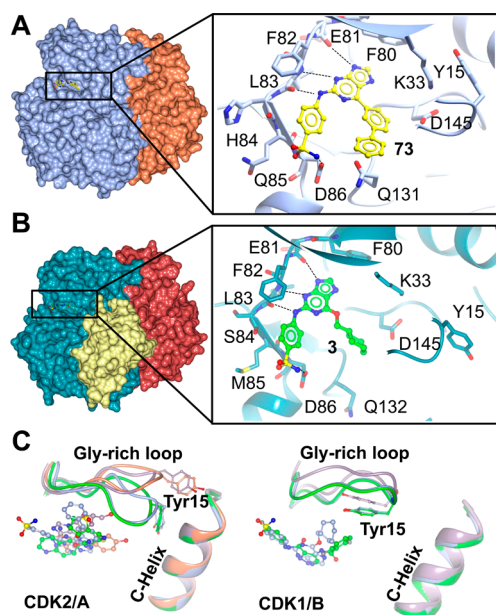


Figure 2. Structures of CDK2-cyclin A-73 and CDK1-cyclin B-CKS2-3. (A) Surface representation of CDK2 (ice-blue) bound to cyclin A (coral) in complex with 73 (yellow). The inset shows the interactions of 73 within the ATP site of CDK2-cyclin A. (B) Surface representation of CDK1 (dark-cyan) bound to cyclin B (pale-crimson) and CKS2 (lemon) in complex with 3 (green). The inset shows the interactions of 3 within the ATP site of CDK1-cyclin B-CKS2. In (A) and (B), the protein backbone is rendered in ribbon representation and selected CDK2 and CDK1 residues are drawn respectively with carbon atoms colored ice-blue and dark-cyan. The hydrogen bonds are shown as black dotted lines. (C) The structural conformations of the CDK2-cyclin A and CDK1-cyclin B-CKS2 glycine-rich loops to illustrate the alternative poses of catalytic residue Tyr15. The left-hand side panel compares CDK2-cyclin A in complex with 73 (ice-blue) to other inhibitor bound CDK2-cyclin A complexes (PDB entries 4EOS (green), 3TNW (coral), and 3MY5 (lilac)). The right-hand panel overlays CDK1-cyclin B-CKS2 bound to 3 (ice-blue) with apo CDK1-cyclin B-CKS2 (PDB entry 4YC3, lilac) and CDK1-cyclin B-CKS2 in complex with compound 23 (PDB entry 5HQ0, green). The conformations of the loop when CDK1 is bound to 3 or compound 23 cannot be distinguished. Gly-rich loop, glycine-rich loop.

stabilizes a glycine-rich loop conformation that is preferred in CDK2 but not in CDK1.

CONCLUSIONS

Novel 6-substituted 2-(4'-sulfamoylanilino)purines have been designed as competitive inhibitors acting at the ATP binding site of CDK2 with particular attention being given to abrogating activity against CDK1. A variety of substituents were explored, either attached directly to C-6 or via an oxygen link, in the context of their possible interaction with a lipophilic pocket close to the ATP ribose binding site. The relationship between the size of a 6-alkoxy group and CDK2-inhibitory activity was found to parallel that previously observed for the corresponding 2-amino-6-alkoxypurines. In general, the new compounds were significantly less potent (typically 10–80×) against CDK1 than CDK2. Most impressive was 4-((6-([1,1'-biphenyl]-3-yl)-9H-purin-2-yl)amino) benzenesulfonamide that was ~2000-fold less active toward CDK1 (IC_{50} 86 μ M), while retaining high potency against CDK2 (0.044 μ M). Compounds substituted with relatively large conformationally constrained groups, e.g. bicyclic and tricyclic aromatic systems, showed

greatly reduced inhibitory activity, indicating poorly accommodation of these substituents in the lipophilic binding site. Analogues of 6-cyclohexylmethoxy-2-(4'-sulfamoylanilino)-purine, in which the purine ring was replaced by a triazolopyrimidine, pyrazolopyrimidine or pyrrolopyrimidine were only marginally less active against CDK2 than the parent purine. However, replacement with an imidazopyridine or imidazopyrimidine gave much less potent derivatives. Co-crystal structures of inhibitors bound to CDK2 and CDK1 revealed that the binding mode of the purine-based inhibitor series is conserved. We show that inhibitor binding to CDK2 stabilizes a glycine-rich loop conformation that shapes the ATP ribose binding pocket, resulting in effective inhibition of CDK2. We propose that this region of the active site might be the basis of the design of further inhibitors differentiating between CDK1 and CDK2.

EXPERIMENTAL SECTION

General Synthetic Procedures. Chemicals and solvents were obtained from standard suppliers. Solvents were either dried by standard techniques or purchased as anhydrous. Reactions needing microwave irradiation were carried out in an Initiator Sixty Biotage apparatus. Petrol refers to petroleum ether (bp 40–60 °C, reagent grade, Fisher Scientific). All reactions that required inert or dry atmosphere were carried out under a blanket of nitrogen, which was dried by passage through a column of phosphorus pentoxide. Glassware was dried in an oven prior to use. Column chromatography was carried out using 40–60 μ m mesh silica in glass columns under medium pressure or with a Biotage SP4 flash purification system using KP-Si. Thin layer chromatography (TLC) was performed on 20 mm precoated plates of silica gel (Merck, silica gel 60F254); visualization was achieved using ultraviolet light (254 nm). NMR spectra were recorded on a Bruker Spectrospin AC 300E (300 MHz) NMR Spectrometer or Bruker BioSpin UltraShield Plus 500 MHz using deuterated solvent as a lock. IR spectra were recorded on a Bio-Rad FTS 3000MX diamond ATR, and UV analysis was performed using a Hitachi U-2000 spectrophotometer. LC-MS analysis was carried out on a Micromass Platform instrument operating in positive and negative ion electrospray mode, employing a 50 mm $\times$ 4.6 mm C18 column (Supelco Discovery or Waters Symmetry) and a 15 min gradient elution of 0.05% formic acid and methanol (10–90%). HRMS were measured using a Finnigan MAT 95 XP or a Finnigan MAT 900 XLT by the EPSRC National Mass Spectrometry Service Centre (Swansea). The purity of final compounds was assessed by reversed-phase HPLC; all tested compounds were >95% purity. HPLC instrument, Agilent 1200 equipped with a photodiode array detector (190–400 nm). Sample temperature, ambient; injection volume, 5 μ L; flow rate, 1 mL/min. 5–100% MeCN gradient over 9 min and an isocratic hold at 100% MeCN for 2.5 min, before returning to initial conditions. Mobile phase A = 0.1% ammonia in water or 0.1% formic acid in water, mobile phase B = MeCN. Column: Waters XSELECT CSH C18, 3.5 μ m, 4.6 mm $\times$ 150 mm or Waters XTerra RP18, 5 μ m, 4.6 mm $\times$ 150 mm. Column maintained at ambient temperature.

General Procedure A. To a stirred suspension of the appropriate haloheterocycle (1.0 mol equiv) and 4-aminobenzenesulfonamide (2.0 mol equiv) in TFE (25 mL/g of haloheterocycle) was added TFA (2.5 mL/g of haloheterocycle) dropwise. The resulting solution was heated under reflux for 12–48 h under a nitrogen atmosphere. The solvent was removed in vacuo, and the residue was redissolved in EtOAc (10–20 mL). The solution was washed with saturated aqueous sodium bicarbonate solution (3 $\times$ 10 mL), and the aqueous extracts were combined and washed with EtOAc (2 $\times$ 15 mL). The combined organic layers were dried (Na_2SO_4) and the solvent removed under reduced pressure to give a residue that was purified in the manner indicated.

6-Chloro-2-fluoro-9H-purine (10).⁴⁷ To a stirred solution of HBF₄ (48% aqueous, 120 mL) at 0 °C was added 2-amino-6-chloropurine (9) (6.0 g, 35.0 mmol). Over 20 min, a solution of NaNO₂ (4.9 g, 70.0

mmol) in water (200 mL) was added dropwise, ensuring the temperature remained close to 0 °C. The pale-yellow solution was raised to room temperature and stirred for 18 h. The resulting solution was neutralized to pH 7 in an ice bath at 0 °C by addition of Na₂CO₃ (6.00 g) in water (200 mL). Solvents were removed in vacuo, and the residual solid was redissolved in MeOH (100 mL) and adsorbed onto silica (250 mL). The crude material was purified by chromatography (silica; 10% MeOH:DCM) to afford **10** as a white crystalline solid (4.52 g, 75%); mp 171–173 °C (lit.⁴⁷ mp 174 °C); UV λ_{max} (EtOH) 393 nm. IR (cm⁻¹) 2964, 2785, 1735, 1581. ¹H NMR (500 MHz, DMSO-*d*₆) δ 8.60 (1H, s, H-8), 13.9 (1H, s, NH). LRMS (ES⁺) *m/z* 172.6 [M + H]⁺.

2-Fluoro-9H-purine (11).^{48,49} To a stirred suspension of 6-chloro-2-fluoropurine (**10**) (0.30 g, 1.74 mmol) and palladium hydroxide on carbon (20% w/w, 0.30 g) in methanol (15 mL) was added ammonium formate (0.34 g, 5.35 mmol). The suspension was heated under reflux for 1 h before filtering through a pad of Celite, eluting with methanol (20 mL). Removal of volatiles under reduced pressure afforded **11** as a white solid (240 mg, 100%); mp 219 °C (dec.) (lit.⁴⁸ decomposed at 216 °C). ¹H NMR (300 MHz, DMSO-*d*₆) δ 8.60 (1H, s, H-8), 9.01 (1H, s, H-6), 13.9 (1H, s, NH). LRMS (ES⁺) *m/z* 139.2 [M + H]⁺.

Phenyl-(9H-purin-2-yl)amine (12). Prepared from aniline (0.18 mL, 2 mmol) and 2-fluoro-9H-purine (**11**, 0.138 g, 1.0 mmol) in accordance with general procedure A. The reaction mixture was stirred under reflux for overnight. After removal of the solvent in vacuo, the residue was dissolved in EtOAc (10–30 mL), washed with saturated aqueous NaHCO₃ (3 × 30 mL), and dried (MgSO₄). The crude material was purified by chromatography (silica; 0–20% MeOH:DCM), followed by recrystallization from DCM to afford **12** as an off-white solid (65 mg, 31%); *R*_f 0.20 (5% MeOH:DCM); mp 200–201 °C; UV λ_{max} (EtOH) 328, 270, 239, 206 nm. IR (cm⁻¹) 3234, 3103, 2804, 1624, 1602, 1539, 1402, 1292, 1217. ¹H NMR (300 MHz, DMSO-*d*₆) δ 6.91 (1H, t, *J* = 7.5 Hz, H-4'), 7.28 (2H, t, *J* = 7.5 Hz, H-3' and H-5'), 7.83 (2H, d, *J* = 9.0 Hz, H-2' and H-6'), 8.82 (1H, s, H-6), 8.24 (1H, s, H-8), 9.53 (1H, s, NH), 12.9 (1H, s, NH). LRMS (ES⁺) *m/z* 212.0 [M + H]⁺. HRMS calcd for C₁₁H₁₀N₅ [M + H]⁺ 212.0931, found 212.0933.

4-(9H-Purin-2-ylamino)benzenesulfonamide (13). Prepared following general procedure A from 2-fluoro-9H-purine (**11**, 0.10 g, 0.725 mmol). Recrystallization from MeOH:H₂O gave **13** as a pale-white solid (74 mg, 35%); *R*_f 0.40 (20% MeOH:DCM); mp >320 °C (dec); UV λ_{max} (EtOH) 287, 213 nm. IR (cm⁻¹) 3223, 1585, 1481, 1307, 1251, 1148, 1091. ¹H NMR (300 MHz, DMSO-*d*₆) δ 7.17 (2H, s, SO₂NH₂), 7.73 (2H, d, *J* = 9.0 Hz, H-3' and H-5'), 7.99 (2H, d, *J* = 9.0 Hz, H-2' and H-6'), 8.33 (1H, s, H-6), 8.00 (1H, s, H-8), 10.00 (1H, s, NH). LRMS (ES⁺) *m/z* 291.0 [M + H]⁺. HRMS calcd for C₁₁H₁₁N₆O₂S [M + H]⁺ 291.0659, found 291.0659.

N²-Phenylguanine 2,2,2-trifluoroacetate (15). The title compound was prepared following general procedure A using 2-bromohypoxanthine (**14**, 1.00 g, 4.7 mmol) and aniline (0.9 mL, 9.40 mmol) to yield **15** as a white solid (1.17 g, 73%). The isolated compound was pure by analytical HPLC without the need for further purification; mp 229–231 °C; UV λ_{max} (EtOH) 273 nm. IR (cm⁻¹) 3332, 3128, 2943, 2756, 2555, 2387, 1678, 1572. ¹H NMR (300 MHz, DMSO-*d*₆) δ 7.07 (1H, t, *J* = 7.5 Hz, H-4'), 7.36 (2H, dd, *J* = 7.5, 8.0 Hz, H-3' and H-5'), 7.62 (2H, d, *J* = 8.0 Hz, H-2' and H-6'), 7.94 (1H, s, H-8), 8.46 (1H, br s, NH), 9.00 (1H, br s, NH). ¹³C NMR (75 MHz, CDCl₃) δ 113, 120, 123, 129, 138, 139, 150, 152, 155. LRMS (ES⁺) *m/z* 228.3 [M + H]⁺. HRMS calcd for C₁₁H₁₀N₅O [M + H]⁺ 228.0881, found 228.0880.

6-Oxo-2-((4-sulfamoylphenyl)amino)-6,9-dihydro-1H-purine 2,2,2-trifluoroacetate (16). Synthesized in accordance with general procedure A from 2-bromohypoxanthine (**14**, 0.10 g, 0.47 mmol) to yield **16** as an amorphous white solid (60 mg, 42%); mp >300 °C; UV λ_{max} (EtOH) 275 nm. IR (cm⁻¹) 3348, 3113, 2990, 2877, 2792, 2333, 1672, 1575, 1365, 1153. ¹H NMR (300 MHz, DMSO-*d*₆) δ 7.28 (2H, s, SO₂NH₂), 7.79 (4H, m, H-2', H-3', H-5' and H-6'), 8.31 (1H, s, H-8), 9.25 (1H, br s, NH), 10.84 (1H, br s, NH). ¹³C NMR (75 MHz,

CDCl₃) δ 113, 120, 123, 127, 138, 139, 150, 152, 156. LRMS (ES⁺) *m/z* 307.3 [M + H]⁺.

1-(2-Amino-9H-purin-6-yl)-4-aza-1-azoniabicyclo[2.2.2]octane (17).³⁴ 1,4-Diazabicyclo[2.2.2]octane (9.90 g, 88.4 mmol) was added to a solution of 2-amino-6-chloropurine **9** (5.00 g, 29.5 mmol) in DMSO (100 mL) over 1 h, and the mixture was stirred at room temperature for 24 h. The resulting white precipitate was filtered and washed with diethyl ether. The solid was suspended in DCM (200 mL) and stirred for 1 h. After filtration and washing with DCM several times, **9** was obtained as a white solid, which was dried in vacuo (7.95 g, 96%) and used without further purification; mp 230 °C (dec) (lit.³⁴ decomposed at 230 °C). ¹H NMR (300 MHz, D₂O) δ 3.31 (6H, t, *J* = 7.5 Hz, N(CH₂)₃), 4.07 (6H, t, *J* = 7.5 Hz, N⁺(CH₂)₃), 8.13 (1H, s, H-8). ¹³C NMR (75 MHz, D₂O) δ 38.7, 53.4, 116.0, 143.7, 151.3, 158.4.

General Procedure B. 2-Amino-6-chloropurine (**9**, 1.0 mol equiv) was added to a solution prepared from metallic sodium (5.0 mol equiv) dissolved in the appropriate alcohol (3.4 mL/mmol). The mixture was stirred at reflux until LCMS analysis indicated the absence of starting materials (3–24 h). After cooling, the reaction mixture was neutralized with glacial AcOH and the volatile material was removed in vacuo. Unless otherwise indicated, purification was achieved either by recrystallization from H₂O or by adding H₂O to the reaction mixture and extracting the product into EtOAc (3 × 100 mL), followed by drying (MgSO₄) and removal of the solvent in vacuo.

General Procedure C. The appropriate alcohol (4.0 mol equiv) was added dropwise to a stirred suspension of NaH (3.0 mol equiv) in DMSO (2.5–3.0 mL/mmol), and the resulting mixture was stirred for 1–2 h. To this was added DABCO-purine (**17**, 1.0 mol equiv) or the appropriate haloheterocycle (1.0 mol equiv), and the mixture was stirred for 24 h with heating as specified. Water (20–200 mL) was added, and the basic emulsion was neutralized with glacial acetic acid. The aqueous phase was extracted with EtOAc (3 × 50–100 mL), and the organic layers were washed with saturated aqueous NaCl (100 mL). The combined organic layers were dried (MgSO₄) and concentrated in vacuo to yield the crude product, which was purified by chromatography on silica and/or recrystallization from an appropriate solvent.

General Procedure D. To a stirred solution of hydrofluoroboric acid (50%, aq, 20.0 mol equiv) cooled below –20 °C was added the appropriate 2-amino-6-alkoxypurine (1.0 mol equiv). While maintaining the temperature at –15 °C, a solution of NaNO₂ (2.0 mol equiv) in H₂O (1–3 mL/mmol NaNO₂) was added dropwise over 10 min. The mixture was stirred at room temperature for 3 h and neutralized at –15 °C by the dropwise addition of 15% (w/v) aqueous Na₂CO₃ solution, and the precipitated solid was collected by filtration and washed with H₂O. The residual solid was triturated with EtOAc (3 × 100 mL) and filtered. The combined filtrates were concentrated under reduced pressure to furnish the product, which was purified as indicated.

General Procedure E. The appropriate 9-(tetrahydro-2H-pyran-2-yl)-9H-purine (1.0 mol equiv) was dissolved in 2-propanol (60 mL/g) and deionized water (20 mL/g). Trifluoroacetic acid (10.0 mol equiv) was added, and the reaction mixture was stirred at 100 °C for 2 h and cooled to room temperature, and the solution was adjusted to pH 8 by addition of conc aqueous ammonia solution. The volume of solvent was reduced by 50% under reduced pressure; if a precipitate resulted, this was collected and washed with 2-propanol (2 × 10 mL). If precipitation was not observed following neutralization, the reaction mixture was partitioned between EtOAc (20 mL) and saturated aqueous NaCl solution (20 mL). The organic layer was dried (Na₂SO₄) and evaporated under reduced pressure, and the product was purified by chromatography as specified.

General Procedure F. Tetrabutylammonium fluoride solution (1.0 M in THF, 1.5 mol equiv) was added dropwise to a stirred solution of the appropriate 6-triisopropylsilylthynylpurine (1.0 mol equiv) in anhydrous THF (10–20 mL), and the mixture was stirred for 5–15 min at ambient temperature under N₂. Volatiles were evaporated under reduced pressure, and the residue was redissolved in EtOAc (100 mL/g) and washed with saturated aqueous NaCl (100 mL/g),

and the organic fraction was removed in vacuo. The product was purified as described.

General Procedure G. An oxygen-free solution of the required boronic acid or potassium trifluoroborate salt (1.09 mmol), the appropriately 2-substituted 6-chloro-9-(tetrahydro-2H-pyran-2-yl)-9H-purine (0.78 mmol), dicyclohexyl(2',6'-dimethoxybiphenyl-2-yl)-phosphine (SPhos) (2.5 mol %), and palladium acetate (1.0 mol %) in toluene (4 mL) was degassed by bubbling nitrogen through the solution in a sealed vial for 5 min. To the pale-yellow solution was added K_3PO_4 (1.56 mmol) and water (approximately 50 μ L). The solution was again degassed for 15 min and subsequently heated to 100 °C for 18 h. The black-brown suspension was filtered through Celite, eluting with MeOH (3 $\times$ 10 mL), and the product was isolated by chromatography as indicated.

2-Amino-6-ethoxypurine (18).^{50,51} Treatment of EtOH (100 mL) with Na (3.38 g, 147.5 mmol), followed by addition of 2-amino-6-chloropurine **9** (5.0 g, 29.5 mmol) according to general procedure B, afforded the crude product. Recrystallization from H_2O gave **18** as a white powder (4.75 g, 90%); R_f 0.26 (10% MeOH:DCM), mp 247 °C (dec) (lit.⁵¹ mp 230 °C (dec), lit.⁵⁰ mp 293 °C (dec)). IR (cm⁻¹) 3321, 2981, 2363, 2143, 1619, 1584, 1508, 1458, 1427, 1396, 1375, 1342, 1285, 1223, 1161, 1118 cm⁻¹. ¹H NMR (300 MHz, DMSO- d_6) δ 1.35 (3H, t, J = 7.1 Hz, CH_3), 4.44 (2H, q, J = 7.1 Hz, CH_2), 6.21 (2H, s, NH_2), 7.81 (1H, s, H-8). ¹³C NMR (75 MHz, DMSO- d_6) δ 14.9, 61.6, 160.1. LRMS (ES⁺) m/z 180.3 [M + H]⁺. Anal. Found: C, 47.11; H, 4.97; N, 38.97. $C_7H_9N_5O$ requires: C, 46.92; H, 5.06; N, 39.09.

2-Amino-6-n-propoxypurine (19).^{50,51} Treatment of 1-propanol (20 mL) with Na (0.68 g, 29.5 mmol), followed by addition of 2-amino-6-chloropurine **9** (1.0 g, 5.9 mmol) according to general procedure B, gave the crude product. Recrystallization from H_2O afforded **19** (1.06 g, 93%) as a yellow solid, R_f 0.21 (10% MeOH:DCM), mp 196–198 °C (lit.⁵¹ mp 199–201 °C, lit.⁵⁰ mp 208 °C). IR (cm⁻¹) 3484, 3389, 3318, 3203, 2965, 2935, 2882, 2522, 2361, 2338, 1578, 1507, 1449, 1398, 1362, 1331, 1276, 1225, 1165, 1119. ¹H NMR (300 MHz, DMSO- d_6) δ 0.97 (3H, t, J = 7.4 Hz, CH_3), 1.76 (2H, tq, J = 7.2, 7.2 Hz, $CH_2CH_2CH_3$), 4.34 (2H, t, J = 6.8 Hz, OCH_2), 6.20 (2H, s, NH_2), 7.82 (1H, s, H-8), 12.37 (1H, br s, NH). ¹³C NMR (75 MHz, DMSO- d_6) δ 10.7, 22.2, 67.3, 138.5, 160.1, 160.3, 162.2. LRMS (ES⁺) m/z 194.17 [M + H]⁺. Anal. Found: C, 46.81; H, 5.95; N, 33.78. $C_8H_{11}N_5O \cdot 0.7H_2O$ requires: C, 46.69; H, 6.07; N, 34.03.

2-Amino-6-isopropoxypurine (20).³¹ Treatment of 2-propanol (0.85 mL, 14.2 mmol) with NaH (0.26 mg, 10.7 mmol) in DMSO (10 mL), followed by addition of **17** (1.0 g, 3.6 mmol) was performed according to general procedure C, to afford the crude product. Purification by chromatography (silica; 5–10% MeOH:DCM) afforded **20** as a yellow oil. Trituration of the oil with diethyl ether afforded **20** (0.38 g, 55%) as an off-white solid; R_f 0.28 (10% MeOH:DCM), mp 205–207 °C (lit.³¹ mp 209–210 °C). IR (cm⁻¹) 3491, 3331, 3195, 2976, 2773, 1618, 1580, 1504, 1445, 1391, 1372, 1321, 1272, 1224, 1180, 1144. ¹H NMR (300 MHz, DMSO- d_6) δ 1.33 (6H, d, J = 6.2 Hz, 2 $\times$ CH_3), 5.47 (1H, septet, J = 6.2 Hz, OCH), 6.17 (2H, s, NH_2), 7.81 (1H, s, H-8), 12.39 (1H, br s, NH). ¹³C NMR (75 MHz, DMSO- d_6) δ 22.3, 68.2, 160.1. LRMS (ES⁺) m/z 194.15 [M + H]⁺. Anal. Found: C, 49.48; H, 5.64; N, 35.71. $C_8H_{11}N_5O \cdot 0.1H_2O$ requires: C, 49.27; H, 5.79; N, 35.91.

2-Amino-6-(2-methyl-1-propoxy)purine (21).³¹ 2-Methyl-1-propanol (1.31 mL, 14.2 mmol) was added to NaH (0.25 mg, 10.7 mmol) in DMSO (10 mL), followed by addition of **17** (1.0 g, 3.6 mmol) according to general procedure C, to afford the crude product. Trituration of the residual yellow oil with petrol furnished a cream solid. Purification by chromatography (silica; 5–10% MeOH:DCM) gave **21** (0.42 g, 57%) as pale-yellow crystals; R_f 0.41 (10% MeOH:DCM), mp 180–182 °C (lit.³¹ mp 89–92 °C). IR (cm⁻¹) 3512, 3473, 3359, 2958, 2875, 1620, 1575, 1507, 1451, 1386, 1363, 1332, 1272, 1223, 1159, 1121. ¹H NMR (300 MHz, DMSO- d_6) δ 0.98 (6H, d, J = 6.7 Hz, 2 $\times$ CH_3), 2.07 (1H, septet, J = 6.7 Hz, OCH_2CH), 4.17 (2H, d, J = 6.8 Hz, OCH_2), 6.21 (2H, s, NH_2), 7.82 (1H, s, H-8). ¹³C NMR

(75 MHz, DMSO- d_6) δ 19.4, 27.8, 71.9, 160.1, 162.2. LRMS (ES⁻) m/z 206.81 [M - H]⁻.

2-Amino-6-(1-methylpropoxy)purine (22).³¹ Treatment of 2-butanol (20 mL) with Na (0.68 g, 29.5 mmol), followed by addition of 2-amino-6-chloropurine **9** (1.0 g, 5.9 mmol) was performed according to general procedure B, to give a pale-yellow solid. H_2O was added, and the product was extracted into EtOAc (3 $\times$ 100 mL), dried ($MgSO_4$), and the solvent removed in vacuo to yield **22** (0.90 g, 74%) as off-white crystals, R_f 0.3 (10% MeOH:DCM); mp 77 °C (dec) (lit.³¹ mp 88–90 °C). IR (cm⁻¹) 3322, 3190, 2971, 2936, 2876, 2772, 1616, 1575, 1504, 1451, 1393, 1375, 1326, 1273, 1222. ¹H NMR (300 MHz, DMSO- d_6) δ 0.91 (3H, t, J = 7.4 Hz, $CH_3CH_2CHCH_3$), 1.29 (3H, d, J = 6.2 Hz, $CH_3CH_2CHCH_3$), 1.57–1.79 (2H, m, $CH_3CH_2CHCH_3$), 5.27–5.38 (1H, m, $CH_3CH_2CHCH_3$), 6.19 (2H, s, NH_2), 7.79 (1H, s, H-8). ¹³C NMR (75 MHz, DMSO- d_6) δ 10.0, 19.8, 28.9, 72.7, 137.7, 160.1. LRMS (ES⁻) m/z 206.08 [M - H]⁻. Anal. Found: C, 51.52; H, 6.43; N, 32.64. $C_9H_{13}N_5O \cdot 0.25H_2O$ requires: C, 51.05; H, 6.43; N, 33.08.

2-Amino-6-(prop-2-ynoxy)purine (23).⁵¹ Treatment of propargyl alcohol (0.83 mL, 14.2 mmol) with NaH (0.25 mg, 10.7 mmol) in DMSO (10 mL), followed by addition of **17** (1.0 g, 3.6 mmol) was performed according to general procedure C, affording the product **23** (0.54 g, 80%) as a beige solid, R_f 0.26 (10% MeOH:DCM); mp 181 °C (dec) (lit.⁵¹ mp 230 °C (dec)). IR (cm⁻¹) 3319, 3190, 2777, 2126, 1651, 1621, 1589, 1510, 1460, 1438, 1400, 1339, 1277, 1221, 1184, 1142. ¹H NMR (300 MHz, DMSO- d_6) δ 3.58 (1H, t, J = 2.4 Hz, OCH_2CCH), 5.10 (2H, d, J = 2.4 Hz, OCH_2CCH), 6.35 (2H, s, NH_2), 7.85 (1H, s, H-8). ¹³C NMR (75 MHz, DMSO- d_6) δ 53.0, 78.0, 79.6, 113.7, 138.5, 155.7, 159.1, 159.9. LRMS (ES⁺) m/z 190.00 [M + H]⁺.

2-Fluoro-6-ethoxypurine (24). Prepared following general procedure D from **18** (1.0 g, 5.6 mmol) to give **24** as a white solid (0.22 g, 22%); R_f 0.39 (10% MeOH:DCM); mp 225–227 °C. IR (cm⁻¹) 3117, 2997, 2740, 2669, 2506, 2363, 2339, 1836, 1760, 1605, 1487, 1425, 1373, 1343, 1281, 1219, 1151, 1111, 1038, 1007. ¹H NMR (300 MHz, DMSO- d_6) δ 1.41 (3H, t, J = 7.1 Hz, CH_3), 4.57 (2H, q, J = 7.1 Hz, CH_2), 8.39 (1H, s, H-8), 13.55 (1H, br s, NH). ¹³C NMR (75 MHz, DMSO- d_6) δ 14.6 (CH_3), 64.0 (CH_2), 144.0 (C^8). LRMS (ES⁺) m/z 183.1 [M + H]⁺. Anal. Found: C, 44.80; H, 3.67; N, 29.48. $C_8H_7N_4OF \cdot 0.3H_2O$ requires: C, 44.83; H, 4.08; N, 29.87.

2-Fluoro-6-n-propoxypurine (25). Synthesized in accordance with general procedure D from **19** (1.0 g, 5.2 mmol) to yield **25** (0.58 g, 57%) as a white solid; R_f 0.18 (70% EtOAc:petrol); mp 197–199 °C. IR (cm⁻¹) 3105, 2973, 2944, 2886, 2741, 2680, 2523, 2362, 2338, 1772, 1607, 1422, 1368, 1347, 1269, 1213, 1149, 1038. ¹H NMR (300 MHz, DMSO- d_6) δ 1.00 (3H, t, J = 7.4 Hz, CH_3), 1.82 (2H, tq, J = 7.1, 7.1 Hz, $CH_2CH_2CH_3$), 4.47 (2H, t, J = 6.7 Hz, $CH_2CH_2CH_3$), 8.39 (1H, s, H-8), 13.55 (1H, br s, NH). ¹³C NMR (75 MHz, DMSO- d_6) δ 10.5 (CH_3), 22.0 ($CH_2CH_2CH_3$), 69.4 (O^6CH_2), 144.0 (Ar-C), 156.0 (C^8), 158.8 (Ar-C). LRMS (ES⁺) m/z 154.0 [M - C_3H_7]⁺. HRMS calcd for $C_8H_9FN_4O$ [M⁺] 196.0760, found 196.0757.

2-Fluoro-6-isopropoxypurine (26). Synthesized by general procedure D from **20** (0.30 g, 1.6 mmol) to yield **26** as a white solid (0.16 g, 53%); R_f 0.22 (70% EtOAc:petrol); mp 184 °C (dec). IR (cm⁻¹) 2990, 2536, 1605, 1472, 1424, 1371, 1275, 1034, 1017. ¹H NMR (300 MHz, DMSO- d_6) δ 1.40 (6H, d, J = 6.2 Hz, 2 $\times$ CH_3), 5.49 (1H, septet, J = 6.2 Hz, OCH), 8.37 (1H, s, H-8), 13.52 (1H, br s, NH). ¹³C NMR (75 MHz, DMSO- d_6) δ 22.0 (CH_3), 71.5 (O^6CH), 162.2 (C^8). LRMS (ES⁺) m/z 154.0 [M - C_3H_7]⁺. HRMS calcd for $C_8H_9FN_4O$ [M⁺] 196.0760, found 196.0760.

2-Fluoro-6-(2-methyl-1-propoxy)purine (27). From **21** (0.40 g, 1.9 mmol) in accordance with general procedure D to yield **27** (0.25 g, 62%) as a white solid; R_f 0.25 (70% EtOAc:petrol); mp 202–204 °C. IR (cm⁻¹) 3130, 3059, 2968, 2937, 2884, 2804, 2693, 2528, 1760, 1700, 1603, 1476, 1451, 1418, 1364, 1341, 1267, 1211, 1140, 1113, 1040. ¹H NMR (300 MHz, DMSO- d_6) δ 1.01 (6H, d, J = 6.7 Hz, 2 $\times$ CH_3), 2.13 (1H, septet, J = 6.7 Hz, OCH_2CH), 4.30 (2H, d, J = 6.7 Hz, OCH_2), 8.40 (1H, s, H-8), 13.55 (1H, br s, NH). ¹³C NMR (75 MHz, DMSO- d_6) δ 19.2 (CH_3), 27.7 (O^6CH_2CH), 73.7 (O^6CH_2).

LRMS (ES^+) m/z 154.0 [$\text{M} - \text{C}_3\text{H}_7$] $^+$. HRMS calcd for $\text{C}_9\text{H}_{11}\text{FN}_4\text{O}$ [M^+] 210.0917, found 210.0922.

2-Fluoro-6-(1-methylpropoxy)purine (28). Prepared according to general procedure D from **22** (0.66 g, 3.2 mmol) as a white solid (0.29 g, 43%); R_f 0.21 (70% EtOAc:petrol); mp 187–189 °C. IR (cm^{-1}) 2975, 2940, 1604, 1468, 1421, 1372, 1344, 1277, 1211, 1148, 1038. ^1H NMR (300 MHz, $\text{DMSO}-d_6$) δ 0.93 (3H, t, $J = 7.4$ Hz, $\text{CH}_3\text{CH}_2\text{CHCH}_3$), 1.36 (3H, d, $J = 6.2$ Hz, $\text{CH}_3\text{CH}_2\text{CHCH}_3$), 1.66–1.83 (2H, m, $\text{CH}_3\text{CH}_2\text{CHCH}_3$), 5.29–5.37 (1H, m, $\text{CH}_3\text{CH}_2\text{CHCH}_3$), 8.37 (1H, s, H-8), 13.55 (1H, s, NH). ^{13}C NMR (75 MHz, $\text{DMSO}-d_6$) δ 9.8 ($\text{CH}_3\text{CH}_2\text{CHCH}_3$), 19.5 ($\text{CH}_3\text{CH}_2\text{CHCH}_3$), 28.6 ($\text{CH}_3\text{CH}_2\text{CHCH}_3$), 75.9 ($\text{CH}_3\text{CH}_2\text{CHCH}_3$), 158.8 (C^8). LRMS (ES^-) m/z 209.1 [$\text{M} - \text{H}$] $^-$. Anal. Found: C, 51.71; H, 5.27; N, 26.33. $\text{C}_9\text{H}_{11}\text{N}_4\text{OF}$ requires: C, 51.42; H, 5.27; N, 26.65.

2-Fluoro-6-(prop-2-ynyloxy)purine (29). Synthesized from **23** (0.28 g, 1.5 mmol) in accordance with general procedure D as a pale-yellow solid (77 mg, 27%); R_f 0.18 (70% EtOAc:petrol); mp 201 °C (dec). IR (cm^{-1}) 3261, 3123, 2970, 2864, 2803, 2363, 2336, 2140, 1651, 1607, 1460, 1379, 1341, 1275, 1217, 1142, 1107, 1048. ^1H NMR (300 MHz, $\text{DMSO}-d_6$) δ 3.70 (1H, t, $J = 2.4$ Hz, OCH_2CCH), 5.23 (2H, d, $J = 2.2$ Hz, OCH_2CCH), 8.45 (1H, s, H-8). ^{13}C NMR (75 MHz, $\text{DMSO}-d_6$) δ 55.4 ($\text{O}^6\text{CH}_2\text{CCH}$), 78.5 ($\text{O}^6\text{CH}_2\text{CCH}$), 79.1 ($\text{O}^6\text{CH}_2\text{CCH}$), 162.2 (C^8). LRMS (ES^+) m/z 192.0 [M^+]. HRMS calcd for $\text{C}_8\text{H}_5\text{FN}_4\text{O}$ [M^+] 192.0447, found 192.0448.

2-Sulfanilyl-6-ethoxypurine (30). Prepared from **24** (0.20 g, 1.1 mmol) following general procedure A. Purification by chromatography (silica; 5–20% MeOH:DCM) afforded **30** (39 mg, 11%) as an off-white solid; R_f 0.20 (10% MeOH:DCM); mp 265 °C (dec); UV λ_{max} (EtOH) 257, 241 nm. IR (cm^{-1}) 3473, 3317, 3223, 2984, 2787, 2363, 2338, 1628, 1578, 1540, 1499, 1442, 1376, 1312, 1254, 1142, 1013. ^1H NMR (300 MHz, $\text{DMSO}-d_6$) δ 1.43 (3H, t, $J = 7.1$ Hz, CH_3), 4.59 (2H, q, $J = 7.1$ Hz, CH_2), 7.16 (2H, s, NH_2), 7.71 (2H, d, $J = 8.9$ Hz, H-3' and H-5'), 7.97 (2H, d, $J = 8.9$ Hz, H-2' and H-6'), 8.08 (1H, s, H-8), 9.76 (1H, s, NH), 12.69 (1H, s, NH). ^{13}C NMR (75 MHz, $\text{DMSO}-d_6$) δ 14.9 (CH_3), 62.6 (CH_2), 117.6 (Ar-C), 126.8 (Ar-C), 155.1 (C^8). LRMS (ES^+) m/z 335.0 [$\text{M} + \text{H}$] $^+$. HRMS calcd for $\text{C}_{13}\text{H}_{14}\text{N}_6\text{O}_3\text{S}$ [M^+] 334.0848, found 334.0835.

2-Sulfanilyl-6-propoxypurine (31). Prepared from **25** (0.25 g, 1.3 mmol) according to general procedure A. Purification by chromatography (silica; 50–100% EtOAc:petrol followed by 0–10% MeOH:DCM), afforded **31** (0.25 g, 56%) as a beige solid; R_f 0.10 (70% EtOAc:petrol); mp 207 °C (dec); UV λ_{max} (EtOH) 310 nm. IR (cm^{-1}) 3251, 3115, 2971, 2934, 2878, 2361, 2340, 1700, 1596, 1580, 1528, 1498, 1453, 1389, 1321, 1306, 1243, 1146, 1094. ^1H NMR (300 MHz, $\text{DMSO}-d_6$) δ 1.03 (3H, t, $J = 7.4$ Hz, CH_3), 1.85 (2H, tq, $J = 6.9$, 6.9 Hz, $\text{CH}_3\text{CH}_2\text{CH}_2\text{O}$), 4.50 (2H, t, $J = 6.8$ Hz, $\text{CH}_3\text{CH}_2\text{CH}_2\text{O}$), 7.17 (2H, s, NH_2), 7.72 (2H, d, $J = 8.8$ Hz, H-2' and H-6'), 7.97 (2H, d, $J = 8.8$ Hz, H-3' and H-5'), 8.09 (1H, s, H-8), 9.77 (1H, s, NH), 12.96 (1H, br s, NH). ^{13}C NMR (75 MHz, $\text{DMSO}-d_6$) δ 10.7 (CH_3), 22.2 ($\text{CH}_3\text{CH}_2\text{CH}_2\text{O}$), 68.2 (O^6CH_2), 117.6 (CHCNH), 126.8 (CHCSO $_2$ NH $_2$), 135.7 (Ar-CNH), 144.6 (CSO $_2$ NH $_2$), 155.1 (C^8). LRMS (ES^+) m/z 349.8 [$\text{M} + \text{H}$] $^+$. HRMS calcd for $\text{C}_{14}\text{H}_{16}\text{N}_6\text{O}_3\text{S}$ [M^+] 348.1005, found 348.0994; Anal. Found: C 45.41; H 4.75; N 22.40. $\text{C}_{14}\text{H}_{16}\text{N}_6\text{O}_3\text{S} \cdot 0.3\text{CF}_3\text{CO}_2\text{H}$ requires: C, 45.83; H, 4.29; N, 21.97.

2-Sulfanilyl-6-isopropoxypurine (32). Synthesized in accordance with general procedure A from **26** (0.07 g, 0.4 mmol). Purification by chromatography (silica; 50–100% EtOAc:petrol followed by 0–10% MeOH:DCM), afforded **32** (38 mg, 31%) as a beige solid; R_f 0.03 (70% EtOAc:petrol); mp 177 °C (dec); UV λ_{max} (EtOH) 307 nm. IR (cm^{-1}) 3472, 3312, 3210, 3117, 2982, 2793, 2691, 2643, 2362, 2342, 1715, 1685, 1617, 1597, 1578, 1538, 1500, 1440, 1381, 1306, 1255, 1140, 1117, 1096. ^1H NMR (300 MHz, $\text{DMSO}-d_6$) δ 1.42 (6H, d, $J = 6.2$ Hz, $2 \times \text{CH}_3$), 5.59 (1H, septet, $J = 6.2$ Hz, OCH), 7.16 (2H, s, NH_2), 7.72 (2H, d, $J = 8.8$ Hz, H-2' and H-6'), 7.96 (2H, d, $J = 8.8$ Hz, $2 \times \text{H}-3'$ and H-5'), 8.07 (1H, s, H-8), 9.74 (1H, s, NH), 12.91 (1H, br s, NH). ^{13}C NMR (75 MHz, $\text{DMSO}-d_6$) δ 21.9 (CH_3), 69.0 (O^6CH), 117.1 (CHCNH), 126.5 (CHCSO $_2$ NH $_2$), 135.2 (Ar-CNH),

144.3 (CSO $_2$ NH $_2$), 154.7 (C^8). LRMS (ES^+) m/z 349.8 [$\text{M} + \text{H}$] $^+$. HRMS calcd for $\text{C}_{14}\text{H}_{16}\text{N}_6\text{O}_3\text{S}$ [M^+] 348.1005, found 348.1007.

2-Sulfanilyl-6-(2-methyl-1-propoxy)purine (33). Prepared following general procedure A from **27** (0.12 g, 0.6 mmol). Purification by chromatography (silica; 50–100% EtOAc:petrol followed by 0–10% MeOH:DCM), afforded **33** (0.11 g, 53%) as a light-brown solid; R_f 0.21 (70% EtOAc:petrol); mp 133 °C (dec); UV λ_{max} (EtOH) 308 nm. IR (cm^{-1}) 3285, 3108, 2962, 2360, 2337, 1595, 1576, 1543, 1499, 1438, 1396, 1321, 1249, 1221, 1144, 1096, 1016. ^1H NMR (300 MHz, $\text{DMSO}-d_6$) δ 1.03 (6H, d, $J = 6.7$ Hz, $2 \times \text{CH}_3$), 2.16 (1H, septet, $J = 6.7$ Hz, OCH $_2$ CH), 4.33 (2H, d, $J = 6.8$ Hz, OCH $_2$ CH), 7.17 (2H, s, NH_2), 7.71 (2H, d, $J = 8.9$ Hz, H-2' and H-6'), 7.97 (2H, d, $J = 8.9$ Hz, H-3' and H-5'), 8.09 (1H, s, H-8), 9.77 (1H, s, NH). ^{13}C NMR (75 MHz, $\text{DMSO}-d_6$) δ 19.4 (CH_3), 27.9 ($\text{O}^6\text{CH}_2\text{CH}$), 72.7 (O^6CH_2), 117.6 (CHCNH), 126.8 (CHCSO $_2$ NH $_2$), 135.7 (Ar-CNH), 144.6 (CSO $_2$ NH $_2$), 155.1 (C^8). LRMS (ES^+) m/z 363.8 [$\text{M} + \text{H}$] $^+$. HRMS calcd for $\text{C}_{15}\text{H}_{18}\text{N}_6\text{O}_3\text{S}$ [M^+] 362.1161, found 362.1145.

(R/S)-6-(1-Methylpropoxy)-2-sulfanilylpurine (34). Synthesized from **28** (0.1 g, 0.5 mmol) following general procedure A. Purification by chromatography (silica; 50–100% EtOAc:petrol followed by 0–10% MeOH:DCM), afforded **34** (41 mg, 24%) as a light-brown solid; R_f 0.10 (70% EtOAc:petrol); mp 162 °C (dec); UV (EtOH) λ_{max} 308 nm. IR (cm^{-1}) 3649, 3317, 3219, 3106, 2972, 2938, 2874, 2363, 2340, 1701, 1596, 1579, 1528, 1498, 1453, 1389, 1306, 1243, 1145, 1094. ^1H NMR (300 MHz, $\text{DMSO}-d_6$) δ 0.97 (3H, t, $J = 7.4$ Hz, $\text{CH}_3\text{CH}_2\text{CHCH}_3$), 1.39 (3H, d, $J = 6.2$ Hz, $\text{CH}_3\text{CH}_2\text{CHCH}_3$), 1.68–1.87 (2H, m, $\text{CH}_3\text{CH}_2\text{CHCH}_3$), 5.40–5.46 (1H, m, $\text{CH}_3\text{CH}_2\text{CHCH}_3$), 7.17 (2H, s, NH_2), 7.72 (2H, d, $J = 8.8$ Hz, H-2' and H-6'), 7.96 (2H, d, $J = 8.9$ Hz, H-3' and H-5'), 8.13 (1H, s, H-8), 9.76 (1H, s, NH). ^{13}C NMR (75 MHz, $\text{DMSO}-d_6$) δ 10.0 ($\text{CH}_3\text{CH}_2\text{CHCH}_3$), 19.7 ($\text{CH}_3\text{CH}_2\text{CHCH}_3$), 28.8 ($\text{CH}_3\text{CH}_2\text{CHCH}_3$), 74.2 ($\text{CH}_3\text{CH}_2\text{CHCH}_3$), 117.6 (CHCNH), 126.8 (CHCSO $_2$ NH $_2$), 135.8 (Ar-CNH), 144.6 (CSO $_2$ NH $_2$), 155.2 (C^8). LRMS (ES^+) m/z 363.8 [$\text{M} + \text{H}$] $^+$. HRMS calcd for $\text{C}_{15}\text{H}_{18}\text{N}_6\text{O}_3\text{S}$ [M^+] 362.1161, found 362.1166. Anal. Found: C, 48.32; H, 5.15; N, 22.08. $\text{C}_{15}\text{H}_{18}\text{N}_6\text{O}_3\text{S} \cdot 0.2\text{CF}_3\text{CO}_2\text{H}$ requires: C, 48.02; H, 4.76; N, 21.82.

2-Sulfanilyl-6-(prop-2-ynyloxy)purine (35). Synthesized in accordance with general procedure A from **29** (0.045 g, 0.2 mmol). Purification by chromatography (silica; 50–100% EtOAc:petrol followed by 0–10% MeOH:DCM), yielded **35** (27 mg, 34%) as a beige solid; R_f 0.1 (70% EtOAc:petrol); mp 161 °C (dec); UV (EtOH) λ_{max} 307 nm. IR (cm^{-1}) 3370, 3245, 2971, 2795, 2361, 2340, 1715, 1699, 1619, 1585, 1575, 1540, 1499, 1472, 1454, 1443, 1396, 1321, 1307, 1271, 1143, 1095. ^1H NMR (300 MHz, $\text{DMSO}-d_6$) δ 3.68 (1H, t, $J = 2.3$ Hz, OCH_2CCH), 5.24 (2H, d, $J = 2.3$ Hz, OCH_2CCH), 7.18 (2H, s, NH_2), 7.72 (2H, d, $J = 8.8$ Hz, H-2' and H-6'), 7.98 (2H, d, $J = 8.8$ Hz, H-3' and H-5'), 8.13 (1H, s, H-8), 9.87 (1H, s, NH). ^{13}C NMR (75 MHz, $\text{DMSO}-d_6$) δ 53.6 ($\text{O}^6\text{CH}_2\text{CCH}$), 78.0 ($\text{O}^6\text{CH}_2\text{CCH}$), 79.1 ($\text{O}^6\text{CH}_2\text{CCH}$), 115.1 (Ar-C), 117.5 (CHCNH), 126.5 (CHCSO $_2$ NH $_2$), 135.5 (Ar-CNH), 139.9 (C^2NH), 143.9 (CSO $_2$ NH $_2$), 154.3 (C^8), 158.6 (Ar-C). LRMS (ES^+) m/z 345.8 [$\text{M} + \text{H}$] $^+$. HRMS calcd for $\text{C}_{14}\text{H}_{12}\text{N}_6\text{O}_3\text{S}$ [M^+] 344.0692, found 344.0700.

6-Chloro-2-fluoro-9-(tetrahydro-2H-pyran-2-yl)-9H-purine (36). 3,4-Dihydropyran (60 μL , 0.58 mmol) was added dropwise over 10 min to a vigorously stirred solution of **10** (0.10 g, 0.58 mmol) and (*rac*)-camphorsulfonic acid (5 mg, 0.02 mmol) in EtOAc (50 mL) at 65 °C. The temperature was maintained at 65 °C for 18 h. The resulting bright-yellow solution was neutralized to pH 7 by careful addition of aqueous NH_3 solution until a cloudy suspension persisted. The crude mixture was washed with brine (2×30 mL) and the aqueous phase was re-extracted with EtOAc (2×30 mL). The combined organic extracts were dried (Na_2SO_4) and purified by chromatography (silica; 30% EtOAc:petrol) to afford **36** as a pale-yellow oil which became a white waxy solid on refrigeration (110 mg, 75%); mp 92–94 °C; UV λ_{max} (EtOH) 269 nm. IR (cm^{-1}) 3132, 2955, 2876, 1574. ^1H NMR (500 MHz, $\text{DMSO}-d_6$) δ 1.57–1.64 (2H, m, CH_2), 1.70–1.80 (1H, m, CH), 1.94–2.02 (2H, m, CH_2), 2.23–2.32 (1H, m, CH), 3.70–3.77 (1H, m, CH), 4.00–4.06 (1H, m, CH), 5.71 (1H, dd, $J = 2.3$, 10.9 Hz, NCH), 8.92 (1H, s, H-8). ^{13}C NMR

(125 MHz, DMSO- d_6) δ 22.0 (N^9 -CHCH $_2$ CH $_2$), 24.3 (N^9 -CHOCH $_2$ CH $_2$), 29.6 (N^9 -CHCH $_2$ CH $_2$), 67.7 (N^9 -CHOCH $_2$ CH $_2$), 81.7 (N^9 -CH), 130.1 (d, J_{CF} = 4.8 Hz, Ar-C), 146.4 (d, J_{CF} = 2.8 Hz, Ar-C), 150.6 (d, J_{CF} = 18.5 Hz, Ar-C), 153.2 (d, J_{CF} = 17.6 Hz, Ar-C), 156.2 (d, J_{CF} = 213.7 Hz, Ar-C $_2$). LRMS (ES $^+$) m/z 257.2 [M + H] $^+$.

2-Fluoro-9-(tetrahydro-2H-pyran-2-yl)-6-((triisopropylsilyl)ethynyl)-9H-purine (37). An oxygen-free solution of **36** (0.050 g, 1.95 mmol), bis(triphenylphosphine)palladium(II) chloride (41 mg, 3.0 mol %) and copper iodide (7 mg, 2.0 mol %) in THF (10 mL) was degassed by bubbling nitrogen through the solution in a sealed vial for 5 min. Triisopropylsilylacetylene (0.50 mL, 2.2 mmol) and triethylamine (0.70 mL, 4.9 mmol) were added to the mixture, which was again degassed for 15 min. After stirring at room temperature for 18 h, the black-brown suspension was filtered through Celite, eluting with MeOH (3 $\times$ 20 mL). Purification by chromatography (silica; 10% EtOAc:petrol) furnished **37** as a viscous yellow oil (78 mg, 99%); R_f 0.30 (10% EtOAc:petrol); UV λ_{max} (EtOH) 303 nm. IR (cm $^{-1}$) 3433, 2946, 2866, 2706, 2361, 1703. 1H NMR (500 MHz, DMSO- d_6) δ 1.12–1.25 (21H, m, Si(CH(CH $_3$) $_2$) $_3$), 1.56–1.63 (2H, m, CH $_2$), 1.71–1.80 (1H, m, CH), 1.94–2.03 (2H, m, CH $_2$), 2.20–2.30 (1H, m, CH), 3.70–3.78 (1H, m, CH), 4.01–4.07 (1H, m, CH), 5.70 (1H, dd, J = 2.2, 10.8 Hz, NCH), 8.89 (1H, s, H-8). ^{13}C NMR (125 MHz, DMSO- d_6) δ 10.5 (Si(CH(CH $_3$) $_2$) $_3$), 18.4 (Si(CH(CH $_3$) $_2$) $_3$), 22.0 (N^9 -CHCH $_2$ CH $_2$), 24.4 (N^9 -CHOCH $_2$ CH $_2$), 29.7 (N^9 -CHCH $_2$ CH $_2$), 67.7 (N^9 -CHOCH $_2$ CH $_2$), 81.4 (N^9 -CH), 100.2 (C $\equiv$ C), 102.8 (C $\equiv$ C), 133.9 (d, J_{CF} = 4.6 Hz, Ar-C), 140.8 (d, J_{CF} = 17.9 Hz, Ar-C), 146.6 (d, J_{CF} = 2.8 Hz, Ar-C), 153.8 (d, J_{CF} = 17.8 Hz, Ar-C), 157.4 (d, J_{CF} = 210.6 Hz, Ar-C $_2$). HRMS calcd for C $_{21}$ H $_{32}$ FN $_4$ O $_5$ Si [M + H] $^+$ 403.2324, found 403.2324.

2-Fluoro-6-((triisopropylsilyl)ethynyl)-9H-purine (38). The title compound was prepared according to general procedure E from **37** (0.29 g, 0.72 mmol). Purification by chromatography (silica; 30% EtOAc:petrol) gave **38** as a pale-yellow oil (22 mg, 96%); R_f 0.25 (30% EtOAc:petrol); UV λ_{max} (EtOH) 302 nm. IR (cm $^{-1}$) 2945, 2866, 2361, 2000, 1584. 1H NMR (500 MHz, DMSO- d_6) δ 1.12–1.21 (21H, m, Si(CH(CH $_3$) $_2$) $_3$), 8.68 (1H, s, H-8), 13.89 (1H, br, NH). ^{13}C NMR (125 MHz, CDCl $_3$) δ 11.1 (Si(CH(CH $_3$) $_2$) $_3$), 18.5 (Si(CH(CH $_3$) $_2$) $_3$), 99.5 (C $\equiv$ C), 106.1 (C $\equiv$ C), 131.2 (m, Ar-C), 141.6 (m, Ar-C), 145.3 (m, Ar-C), 156.6 (m, Ar-C), 158.6 (d, J_{CF} = 217.4 Hz, Ar-C $_2$). HRMS calcd for C $_{16}$ H $_{24}$ FN $_4$ Si [M + H] $^+$ 319.1749, found 319.1752.

6-(2-(Triisopropylsilyl)ethynyl)-N-phenyl-9H-purin-2-amine (39). Prepared in accordance with general procedure A from **38** (0.455 g, 1.43 mmol). Purified by chromatography (silica; 5% MeOH:DCM) to give the title compound as a yellow gum (0.39 g, 1.00 mmol, 70%); R_f 0.39 (5% MeOH:DCM); UV λ_{max} (EtOH) 276 nm. IR (cm $^{-1}$) 3389, 2361, 2021. 1H NMR (500 MHz, CDCl $_3$) δ 1.09–1.22 (21H, m, Si(CH(CH $_3$) $_2$) $_3$), 7.00–7.04 (1H, m, H-4'), 7.22 (1H, s, NH), 7.29 (2H, dd, J = 7.7, 8.1 Hz, H-3' and H-5'), 7.46 (1H, s, H-8), 7.52 (2H, dd, J = 1.7, 7.7 Hz, H-2' and H-6'), 10.42 (1H, s, NH). ^{13}C NMR (125 MHz, CDCl $_3$) δ 11.3 (Si(CH(CH $_3$) $_2$) $_3$), 18.7 (Si(CH(CH $_3$) $_2$) $_3$), 100.8 (C $\equiv$ C), 101.8 (C $\equiv$ C), 120.7 (Ar-C), 123.7 (Ar-C), 129.4 (Ar-C), 139.2 (Ar-C), 141.0 (Ar-C), 142.6 (Ar-C), 153.4 (Ar-C), 156.5 (Ar-C). LRMS (ES $^+$) m/z 392.0 [M + H] $^+$.

4-(6-((Triisopropylsilyl)ethynyl)-9H-purin-2-ylamino)-benzenesulfonamide (40). Following general procedure A employing **38** (0.156 g, 0.49 mmol). Purification by reversed phase column chromatography (C18 silica; 25–95% MeCN/water + 0.1% HCOOH), afforded **40** as a yellow solid (70 mg, 30%); mp 163–165 $^{\circ}C$; UV λ_{max} (EtOH) 361 nm. IR (cm $^{-1}$) 3327, 2944, 2867, 1569, 1531, 1368, 1149. 1H NMR (500 MHz, DMSO- d_6) δ 1.15 (21H, m, Si(CH(CH $_3$) $_2$) $_3$), 7.17 (2H, s, SO $_2$ NH $_2$), 7.70 (2H, d, J = 9.0 Hz, H-2' and H-6'), 7.95 (2H, d, J = 9.0 Hz, H-3' and H-5'), 8.33 (1H, s, H-8). ^{13}C NMR (125 MHz, DMSO- d_6) δ 10.6, 18.4, 90.6, 98.4, 117.1, 123.4, 126.5, 135.6, 144.0, 155.5, 165.0, 166.3, 166.5, 291.0, 286.5, 215.5. LRMS (ES $^+$) m/z 471.4 [M + H] $^+$. HRMS calcd for C $_{22}$ H $_{30}$ N $_6$ O $_2$ SSi [M + H] $^+$ 471.1987, found 471.1942.

6-Ethynyl-N-phenyl-9H-purin-2-amine (41). Treatment of **39** (0.330 g, 0.84 mmol) according to general procedure F, followed by purification by chromatography (silica; EtOAc), yielded **41** as a yellow solid (201 mg, 100%); R_f 0.27 (100% EtOAc); mp 140–160 $^{\circ}C$ (dec);

UV λ_{max} (EtOH) 243 nm. IR (cm $^{-1}$) 3414, 3111, 3072, 2920, 2110, 1704. 1H NMR (500 MHz, DMSO- d_6) δ 4.86 (1H, s, C $\equiv$ CH), 6.91–6.96 (1H, m, H-4'), 7.29 (2H, dd, J = 7.6, 8.0 Hz, H-3' and H-5'), 7.80 (2H, dd, J = 2.0, 8.0 Hz, H-2' and H-6'), 8.30 (1H, s, H-8), 9.70 (1H, s, NH), 13.17 (1H, br, NH). ^{13}C NMR (125 MHz, DMSO- d_6) δ 87.0 (C $\equiv$ CH), 118.3 (Ar-C), 120.0 (Ar-C), 128.4 (Ar-C), 140.8 (Ar-C), 156.2 (Ar-C). HRMS calcd for C $_{13}$ H $_{10}$ N $_5$ [M + H] $^+$ 236.0931, found 236.0934.

4-(6-Ethynyl-9H-purin-2-ylamino)benzenesulfonamide (42). Deprotection of **40** (0.050 g, 0.11 mmol) in accordance with general procedure F, followed by purification by chromatography (silica; 10% MeOH:DCM), afforded **42** as a beige solid (18 mg, 50%); mp 156–158 $^{\circ}C$. IR (cm $^{-1}$) 3347, 3255, 2920, 2848, 2118, 1568, 1529, 1477, 1128. UV λ_{max} (EtOH) 356, 292, 215 nm. 1H NMR (500 MHz, DMSO- d_6) δ 4.90 (1H, s, C $\equiv$ CH), 7.16 (2H, s, SO $_2$ NH $_2$), 7.17 (2H, d, J = 9.0 Hz, H-2' and H-6'), 7.93 (2H, d, J = 9.0 Hz, H-3' and H-5'), 8.36 (1H, s, H-8). LRMS (ES $^+$) m/z 315.1 [M + H] $^+$. HRMS calcd for C $_{13}$ H $_{10}$ N $_6$ O $_2$ S [M + H] $^+$ 315.0664, found 315.0687. Note: Insufficient material for ^{13}C NMR analysis.

6-Ethynyl-2-fluoro-9-(tetrahydro-2H-pyran-2-yl)-9H-purine (43). Deprotection of **37** (68 mg, 0.169 mmol) in accordance with general procedure F, but using 1.2 mol equiv of TBAF, followed by purification by chromatography (silica; 15–100% EtOAc:petrol), afforded **43** as an off-white solid (41 mg, 100%); R_f 0.32 (50% EtOAc:petrol). 1H NMR (500 MHz, CDCl $_3$) δ 1.58–1.63 (1H, m, CH), 1.63–1.78 (2H, m, CH $_2$), 1.90–1.99 (1H, m, CH), 1.99–2.06 (1H, m, CH), 2.07–2.12 (1H, m, CH), 3.68–3.74 (1H, m, CH), 3.71 (1H, s, C $\equiv$ CH), 4.09–4.14 (1H, m, CH), 5.65 (1H, dd, J = 2.6, 10.8 Hz, NCH), 8.26 (1H, s, H-8). LRMS (ES $^+$) m/z 247.0 [M + H] $^+$. Note: Insufficient material for ^{13}C NMR analysis.

6-Chloro-N-phenyl-9H-purin-2-amine (44). A mixture of N^2 -phenylguanine trifluoroacetate (**15**) (2.00 g, 5.87 mmol) and N,N -diethylaniline (1.9 mL, 11.73 mmol) in POCl $_3$ (30 mL) was heated at 115 $^{\circ}C$ for 60 min. The resultant yellow solution was carefully added dropwise to crushed ice–water with stirring, and the aqueous solution was neutralized by cautious addition of aqueous NaOH solution (1.0 M). The aqueous mixture was extracted with EtOAc (2 $\times$ 20 mL), and the combined organic extracts were dried (Na $_2$ SO $_4$) and evaporated under reduced pressure. Purification by chromatography (silica; 50% EtOAc:petrol) gave **44** as a white powder (670 mg, 46%); mp 172–174 $^{\circ}C$; UV λ_{max} (EtOH) 329, 272 nm. IR (cm $^{-1}$) 3399, 3289, 1627, 1601, 1571, 1540. 1H NMR (500 MHz, DMSO- d_6) δ 6.89–6.92 (1H, t, J = 7.5 Hz, H-4'), 7.23–7.26 (2H, dd, J = 7.4, 7.5 Hz, H-3' and H-5'), 7.71–7.73 (2H, d, J = 7.4 Hz, H-2' and H-6'), 8.20 (1H, s, H-8), 9.81 (1H, br s, NH), 13.20 (1H, br s, NH). ^{13}C NMR (125 MHz, DMSO- d_6) δ 118.6, 121.5, 128.5, 140.3, 142.4, 155.5. LRMS (ES $^+$) m/z 246.1 [M + H] $^+$.

6-Chloro-N-phenyl-9-(tetrahydro-2H-pyran-2-yl)-9H-purin-2-amine (45). To a solution of **44** (0.2 g, 0.82 mmol) and CSA (0.01 g, 0.041 mmol) in EtOAc (25 mL) was added DHP (83 μ L, 0.90 mmol), and the mixture was heated at 75 $^{\circ}C$ for 17 h. After cooling, the pH was adjusted to 8–9 with concentrated aqueous ammonia, and the volatiles were removed in vacuo. Purification by chromatography (silica; 1/2 to 2/1 EtOAc:petrol) furnished **45** as a waxy solid (194 mg, 72%); mp 194–196 $^{\circ}C$; UV λ_{max} (EtOH) 330, 275 nm. IR (cm $^{-1}$) 3274, 3106, 2934, 2849, 1579, 1573, 1535. 1H NMR (500 MHz, DMSO- d_6) δ 1.38–1.39 (2H, m, CH $_2$), 1.49–1.54 (1H, m, CH), 1.76–1.80 (2H, m, CH $_2$), 2.09–2.17 (1H, m, CH), 3.43–3.48 (1H, m, CH), 3.81–3.83 (1H, m, CH), 5.38–5.40 (1H, dd, J = 2.0, 11.0 Hz, NCH), 6.74–6.77 (1H, t, J = 8.5 Hz, H-4'), 7.07–7.11 (2H, dd, J = 7.5, 8.5 Hz, H-3' and H-5'), 7.56–7.58 (2H, d, J = 7.5 Hz, H-2' and H-6'), 8.26 (1H, s, H-8), 9.79 (1H, br s, NH), 13.21 (1H, br s, NH). ^{13}C NMR (125 MHz, DMSO- d_6) δ 24.5, 29.4, 54.9, 67.7, 81.6, 118.7, 121.7, 124.7, 128.6, 140.1, 142.4, 149.3, 152.7, 155.3. LRMS (ES $^+$) m/z 329.7 [M + H] $^+$.

N-Phenyl-6-(prop-1-yn-1-yl)-9-(tetrahydro-2H-pyran-2-yl)-9H-purin-2-amine (46). A suspension of **45** (0.1 g, 0.30 mmol), PdCl $_2$ (CH $_3$ CN) $_2$ (0.8 mg, 0.003 mmol), dicyclohexylphosphino-2',4',6'-triisopropylbiphenyl (4.3 mg, 0.009 mmol), and Cs $_2$ CO $_3$ (0.26 g, 0.79 mmol) in acetonitrile (1 mL) was degassed for 10 min

under a nitrogen atmosphere at room temperature. While degassing, an excess of methylacetylene was condensed and added to the reaction mixture via a cannula, and the reaction mixture was heated and stirred at 40 °C for 2.5 h. After cooling to room temperature, water (5 mL) was added and the mixture was extracted with EtOAc (3 × 15 mL). The combined organic extracts were washed with brine (10 mL), dried (MgSO₄), and the solvent was evaporated under reduced pressure. Purification by chromatography (silica; 30%–80% EtOAc:petrol) gave **46** as a colorless glassy compound (78 mg, 77%); *R*_f 0.27 (30% EtOAc:petrol). ¹H NMR (500 MHz, CDCl₃) δ 1.62–1.85 (3H, m, CH₂ and CH), 2.06–2.15 (3H, m, CH₂ and CH), 2.23 (3H, s, C≡CCH₃), 3.74–3.81 (1H, m, CH), 4.15–4.21 (1H, m, CH), 5.60–5.67 (1H, m, CH), 7.04 (1H, dd, *J* = 7.4, 7.3 Hz, H-4'), 7.34 (2H, dd, *J* = 8.5, 7.4 Hz, H-2' and H-6'), 7.47 (1H, s, NH), 7.34 (2H, dd, *J* = 8.6, 0.9 Hz, H-3' and H-5'), 8.03 (1H, s, H-8). LRMS (ES⁺) *m/z* 334.0 [M + H]⁺. Note: Insufficient material for ¹³C NMR analysis.

N-Phenyl-6-(phenylethynyl)-9-(tetrahydro-2H-pyran-2-yl)-9H-purin-2-amine (47). To a degassed suspension of **45** (0.1 g, 0.30 mmol), PdCl₂(CH₃CN)₂ (0.8 mg, 0.003 mmol), dicyclohexylphosphino-2',4',6'-triisopropylbiphenyl (4.3 mg, 0.009 mmol), and Cs₂CO₃ (0.26 g, 0.79 mmol) in anhydrous acetonitrile (1 mL) was added an excess of phenylacetylene, and the mixture was heated at 80 °C for 2 h. After cooling, water (5 mL) was added and the suspension was extracted with EtOAc (3 × 15 mL). The combined organic extract was washed with brine (10 mL) and dried (MgSO₄), and the solvent was removed in vacuo. Purification by chromatography (silica; 0–80% EtOAc:petrol) gave **47** as a colorless glassy compound (94 mg, 78%); *R*_f 0.53 (65% EtOAc:petrol). ¹H NMR (500 MHz, CDCl₃) δ 1.64–1.88 (3H, m, CH₂ and CH), 2.07–2.19 (3H, m, CH₂ and CH), 3.74–3.84 (1H, m, CH), 4.16–4.24 (1H, m, CH), 5.63–5.71 (1H, m, CH), 7.03–7.08 (1H, m, H-4'), 7.32–7.46 (5H, m, phenyl), 7.50 (1H, s, NH), 7.70 (2H, d, *J* = 8.3 Hz, H-2' and H-6'), 7.72–7.75 (2H, m, H-3' and H-5'), 8.08 (1H, s, H-8). ¹³C NMR (125 MHz, CDCl₃) δ 22.9, 25.0, 31.5, 68.8, 82.2, 84.1, 97.6, 118.9, 121.5, 122.5, 128.5, 129.0, 129.4, 130.0, 132.9, 139.8, 141.0, 142.3, 152.5, 156.2. LRMS (ES⁺) *m/z* 396.4 [M + H]⁺.

N-Phenyl-6-(prop-1-yn-1-yl)-9H-purin-2-amine (48). Prepared in accordance with general procedure E from **46** (0.061 g, 0.18 mmol). Recrystallization from EtOAc:petrol gave **48** as a pale-yellow solid (18 mg, 40%); mp 155–158 °C (dec); UV λ_{max} (EtOH) 272 nm. IR (cm⁻¹) 1529, 1577, 2236, 2919. ¹H NMR (500 MHz, MeOD-*d*₄) δ 2.12 (3H, s, C≡CCH₃), 6.87 (1H, t, *J* = 7.3 Hz, H-4'), 7.18 (2H, t, *J* = 7.8 Hz, H-2' and H-6'), 7.63 (2H, d, *J* = 7.9 Hz, H-3' and H-5'), 8.04 (1H, s, H-8). ¹³C NMR (125 MHz, MeOD-*d*₄) δ 4.3, 116.9, 119.2, 120.1, 122.9, 129.6, 163.2, 163.5. LRMS (ES⁺) *m/z* 250.1 [M + H]⁺. HRMS calcd for C₁₄H₁₂N₅ [M + H]⁺ 250.1087, found 250.1085.

N-Phenyl-6-(phenylethynyl)-9H-purin-2-amine (49). Prepared in accordance with general procedure E from **47** (0.055 g, 0.14 mmol). Recrystallization from EtOAc:petrol gave **49** as a yellow solid (22 mg, 51%); mp 206–208 °C (dec); UV λ_{max} (EtOH) 281 nm. IR (cm⁻¹) 1532, 1574, 2208, 3226. ¹H NMR (500 MHz, MeOD-*d*₄) δ 6.79 (1H, t, *J* = 7.3 Hz, H-4'), 7.11 (2H, dd, *J* = 7.8, 8.1 Hz, H-2' and H-6'), 7.23–7.32 (3H, m, H-3', H-5' and Ar-H), 7.54–7.58 (4H, m, 4 × ArH), 8.02 (1H, s, H-8). ¹³C NMR (125 MHz, MeOD-*d*₄) δ 97.3, 120.2, 122.8, 122.9, 129.6, 129.8, 131.2, 133.5, 141.8, 158.6. LRMS (ES⁺) *m/z* 312.2 [M + H]⁺. HRMS calculated for C₁₉H₁₄N₅ [M + H]⁺ 312.1244, found 312.1242.

6-Ethyl-2-fluoro-9-(tetrahydro-2H-pyran-2-yl)-9H-purine (50). To a solution of **43** (0.05 g, 0.20 mmol) and quinoline (20 μL) in EtOAc (5 mL) was added Lindlar's catalyst (10 mg, 20% w/w), and the mixture was stirred for 2 h under an atmosphere of H₂. The suspension was filtered through Celite, eluting with methanol (30 mL), and volatiles were removed in vacuo. Purification by chromatography (silica; 50% EtOAc:petrol) yielded **50** as a colorless oil (50 mg, 99%); UV λ_{max} (EtOH) 264 nm. IR (cm⁻¹) 2946, 2860, 2364, 2338, 1604. ¹H NMR (500 MHz, DMSO-*d*₆) δ 1.35 (3H, t, *J* = 7.5 Hz, CH₃CH₂), 1.59 (2H, m, CH₂), 1.97 (1H, m, CH), 1.99 (2H, d, *J* = 10.5 Hz, CH₂), 2.50 (1H, m, CH), 3.01 (2H, q, *J* = 7.5 Hz, CH₃CH₂), 3.73 (1H, m, CH), 4.04 (1H, d, *J* = 10.5 Hz, CH), 5.69 (1H, d, *J* = 10.5 Hz, CH), 8.74 (1H, s, H-8). ¹³C NMR (500 MHz,

DMSO-*d*₆) δ 12.3, 22.7, 24.8, 26.5, 31.8, 68.9, 82.1, 142.1, 157.9, 159.6, 167.4, 167.5. LRMS (ES⁺) *m/z* 251.0 [M + H]⁺.

6-Ethyl-2-fluoro-9H-purine (51). Prepared in accordance with general procedure E from **50** (0.18 g, 0.72 mmol). Purification by chromatography (silica; 25% EtOAc:petrol) yielded **51** as a white solid (0.117 g, 98%); mp 146–148 °C; UV λ_{max} (EtOH) 269 nm. IR (cm⁻¹) 1676, 1616, 1573. ¹H NMR (400 MHz, DMSO-*d*₆) δ 1.30 (3H, t, *J* = 7.5 Hz, CH₃CH₂), 3.00 (2H, q, *J* = 7.5 Hz, CH₃CH₂), 8.17 (1H, s, H-8). ¹³C NMR (125 MHz, DMSO-*d*₆) δ 11.7, 25.9, 68.3, 116.0, 118.4, 146.4, 158.0 (d, *J*_{C-F} = 206.0 Hz), 158.3 (d, *J*_{C-F} = 31.5 Hz). LRMS (ES⁺) *m/z* 167.7 [M + H]⁺.

4-(6-Ethyl-9H-purin-2-ylamino)benzenesulfonamide (52). Prepared by general procedure A from **51** (0.081 g, 0.49 mmol), following purification by chromatography (silica; 50% EtOAc:petrol), as a white solid (33 mg, 21%); mp 291–293 °C; UV λ_{max} (EtOH) 318, 287, 212 nm. IR (cm⁻¹) 3377, 3060, 2852, 1388, 1158. ¹H NMR (500 MHz, DMSO-*d*₆) δ 1.35 (3H, t, *J* = 6.0 Hz, CH₃CH₂), 3.00 (2H, q, *J* = 6.0 Hz, CH₃CH₂), 7.14 (2H, br s, SO₂NH₂), 7.69 (2H, d, *J* = 7.5 Hz, H-2' and H-6'), 7.99 (2H, d, *J* = 7.5 Hz, H-3' and H-5'), 8.17 (1H, s, H-8), 9.89 (1H, br s, NH). ¹³C NMR (125 MHz, DMSO-*d*₆) δ 12.3, 26.0, 117.0, 118.9, 125.4, 126.4, 135.2, 141.2, 144.4, 154.0, 155.5. LRMS (ES⁺) *m/z* 318.6 [M + H]⁺. HRMS calcd for C₁₃H₁₄N₆O₂S [M + H]⁺ 319.097076, found 319.09788.

6-Cyclopropyl-2-fluoro-9-(tetrahydro-2H-pyran-2-yl)-9H-purine (53). The title compound was prepared from **36** (0.04 g, 0.27 mmol) and potassium cyclopropyltrifluoroborate (50 mg, 0.19 mmol) following general procedure G. Purification by chromatography (silica; 50% EtOAc:petrol) afforded **53** as a colorless oil (21 mg, 42%); UV λ_{max} (EtOH) 211, 325, 391 nm. IR (cm⁻¹) 2986. ¹H NMR (400 MHz, DMSO-*d*₆) δ 1.25 (4H, d, *J* = 6.5 Hz, cyclopropyl 2 × CH₂), 1.55 (2H, m, CH₂), 1.72 (1H, m, CH), 1.95 (2H, d, *J* = 10.5 Hz, CH₂), 2.27 (1H, m, CH), 2.67 (1H, m, cyclopropyl CH), 3.70 (1H, m, CH), 3.99 (1H, d, *J* = 10.5 Hz, CH), 5.63 (1H, d, *J* = 10.5 Hz, CH), 8.70 (1H, s, H-8). ¹³C NMR (500 MHz, DMSO-*d*₆) δ 11.9, 13.2, 22.2, 24.4, 29.7, 67.7, 81.3, 130.9, 144.1, 151.8, 157.6, 159.8, 176.9. LRMS (ES⁺) *m/z* 262.4 [M + H]⁺.

6-Cyclopropyl-2-fluoro-9H-purine (54). The title compound was prepared from **53** (0.19 g, 0.72 mmol) employing general procedure E. Purification by chromatography (silica; EtOAc) afforded **54** as a white solid (0.12 g, 95%); mp 137–139 °C; UV λ_{max} (EtOH) 323, 213 nm. IR (cm⁻¹) 3062. ¹H NMR (400 MHz, DMSO-*d*₆) δ 1.11 (4H, d, *J* = 6.5 Hz, cyclopropyl 2 × CH₂), 2.51 (1H, m, cyclopropyl CH), 8.40 (1H, s, H-8). ¹³C NMR (400 MHz, DMSO-*d*₆) δ 11.6, 13.1, 131.2, 144.2, 151.8, 157.6, 159.8. LRMS (ES⁺) *m/z* 179.0 [M + H]⁺.

4-(6-Cyclopropyl-9H-purin-2-ylamino)benzenesulfonamide (55). Synthesized following general procedure A from **54** (0.086 g, 0.49 mmol). Purification by chromatography (silica; EtOAc) afforded **55** as a pale-yellow solid (24 mg, 15%); mp 291–293 °C; UV λ_{max} (EtOH) 315.5, 289, 213 nm. IR (cm⁻¹) 3389, 3062, 2921, 2851, 1727, 1604, 1579, 1531, 1379, 1149. ¹H NMR (500 MHz, DMSO-*d*₆) δ 1.23 (4H, d, *J* = 6.5 Hz, cyclopropyl 2 × CH₂), 1.35 (1H, m, cyclopropyl CH), 7.18 (2H, br s, SO₂NH₂), 7.74 (2H, d, *J* = 8.0 Hz, H-2' and H-6'), 7.99 (2H, d, *J* = 8.0 Hz, H-3' and H-5'), 8.22 (1H, s, H-8), 9.75 (1H, br s, NH). ¹³C NMR (125 MHz, DMSO-*d*₆) δ 10.5, 13.0, 117.1, 126.4, 126.6, 141.1, 144.4, 151.8, 155.5, 162.9. LRMS (ES⁺) *m/z* 331.1 [M + H]⁺. HRMS calcd for C₁₄H₁₅N₆O₂S [M + H]⁺ 331.0977, found 331.0980.

2-Fluoro-6-phenyl-9-(tetrahydro-2H-pyran-2-yl)-9H-purine (56). Synthesized in accordance with general procedure G from **36** (0.20 g, 0.78 mmol) and phenylboronic acid (0.133 g, 1.09 mmol). Purification by chromatography (silica; 20% EtOAc:petrol) gave **56** as a white solid (0.30 g, 95%); mp 138–140 °C; UV λ_{max} (EtOH) 306, 269, 203 nm. IR (cm⁻¹) 3117, 2956, 2858, 1586. ¹H NMR (300 MHz, DMSO-*d*₆) δ 1.60 (2H, m, CH₂), 1.76 (1H, m, CH), 2.02 (2H, m, CH₂), 2.29 (1H, s, CH), 3.77 (1H, t, *J* = 12.0 Hz, CH), 4.04 (1H, d, *J* = 12.0 Hz, CH), 5.74 (1H, d, *J* = 12.0 Hz, CH), 7.64, (3H, m, H-3", H-4" and H-5"), 8.79 (2H, dd, *J* = 3.0, 7.5 Hz, H-2" and H-6"), 8.25 (1H, s, H-8). ¹³C NMR (125 MHz, DMSO-*d*₆) δ 22.2, 24.4, 29.7, 67.7, 81.3, 128.9, 129.3, 129.5, 132.0, 134.0, 145.4, 154.3 (d, *J*_{C-F} = 17.5

Hz), 155.5 (d, J_{C-F} = 144.0 Hz), 157.0, 158.7. LRMS (ES^+) m/z 299.3 [$M + H$] $^+$.

2-Fluoro-6-(3-methoxyphenyl)-9-(tetrahydro-2H-pyran-2-yl)-9H-purine (57). Prepared according to general procedure G from **36** (0.20 g, 0.78 mmol) and 4-methoxyphenylboronic acid (0.16 g, 1.09 mmol). Purification by chromatography (silica; 30% EtOAc:petrol) gave **57** as a white solid (0.29 g, 80%); mp 133–135 °C; UV λ_{max} (EtOH) 306, 271, 204 nm. IR (cm^{-1}) 3089, 2938, 2857, 1761, 1586, 1516. 1H NMR (300 MHz, DMSO- d_6) δ 1.60 (2H, m, CH_2), 1.76 (1H, m, CH), 2.02 (2H, m, CH_2), 2.29 (1H, s, CH), 3.77 (1H, t, J = 12.0 Hz, CH), 3.89 (3H, s, OCH_3), 4.04 (1H, d, J = 12.0 Hz, CH), 5.74 (1H, d, J = 12.0 Hz, CH), 7.18 (1H, m, H-2"), 7.50–7.55 (1H, m, H-5"), 8.45–8.49 (2H, m, H-4", H-6"), 8.78 (1H, s, H-8). ^{13}C NMR (125 MHz, DMSO- d_6) δ 22.2, 24.4, 29.7, 55.3, 67.7, 81.3, 114.4, 117.9, 122.0, 129.4, 130.0, 135.3, 145.4, 154.3 (d, J_{C-F} = 17.5 Hz), 155.0 (d, J_{C-F} = 160.0 Hz), 156.9, 158.6, 159.4. LRMS (ES^+) m/z 334.2 [$M + H$] $^+$.

2-Fluoro-6-(4-methoxyphenyl)-9-(tetrahydro-2H-pyran-2-yl)-9H-purine (58). Synthesized in accordance with general procedure G from **36** (0.20 g, 0.78 mmol) and 4-methoxyphenylboronic acid (0.16 g, 1.09 mmol). Purification by chromatography (silica; 30% EtOAc:petrol) afforded **58** as a white solid (0.35 g, 98%); mp 131–133 °C; UV λ_{max} (EtOH) 306, 271, 204 nm. IR (cm^{-1}) 3089, 2938, 2857, 1761, 1586, 1516. 1H NMR (300 MHz, DMSO- d_6) δ 1.60 (2H, m, CH_2), 1.76 (1H, m, CH), 2.02 (2H, m, CH_2), 2.29 (1H, s, CH), 3.77 (1H, t, J = 12.0 Hz, CH), 3.88 (3H, s, OCH_3), 4.04 (1H, d, J = 12.0 Hz, CH), 5.74 (1H, d, J = 12.0 Hz, CH), 7.15–7.18 (2H, d, J = 9.0 Hz, H-3" and H-5"), 8.79–8.81 (2H, d, J = 9.0 Hz, H-2" and H-6"), 8.85 (1H, s, H-8). ^{13}C NMR (125 MHz, DMSO- d_6) δ 22.2, 24.4, 29.7, 55.5, 67.7, 81.2, 114.3, 126.5, 128.5, 131.5, 144.7, 153.9 (d, J_{C-F} = 17.5 Hz), 155.2 (d, J_{C-F} = 160.0 Hz), 157.0, 158.7, 162.4. LRMS (ES^+) m/z 329.2 [$M + H$] $^+$.

6-(3-Biphenyl)-2-fluoro-9-(tetrahydro-2H-pyran-2-yl)-9H-purine (59). Prepared following general procedure G from **36** (0.20 g, 0.78 mmol) and 3-biphenylboronic acid (0.22 g, 1.09 mmol). Purification by chromatography (silica; 20% EtOAc:petrol) furnished **59** as a pale-yellow solid (0.33 g, 80%); mp 221–223 °C; UV λ_{max} (EtOH) 346, 274 nm. IR (cm^{-1}) 2992, 2844, 2601, 1589. 1H NMR (300 MHz, DMSO- d_6) δ 1.60 (2H, m, CH_2), 1.76 (1H, m, CH), 2.02 (2H, m, CH_2), 2.29 (1H, s, CH), 3.77 (1H, t, J = 12.0 Hz, CH), 4.04 (1H, d, J = 12.0 Hz, CH), 5.79 (1H, d, J = 12.0 Hz, CH), 7.45 (1H, d, J = 7.0 Hz, H-6"), 7.56 (1H, dd, J = 7.0, 7.1 Hz, H-5"), 7.78 (SH, m, Ph), 8.40 (1H, s, H-2"), 8.84 (1H, d, J = 7.1 Hz, H-4"), 9.18 (1H, s, H-8). ^{13}C NMR (125 MHz, DMSO- d_6) δ 22.2, 24.4, 29.7, 67.7, 81.3, 126.8, 127.8, 127.9, 128.4, 129.1, 129.5, 129.6, 130.3, 134.7, 139.6, 140.8, 145.6, 154.4 (d, J_{C-F} = 17.5 Hz), 155.2 (d, J_{C-F} = 116.0 Hz), 157.0, 158.5. LRMS (ES^+) m/z 375.1 [$M + H$] $^+$.

2-Fluoro-6-(3-piperonyl)-9-(tetrahydro-2H-pyran-2-yl)-9H-purine (60). Synthesized in accordance with general procedure G from **36** (0.20 g, 0.78 mmol) and 3,4-(methylenedioxy)phenylboronic acid (0.18 g, 1.09 mmol). Purification by chromatography (silica; 30% EtOAc:petrol) yielded **60** as a white solid (0.21 g, 56%); mp 151–153 °C; UV λ_{max} (EtOH) 351, 284 nm. IR (cm^{-1}) 3011, 2917, 1721, 1616, 1511. 1H NMR (300 MHz, DMSO- d_6) δ 1.60 (2H, m, CH_2), 1.76 (1H, m, CH), 2.02 (2H, m, CH_2), 2.29 (1H, s, CH), 3.73 (1H, t, J = 12.0 Hz, CH), 4.04 (1H, d, J = 12.0 Hz, CH), 5.73 (1H, d, J = 12.0 Hz, CH), 6.17 (2H, s, OCH_2O), 7.14–7.17 (1H, d, J = 8.0 Hz, H-5"), 8.28 (1H, s, H-2"), 8.48–8.51 (1H, d, J = 8.0 Hz, H-6"), 8.82 (1H, s, H-8). ^{13}C NMR (125 MHz, DMSO- d_6) δ 22.2, 24.4, 29.7, 67.7, 81.2, 101.9, 108.6, 125.4, 128.0, 128.6, 144.9, 147.8, 150.6, 154.1, 154.8, 156.9, 158.7. LRMS (ES^+) m/z 343.1 [$M + H$] $^+$.

6-(4-Dibenzofuryl)-2-fluoro-9-(tetrahydro-2H-pyran-2-yl)-9H-purine (61). Prepared according to general procedure G from **36** (0.20 g, 0.78 mmol) and 4-dibenzofuranboronic acid (0.23 g, 1.09 mmol). Purification by chromatography (silica; 30% EtOAc:petrol) yielded **61** as a cream solid (0.39 g, 92%); mp 168–170 °C; UV λ_{max} (EtOH) 398, 256, 202 nm. IR (cm^{-1}) 3092, 2589, 1579. 1H NMR (300 MHz, DMSO- d_6) δ 1.60 (2H, m, CH_2), 1.76 (1H, m, CH), 2.02 (2H, m, CH_2), 2.29 (1H, s, CH), 3.77 (1H, t, J = 12 Hz, CH), 4.04 (1H, d, J = 12.0 Hz, CH), 5.79 (1H, d, J = 12.0 Hz, CH), 7.45–7.47 (1H, dd, J = 7.0, 7.1 Hz, H-2"), 7.52–7.62 (1H, dd, J = 8.0, 8.1 Hz, H-7"), 7.61–

7.71 (1H, dd, J = 8.0, 8.1 Hz, H-8"), 7.76–7.79 (1H, d, J = 8.0 Hz, H-9"), 8.25–8.27 (1H, d, J = 7.1 Hz, H-3"), 8.42–8.44 (1H, d, J = 8.0 Hz, H-6"), 8.55–8.58 (1H, d, J = 7.5 Hz, H-1"), 8.93 (1H, s, H-8). ^{13}C NMR (125 MHz, DMSO- d_6) δ 22.2, 24.5, 29.7, 67.8, 68.3, 81.4, 111.9, 119.3, 121.3, 122.9, 123.1, 123.4, 124.3, 125.2, 128.1, 130.1, 145.6, 153.1, 154.7 (d, J_{C-F} = 18.0 Hz), 155.7 (d, J_{C-F} = 160.0 Hz), 156.9, 158.5. LRMS (ES^+) m/z 389.4 [$M + H$] $^+$.

2-Fluoro-6-(1-thianthrenyl)-9-(tetrahydro-2H-pyran-2-yl)-9H-purine (62). Prepared in accordance with general procedure G from **36** (0.20 g, 0.78 mmol) and 1-thianthreneboronic acid (0.28 g, 1.09 mmol). Purification by chromatography (silica; 30% EtOAc:petrol) afforded **62** as an orange solid (0.43 g, 90%); mp 164–166 °C; UV λ_{max} (EtOH) 396, 224 nm. IR (cm^{-1}) 3092, 2814, 2589, 1611, 1497. 1H NMR (300 MHz, DMSO- d_6) δ 1.60 (2H, m, CH_2), 1.76 (1H, m, CH), 2.02 (2H, m, CH_2), 2.29 (1H, s, CH), 3.77 (1H, t, J = 12.0 Hz, CH), 4.04 (1H, d, J = 12.0 Hz, CH), 5.79 (1H, d, J = 12.0 Hz, CH), 7.21–7.30 (1H, dd, J = 7.4, 7.5 Hz, H-3"), 7.31–7.39 (1H, dd, J = 7.5, 7.6 Hz, H-7"), 7.37–7.45 (1H, d, J = 7.5 Hz, H-9"), 7.51–7.58 (1H, dd, J = 7.5, 7.6 Hz, H-8"), 7.59–7.65 (1H, d, J = 7.5 Hz, H-6"), 7.79 (1H, d, J = 7.4 Hz, H-2"), 7.89 (1H, d, J = 7.5 Hz, H-4"), 8.93 (1H, s, H-8). ^{13}C NMR (125 MHz, DMSO- d_6) δ 22.2, 24.4, 29.5, 67.7, 68.3, 81.5, 127.5, 128.2, 128.5, 128.7, 128.9, 130.6, 130.9, 131.4, 134.5, 134.7, 135.1, 135.5, 136.4, 146.2, 153.8 (d, J_{C-F} = 17.5 Hz), 156.2, 155.0 (d, J_{C-F} = 150.0 Hz), 157.2. LRMS (ES^+) m/z 437.2 [$M + H$] $^+$.

2-Fluoro-6-phenyl-9H-purine (63). Prepared from **56** (0.22 g, 0.72 mmol) following general procedure E. Repeated trituration of the precipitated solid with water gave **63** (0.14 g, 90%); mp 273–275 °C; UV λ_{max} (EtOH) 349 nm. IR (cm^{-1}) 2820, 1599, 1560. 1H NMR (300 MHz, DMSO- d_6) δ 7.62 (3H, m, H-3", H-4" and H-5"), 8.68 (1H, s, H-8), 8.81 (2H, m, H-2" and H-6"). ^{13}C NMR (125 MHz, DMSO- d_6) δ 128.8, 129.4, 131.7, 134.4, 145.7, 154.4, 156.0, 157.1, 158.7. LRMS (ES^+) m/z 214.6 [$M + H$] $^+$.

2-Fluoro-6-(3-methoxyphenyl)-9H-purine (64). Deprotection of **57** (0.24 g, 0.72 mmol) in accordance with general procedure E afforded the title compound **64** (0.17 g, 98%) following repeated trituration of the precipitated solid with water; mp 237–238 °C; UV λ_{max} (EtOH) 251 nm. IR (cm^{-1}) 2837, 1864, 1587, 1565. 1H NMR (300 MHz, DMSO- d_6) δ 3.89 (3H, s, OCH_3), 7.18 (1H, s, H-2"), 7.50–7.55 (1H, m, H-5"), 8.37 (1H, s, H-8), 8.45–8.49 (2H, m, H-4", H-6"). ^{13}C NMR (125 MHz, DMSO- d_6) δ 55.3, 114.2, 117.6, 121.8, 130.0, 135.7, 146.3, 154.3, 157.2, 158.6, 159.4. LRMS (ES^+) m/z 245.5 [$M + H$] $^+$.

2-Fluoro-6-(4-methoxyphenyl)-9H-purine (65). Synthesized in accordance with general procedure E from **58** (0.24 g, 0.72 mmol). Trituration of the precipitated solid with water furnished **65** as a pale-yellow solid (0.17 g, 95%); mp 212–214 °C; UV λ_{max} (EtOH) 204 nm. IR (cm^{-1}) 2845, 1586. 1H NMR (300 MHz, DMSO- d_6) δ 3.88 (3H, s, OCH_3), 7.15–7.18 (2H, d, J = 9.0 Hz, H-3" and H-5"), 8.60 (1H, s, H-8), 8.79–8.81 (2H, d, J = 9.0 Hz, H-2" and H-6"). ^{13}C NMR (500 MHz, DMSO- d_6) δ 55.4, 114.1, 114.3, 126.8, 127.5, 131.3, 145.3, 157.1, 158.8, 162.2. LRMS (ES^+) m/z 245.2 [$M + H$] $^+$.

6-(Biphenyl-3-yl)-2-fluoro-9H-purine (66). The title compound was prepared according to general procedure E from **59** (0.27 g, 0.72 mmol). Repeated washing of the precipitated solid with water gave **66** (0.20 g, 97%); mp 245–247 °C; UV λ_{max} (EtOH) 349 nm. IR (cm^{-1}) 1607, 1568. 1H NMR (300 MHz, DMSO- d_6) δ 7.43–7.45 (1H, d, J = 7.0 Hz, H-6"), 7.54–7.58 (1H, dd, J = 6.9, 7.0 Hz, H-5"), 7.74–7.81 (SH, m, Ph), 8.40 (1H, s, H-2"), 8.80–8.85 (1H, d, J = 6.9 Hz, H-4"), 8.91 (1H, s, H-8). ^{13}C NMR (125 MHz, DMSO- d_6) δ 126.8, 127.7, 127.9, 128.2, 129.1, 129.5, 130.0, 135.0, 139.7, 140.7, 146.1 (m, J_{C-F}), 157.1, 158.7. LRMS (ES^+) m/z 291.1 [$M + H$] $^+$.

6-(Benzo[d][1,3]dioxol-5-yl)-2-fluoro-9H-purine (67). Deprotection of **60** (0.25 g, 0.72 mmol) was achieved following general procedure E. Trituration of the white solid that precipitated yielded **67** (0.16 g, 87%); mp 165–167 °C; UV λ_{max} (EtOH) 351, 292 nm. IR (cm^{-1}) 2955, 2859, 1601, 1570, 1504. 1H NMR (300 MHz, DMSO- d_6) δ 6.17 (2H, s, OCH_2O), 7.18–7.18 (1H, d, J = 8.0 Hz, H-5"), 8.28 (1H, s, H-2"), 8.48–8.51 (1H, d, J = 8.0 Hz, H-6"), 8.63 (1H, s, H-8). ^{13}C NMR (125 MHz, DMSO- d_6) δ 101.9, 108.6, 125.1, 128.4, 145.3,

147.8, 150.3, 153.9 (d, J_{C-F} = 17.5 Hz), 156.0 (d, J_{C-F} = 150.0 Hz), 157.0, 158.6. LRMS (ES⁺) m/z 259.2 [M + H]⁺.

6-(4-Dibenzofuranyl)-2-fluoro-9H-purine (68). The title compound was prepared in accordance with general procedure E from **61** (0.28 g, 0.72 mmol) and purified by repeated washing with water to afford **68** as an off-white solid (0.20 g, 94%); mp 266–268 °C; UV $\lambda_{\max}$ (EtOH) 250 nm. IR (cm⁻¹) 3020, 2923, 2832, 1867, 1611, 1568. ¹H NMR (300 MHz, DMSO-*d*₆) δ 7.45–7.48 (1H, dd, J = 7.0, 7.1 Hz, H-2''), 7.55–7.63 (1H, dd, J = 7.9, 8.0 Hz, H-7''), 7.61–7.68 (1H, dd, J = 7.9, 8.0 Hz, H-8''), 7.76–7.79 (1H, d, J = 8.0 Hz, H-9''), 8.26–8.29 (1H, d, J = 7.1 Hz, H-3''), 8.43–8.45 (1H, d, J = 8.0 Hz, H-6''), 8.55–8.58 (1H, d, J = 7.0 Hz, H-1''), 8.78 (1H, s, H-8). ¹³C NMR (500 MHz, DMSO-*d*₆) δ 112.1, 121.3, 123.0, 123.5, 124.0, 125.0, 128.0, 130.0, 155.7. LRMS (ES⁺) m/z 305.1 [M + H]⁺.

2-Fluoro-6-(thianthren-1-yl)-9H-purine (69). Synthesized following general procedure E from **62** (0.32 g, 0.72 mmol) and purified by repeated trituration with water to give **69** (0.36 g, 93%); mp 166–168 °C; UV $\lambda_{\max}$ (EtOH) 250 nm. IR (cm⁻¹) 3068, 2817, 2572, 1609, 1511. ¹H NMR (300 MHz, DMSO-*d*₆) δ 7.21–7.29 (1H, dd, J = 7.4, 7.5 Hz, H-3''), 7.31–7.38 (1H, dd, J = 7.5, 7.6 Hz, H-7''), 7.39–7.46 (1H, d, J = 7.5 Hz, H-9''), 7.49–7.55 (1H, dd, J = 7.5, 7.6 Hz, H-8''), 7.58–7.64 (1H, d, J = 7.5 Hz, H-6''), 7.79 (1H, d, J = 7.5 Hz, H-2''), 7.88–7.91 (1H, d, J = 7.4 Hz, H-4''), 8.31 (1H, s, H-8). ¹³C NMR (125 MHz, DMSO-*d*₆) δ 124.6, 127.5, 128.2, 128.4, 128.7, 128.9, 130.5, 131.1, 134.7, 134.9, 156.4, 159.9, 160.5. LRMS (ES⁺) m/z 353.1 [M + H]⁺.

4-(6-Phenyl-9H-purin-2-ylamino)benzenesulfonamide (70). The title compound was synthesized according to general procedure A from **63** (0.10 g, 0.49 mmol). Purified by chromatography (silica; 50% EtOAc:petrol) to give **70** as a white solid (80 mg, 45%); mp 153–155 °C; UV $\lambda_{\max}$ (EtOH) 349, 294 nm. IR (cm⁻¹) 3313, 2922, 2852, 1706, 1589, 1569, 1524, 1373, 1144. ¹H NMR (300 MHz, DMSO-*d*₆) δ 7.17 (2H, br s, SO₂NH₂), 7.60 (3H, m, H-3'', H-4'' and H-5''), 7.75–7.78 (2H, d, J = 8.5 Hz, H-2' and H-6'), 8.03–8.06 (2H, d, J = 8.5 Hz, H-3' and H-5'), 8.37 (1H, s, H-8), 8.94 (2H, m, H-2'' and H-6''), 9.91 (1H, br s, NH). ¹³C NMR (125 MHz, DMSO-*d*₆) δ 117.2, 125.4, 126.5, 128.5, 129.3, 130.8, 135.4, 135.8, 142.6, 144.3, 153.0, 154.6, 155.3. LRMS (ES⁺) m/z 367.2 [M + H]⁺. HRMS calcd for C₁₇H₁₄N₆O₂S [M + H]⁺ 367.0972, found 367.0974.

4-(6-(3-Methoxyphenyl)-9H-purin-2-ylamino)benzenesulfonamide (71). Prepared in accordance with general procedure A from **64** (0.12 g, 0.49 mmol) and purified by chromatography (silica; 50% EtOAc:petrol) to furnish **71** as a white solid (64 mg, 33%); mp 288–290 °C; UV $\lambda_{\max}$ (EtOH) 294 nm. IR (cm⁻¹) 3377, 3097, 3020, 2838, 1729, 1597, 1566, 1531, 1365, 1145. ¹H NMR (300 MHz, DMSO-*d*₆) δ 3.89 (3H, s, OCH₃), 7.17 (2H, s, SO₂NH₂), 7.18 (1H, s, H-2''), 7.50–7.55 (1H, m, H-5''), 7.74–7.77 (2H, d, J = 8.0 Hz, H-2' and H-6'), 8.03–8.06 (2H, d, J = 8.0 Hz, H-3' and H-5'), 8.37 (1H, s, H-8), 8.45–8.49 (2H, m, H-4'' and H-6''). ¹³C NMR (125 MHz, DMSO-*d*₆) δ 55.2, 114.2, 116.6, 117.2, 121.8, 126.5, 129.6, 135.4, 137.1, 144.3, 146.3, 155.2, 115.6, 159.3. LRMS (ES⁺) m/z 397.3 [M + H]⁺. HRMS calcd for C₁₈H₁₆N₆O₃S [M + H]⁺ 397.1077, found 397.1071.

4-(6-(4-Methoxyphenyl)-9H-purin-2-ylamino)benzenesulfonamide (72). Synthesized following general procedure A from **65** (0.12 g, 0.49 mmol) and isolated by chromatography (silica; 50% EtOAc:petrol) to yield **72** as a white solid (90 mg, 46%); mp 172–174 °C; UV $\lambda_{\max}$ (EtOH) 301, 203 nm. IR (cm⁻¹) 3322, 2133, 2005, 1594, 1533, 1514, 1352, 1147. ¹H NMR (300 MHz, DMSO-*d*₆) δ 3.88 (3H, s, OCH₃), 7.17 (2H, br s, SO₂NH₂), 7.18 (2H, d, J = 8.5 Hz, H-3'' and H-5''), 7.74–7.77 (2H, d, J = 8.0 Hz, H-2' and H-6'), 8.02–8.05 (2H, d, J = 8.0 Hz, H-3' and H-5'), 8.32 (1H, s, H-8), 8.84–8.87 (2H, d, J = 8.5 Hz, H-2'' and H-6''). ¹³C NMR (125 MHz, DMSO-*d*₆) δ 55.4, 114.0, 117.2, 124.8, 126.5, 128.2, 131.0, 135.3, 142.0, 144.4, 152.8, 154.3, 155.3, 161.5. LRMS (ES⁺) m/z 397.3 [M + H]⁺. HRMS calcd for C₁₈H₁₆N₆O₃S [M + H]⁺ 397.1077, found 397.1071.

4-((6-([1,1'-Biphenyl]-3-yl)-9H-purin-2-yl)amino)benzenesulfonamide (73). Treatment of **66** (0.14 g, 0.49 mmol) in accordance with general procedure A with purification by chromatog-

raphy (silica; EtOAc) afforded **73** as a white solid (46 mg, 21%); mp 153–155 °C; UV $\lambda_{\max}$ (EtOH) 349, 294 nm. IR (cm⁻¹) 3313, 2922, 2852, 1706, 1589, 1569, 1524, 1373, 1144. ¹H NMR (300 MHz, DMSO-*d*₆) δ 7.18 (1H, br s, SO₂NH₂), 7.39–7.50 (1H, d, J = 7.0 Hz, H-6''), 7.50–7.62 (1H, dd, J = 7.0, 7.1 Hz, H-5''), 7.66–7.83 (5H, m, Ph), 7.83–7.93 (2H, d, J = 8.0 Hz, H-2' and H-6'), 8.00–8.13 (2H, d, J = 8.0 Hz, H-3' and H-5'), 8.35–8.40 (1H, s, H-2''), 8.78–8.93 (1H, d, J = 7.1 Hz, H-4''), 9.18 (1H, s, H-8), 10.00 (1H, br s, NH). LRMS (ES⁺) m/z 367.2 [M + H]⁺. HRMS calcd for C₂₃H₁₈N₆O₂S [M + H]⁺ 367.0972, found 367.0974. HPLC: >96%, XSELECT column, retention time 7.48 min, reverse phase, H₂O/MeCN/HCOOH gradient run, 1 mL min⁻¹.

4-(6-(Benzo[d][1,3]dioxol-5-yl)-9H-purin-2-ylamino)benzenesulfonamide (74). The title compound was prepared following general procedure A from **67** (0.13 g, 0.49 mmol). Purification by chromatography (silica; EtOAc) gave **74** as a pale-yellow solid (30 mg, 15%); mp 284–286 °C; UV $\lambda_{\max}$ (EtOH) 352, 295 nm. IR (cm⁻¹) 3174, 1573, 1526, 1408, 1350, 1153. ¹H NMR (300 MHz, DMSO-*d*₆) δ 6.14 (2H, s, OCH₂O), 7.15 (2H, s, SO₂NH₂), 7.73–7.75 (2H, d, J = 8.0, H-2' and H-6'), 7.98–8.01 (2H, d, J = 8.0 Hz, H-3' and H-5'), 8.31 (1H, s, H-8), 8.37 (1H, s, H-2''), 8.55 (2H, m, H-5'' and H-6''), 9.89 (1H, br s, NH). ¹³C NMR (125 MHz, DMSO-*d*₆) δ 101.6, 108.4, 108.6, 117.2, 124.7, 126.5, 129.9, 133.2, 135.5, 139.8, 142.2, 144.3, 147.6, 149.6, 151.9, 152.4, 154.4, 155.2. LRMS (ES⁺) m/z 443.3 [M + H]⁺. HRMS calcd for C₁₈H₁₄N₆O₄S [M + H]⁺ 443.1285, found 443.1289.

4-(6-(4-Dibenzofuryl)-9H-purin-2-ylamino)benzenesulfonamide (75). Synthesized in accordance with general procedure A from **68** (0.15 g, 0.49 mmol). Purification by chromatography (silica; 50% EtOAc:petrol) gave **75** as a white solid (0.10 g, 47%); mp 213–215 °C; UV $\lambda_{\max}$ (EtOH) 292, 250 nm. IR (cm⁻¹) 3275, 3199, 3104, 2921, 2852, 1714, 1593, 1574, 1535, 1500, 1355, 1146. ¹H NMR (300 MHz, DMSO-*d*₆) δ 7.16 (2H, br s, SO₂NH₂), 7.44–7.53 (1H, dd, J = 6.9, 7.0 Hz, H-2''), 7.54–7.68 (2H, m, H-7'' and H-8''), 7.68–7.74 (1H, d, J = 8.0 Hz, H-9''), 7.69–7.79 (2H, d, J = 8.0 Hz, H-2' and H-6'), 8.18–8.26 (2H, d, J = 8.0 Hz, H-3' and H-5'), 8.23–8.32 (1H, d, J = 6.9 Hz, H-3''), 8.38 (1H, s, H-8), 8.38–8.44 (1H, d, J = 7.5 Hz, H-6''), 8.49–8.58 (1H, d, J = 7.0 Hz, H-1''), 10.07 (1H, br s, NH), 13.22 (1H, br s, NH). ¹³C NMR (125 MHz, DMSO-*d*₆) δ 111.5, 117.3, 121.0, 121.3, 122.9, 123.1, 123.2, 123.4, 124.9, 126.4, 128.0, 130.0, 135.4, 143.0, 144.4, 152.2, 153.0, 154.4, 155.3, 155.7. LRMS (ES⁺) m/z 457.3 [M + H]⁺. HRMS calcd for C₂₃H₁₆N₆O₃S [M + H]⁺ 457.1077, found 457.1075.

6-(4-Thianthren-1-yl)-9H-purin-2-ylamino)benzenesulfonamide (76). Synthesized following general procedure A from **69** (0.17 g, 0.49 mmol), and purified by chromatography (silica; 50% EtOAc:petrol) to furnish **76** as an orange solid (81 mg, 33%); mp 170–172 °C; UV $\lambda_{\max}$ (EtOH) 339, 294, 250 nm. IR (cm⁻¹) 3317, 2922, 2851, 1706, 1588, 1524, 1371, 1146. ¹H NMR (300 MHz, DMSO-*d*₆) δ 7.16 (2H, s, SO₂NH₂), 7.21–7.28 (1H, dd, J = 7.4, 7.5 Hz, H-3''), 7.30–7.36 (1H, dd, J = 7.5, 7.6 Hz, H-7''), 7.38–7.43 (1H, d, J = 7.5 Hz, H-9''), 7.48–7.53 (1H, dd, J = 7.5, 7.6 Hz, H-8''), 7.58–7.63 (1H, d, J = 7.5 Hz, H-6''), 7.73 (2H, d, J = 8.5 Hz, H-2' and H-6'), 7.77–7.80 (1H, d, J = 7.4 Hz, H-2''), 7.88–7.91 (1H, d, J = 7.5 Hz, H-4''), 8.06 (2H, d, J = 8.5 Hz, H-3' and H-5'), 8.34 (1H, s, H-8). LRMS (ES⁺) m/z 505.2 [M + H]⁺. HRMS calcd for C₂₃H₁₆N₆O₂S₃ [M + H]⁺ 505.0570, found 505.0573.

7-Chloro-3-(4-methoxybenzyl)-3H-imidazo[4,5-b]pyridin-5-amine (78). To a stirred solution of **77** (0.04 g, 0.24 mmol) in anhydrous DMF (1 mL) was added anhydrous K₂CO₃ (36 mg, 0.26 mmol) and 4-methoxybenzyl chloride (34 μ L, 0.25 mmol). The reaction mixture was stirred under N₂ at 65 °C for 18 h, and the solvent was removed under reduced pressure. The residual solid was redissolved in EtOAc (30 mL) and washed with brine (3 $\times$ 30 mL), and the combined organic fractions were dried (Na₂SO₄) and the solvent was removed under reduced pressure. Purification by chromatography (silica; 40% EtOAc:petrol) afforded **78** as a light-yellow solid (29 mg, 42%); R_f 0.21 (40% EtOAc:petrol); mp 105 °C; UV $\lambda_{\max}$ (EtOH) 311 nm. IR (cm⁻¹) 3431, 3312, 3198, 3022, 2940, 2798, 1571, 1513. ¹H NMR (500 MHz, DMSO-*d*₆) δ 3.71 (3H, s,

OCH₃), 5.22 (2H, s, CH₂), 6.24 (2H, br s, NH₂), 6.48 (1H, s, H-5), 6.88 (2H, d, *J* = 8.7 Hz, H-2'' and H-6''), 7.23 (2H, d, *J* = 8.8 Hz, H-3'' and H-5''), 8.07 (1H, s, H-2). ¹³C NMR (125 MHz, DMSO-*d*₆) δ 45.4 (Ar-CH₂-Ar), 55.1 (OCH₃), 103.8 (Ar-C), 114.0 (Ar-C), 124.4 (Ar-C), 128.7 (Ar-C), 129.2 (Ar-C), 133.9 (Ar-C), 140.5 (Ar-C), 146.6 (Ar-C), 157.4 (Ar-C), 158.7 (Ar-C). HRMS calcd for C₁₄H₁₄ClN₄O [M + H]⁺ 289.0851, found 289.0855.

7-(Cyclohexylmethoxy)-3-(4-methoxybenzyl)-3H-imidazo[4,5-*b*]pyridin-5-amine (79). Synthesized following general method C from **78** (0.172 g, 0.60 mmol). The product was isolated by chromatography (silica; 80% EtOAc:petrol) to furnish **79** as a colorless oil (0.16 g, 0.45 mmol, 75%); *R*_f 0.47 (20% EtOAc:petrol); UV λ_{max} (EtOH) 271 nm. IR (cm⁻¹) 3311, 2954, 2899, 2163, 1672, 1598, 1516. ¹H NMR (500 MHz, DMSO-*d*₆) δ 0.90–1.87 (11H, m, cyclohexyl), 3.71 (3H, s, OCH₃), 4.00 (2H, d, *J* = 6.7 Hz, OCH₂), 5.17 (2H, br s, NH₂), 5.24 (2H, s, CH₂), 5.90 (1H, s, H-5), 6.87 (2H, d, *J* = 8.7 Hz, H-2'' and H-6''), 7.20 (2H, d, *J* = 8.8 Hz, H-3'' and H-5''), 7.82 (1H, s, H-2). ¹³C NMR (125 MHz, DMSO-*d*₆) δ 25.2 (CH₂), 26.0 (CH₂), 29.1 (CH₂), 37.0 (CH), 45.0 (Ar-CH₂), 55.1 (CH₃), 73.3 (CH₂), 87.5 (Ar-C), 113.9 (Ar-C), 118.2 (Ar-C), 128.7 (Ar-C), 129.7 (Ar-C), 137.9 (Ar-C), 147.4 (Ar-C), 157.8 (Ar-C), 158.5 (Ar-C), 158.6 (Ar-C). HRMS calcd for C₂₁H₂₇N₄O₂ [M + H]⁺ 367.2129, found 367.2132.

7-(Cyclohexylmethoxy)-3H-imidazo[4,5-*b*]pyridin-5-amine (80). A solution of **79** (0.035 g, 0.10 mmol) in TFA (1 mL) was heated under reflux for 2 h, and the cooled solution was neutralized with saturated aqueous NaHCO₃ (25 mL). The reaction mixture was extracted with EtOAc (3 × 30 mL), and the combined organic fraction was dried (MgSO₄) and evaporated under reduced pressure. Purification by chromatography (silica; 10% MeOH:DCM) to furnish **80** as a white solid (16.5 mg, 70%); mp 99–102 °C; UV λ_{max} (EtOH) 293, 244 nm. IR (cm⁻¹) 3447, 3323, 3187, 2923, 2850, 2795, 2026, 1591, 1527. ¹H NMR (500 MHz, CDCl₃) δ 1.03–2.00 (11H, m, cyclohexyl), 4.03 (2H, d, *J* = 6.1 Hz, OCH₂), 4.37 (2H, br s, NH₂), 5.91 (1H, s, H-5), 7.77 (1H, s, H-2), 9.89 (1H, br s, NH). ¹³C NMR (125 MHz, CDCl₃) δ 25.7, 26.4, 29.8, 37.5, 74.4, 88.1, 157.3. HRMS calcd for C₁₃H₁₉N₄O [M + H]⁺ 247.1553, found 247.1556.

7-(Cyclohexylmethoxy)-5-iodo-3-(4-methoxybenzyl)-3H-imidazo[4,5-*b*]pyridine (81). A solution of **79** (0.112 g, 0.31 mmol) in THF (4 mL) containing diiodomethane (123 μL, 1.50 mmol), CuI (59 mg, 0.31 mmol), and isoamyl nitrite (187 μL, 0.92 mmol) was heated to reflux for 4 h, and the dark-brown solution was filtered (Celite) and extracted with EtOAc (2 × 20 mL). The combined organic fraction was washed sequentially with aqueous sodium thiosulfate solution (10%, 2 × 40 mL) and brine (40 mL) and dried (MgSO₄), and the solvent was removed in vacuo. Purification by chromatography (silica; 30% EtOAc:petrol) afforded **81** as a yellow oil (64 mg, 44%); *R*_f 0.19 (30% EtOAc:petrol); UV λ_{max} (EtOH) 262 nm. IR (cm⁻¹) 3333, 2951, 2891, 2122, 1672, 1584, 1521. ¹H NMR (500 MHz, DMSO-*d*₆) δ 0.98–1.89 (11H, m, cyclohexyl), 3.71 (3H, s, OCH₃), 4.19 (2H, d, *J* = 6.5 Hz, OCH₂), 5.17 (2H, s, CH₂), 6.90 (2H, d, *J* = 8.7 Hz, H-2'' and H-6''), 7.18 (1H, s, H-5), 7.26 (2H, d, *J* = 8.8 Hz, H-3'' and H-5''), 8.31 (1H, s, H-2). ¹³C NMR (125 MHz, DMSO-*d*₆) δ 25.1 (CH₂), 25.9 (CH₂), 28.9 (CH₂), 37.0 (CH), 45.8 (Ar-CH₂), 55.1 (CH₃), 74.4 (CH₂), 112.0 (Ar-C), 112.7 (Ar-C), 114.1 (Ar-C), 124.6 (Ar-C), 128.8 (Ar-C), 129.0 (Ar-C), 142.6 (Ar-C), 148.5 (Ar-C), 157.0 (Ar-C), 158.8 (Ar-C). HRMS calcd for C₂₁H₂₅I₂N₃O₂ [M + H]⁺ 478.0986, found 478.0980.

4-((7-(Cyclohexylmethoxy)-3-(4-methoxybenzyl)-3H-imidazo[4,5-*b*]pyridin-5-yl) amino)benzenesulfonamide (82). To a solution of **81** (0.048 g, 0.10 mmol) in anhydrous MeCN (2 mL) was added sulfanilamide (19 mg, 0.11 mmol), anhydrous K₂CO₃ (28 mg, 0.20 mmol), Pd₂(dba)₃ (2.3 mg, 0.004 mmol), and XPhos (1.9 mg, 0.004 mmol). The reaction mixture was degassed under N₂ for 30 min and stirred at 80 °C for 2 h. The cooled solution was filtered (Celite) extracted with EtOAc (2 × 20 mL), and the combined organic extract was washed sequentially with aqueous HCl (1 M, 20 mL), saturated aqueous NaHCO₃ (20 mL), and brine (30 mL), and dried (Na₂SO₄), and the volatiles were evaporated in vacuo. Purification by chromatography (silica; 70% EtOAc:petrol) yielded **82** as a white solid (23 mg, 43%); *R*_f 0.59 (70% EtOAc:petrol); mp 142 °C; UV λ_{max}

(EtOH) 257 nm. IR (cm⁻¹) 3329, 2933, 2874, 2158, 1662, 1588, 1501. ¹H NMR (500 MHz, CDCl₃) δ 1.00–1.97 (11H, m, cyclohexyl), 3.79 (3H, s, OCH₃), 4.01 (2H, d, *J* = 6.1 Hz, OCH₂), 4.76 (2H, br s, NH₂), 5.28 (2H, s, CH₂), 6.17 (1H, s, H-5), 6.79 (1H, br s, NH), 6.88 (2H, d, *J* = 8.8 Hz, H-2'' and H-6''), 7.24 (2H, d, *J* = 8.7 Hz, H-3'' and H-5''), 7.61 (2H, *J* = 8.8 Hz, H-2' and H-6'), 7.81 (2H, *J* = 8.8 Hz, H-3' and H-5'), 8.11 (1H, s, H-2). ¹³C NMR (125 MHz, CDCl₃) δ 25.7 (CH₂), 26.4 (CH₂), 29.8 (CH₂), 37.6 (CH₂), 46.9 (CH), 55.4 (CH₂), 74.8 (OCH₃), 90.7 (OCH₂), 114.3 (Ar-C), 117.1 (Ar-C), 128.0 (Ar-C), 129.1 (Ar-C), 139.7 (Ar-C), 145.5 (Ar-C), 152.1 (Ar-C). HRMS calcd for C₂₇H₃₂N₅O₄S [M + H]⁺ 522.2170, found 522.2166.

4-((7-(Cyclohexylmethoxy)-3H-imidazo[4,5-*b*]pyridin-5-yl)amino)benzenesulfonamide (83). A solution of **82** (0.022 g, 0.042 mmol) in TFA (1 mL) was heated under reflux for 2 h, and the cooled solution was neutralized with saturated aqueous NaHCO₃ (25 mL). The reaction mixture was extracted with EtOAc (3 × 30 mL), and the combined organic fraction was dried (MgSO₄) and evaporated under reduced pressure. The title compound was purified by preparative TLC on silica, using 5% MeOH:DCM as eluent, to afford **83** as a white solid (16.7 mg, 99%); *R*_f 0.25 (5% MeOH:DCM); mp 165–166 °C; UV λ_{max} (EtOH) 324, 274 nm. IR (cm⁻¹) 3332, 2925, 2852, 2164, 1677, 1583, 1500. ¹H NMR (500 MHz, MeOD-*d*₄) δ 1.07–2.05 (11H, m, cyclohexyl), 4.03 (2H, d, *J* = 6.2 Hz, OCH₂), 6.33 (1H, s, H-5), 7.77 (2H, d, *J* = 8.9 Hz, H-2' and H-6'), 7.92 (2H, d, *J* = 8.8 Hz, H-3' and H-5'), 7.96 (1H, s, H-2). ¹³C NMR (125 MHz, CDCl₃) δ 26.9 (CH₂), 27.6 (CH₂), 30.8 (CH₂), 38.9 (CH), 74.9 (CH₂O), 90.9 (Ar-C), 117.9 (Ar-C), 128.2 (Ar-C). HRMS calcd for C₁₉H₂₄N₅O₃S [M + H]⁺ 522.2170, found 522.2166. HPLC: >98%, XSELECT column, retention time 5.47 min, reverse phase, H₂O/MeCN/HCOOH gradient run, 1 mL min⁻¹.

7-(Cyclohexylmethoxy)-3H-[1,2,3]triazolo[4,5-*d*]pyrimidin-5-amine (86). Prepared in accordance with general procedure C from **84** (1.50 g, 8.82 mmol). Precipitation from aqueous AcOH, and repeated washing with water gave **86** a pale-orange solid (1.52 g, 70%); mp 214–216 °C; UV λ_{max} (EtOH) 285 nm. IR (cm⁻¹) 3494, 3378, 2925, 2780, 2651, 2362, 2329, 2115, 2085, 2013, 1906, 1628, 1608, 1578, 1488. ¹H NMR (500 MHz, DMSO-*d*₆) δ 1.03–1.89 (11H, m, cyclohexyl), 4.30 (2H, d, *J* = 6.5 Hz, OCH₂), 6.91 (2H, br s, NH₂), 15.28 (1H, br s, NH). ¹³C NMR (125 MHz, DMSO-*d*₆) δ 25.1 (CH₂), 25.9 (CH₂), 29.1 (CH₂), 36.6 (CH), 71.2 (CH₂), 162.2 (Ar-C). HRMS calcd for C₁₁H₁₇N₆O [M + H]⁺ 249.1458, found 249.1461.

4-(Cyclohexylmethoxy)-1H-pyrazolo[3,4-*d*]pyrimidin-6-amine (87). Prepared according to the general procedure C from **85** (0.125 g, 0.74 mmol). Purification by chromatography (silica; 10% MeOH:DCM) gave **87** as an off white crystalline solid (68 mg, 37%); *R*_f 0.70 (10% MeOH:DCM), mp 193–195 °C; UV λ_{max} (EtOH) 275 nm. IR (cm⁻¹) 3498, 3353, 3232, 3121, 2960, 2921, 2851, 2750, 2667, 2158, 2038, 1902, 1794, 1633, 1578, 1497, 1469, 1441. ¹H NMR (500 MHz, DMSO-*d*₆) δ 0.98–1.82 (11H, m, cyclohexyl), 4.21 (2H, d, *J* = 6.5 Hz, OCH₂), 6.55 (2H, br s, NH₂), 7.77 (1H, s, CH), 12.78 (1H, br s, NH). ¹³C NMR (125 MHz, DMSO-*d*₆) δ 25.2 (CH₂), 26.0 (CH₂), 29.1 (CH₂), 36.7 (CH), 70.5 (CH₂), 95.4 (Ar-C), 131.6 (Ar-C), 158.9 (Ar-C), 162.0 (Ar-C), 163.4 (Ar-C). HRMS calcd for C₁₂H₁₈N₅O [M + H]⁺ 248.1506, found 248.1508.

5-Chloro-7-(cyclohexylmethoxy)-3H-[1,2,3]triazolo[4,5-*d*]pyrimidine (88). To a solution of LiCl (34 mg, 0.80 mmol) in DMA (0.5 mL) at 0 °C was added **86** (0.05 g, 0.20 mmol), followed by isoamyl nitrite (61 μL, 0.30 mmol) and SOCl₂ (16 μL, 0.22 mmol). The reaction mixture was stirred for 1 h and allowed to warm to ambient temperature over 18 h. After addition of saturated aqueous NaHCO₃ (15 mL), the reaction mixture was extracted with EtOAc (3 × 15 mL), and the combined organic layers were washed with brine (2 × 30 mL) and dried (Na₂SO₄). Evaporation of the solvent under reduced pressure furnished **88** as a brown oil (54 mg, > 99%), which was used directly without further purification.

6-Chloro-4-(cyclohexylmethoxy)-1H-pyrazolo[3,4-*d*]pyrimidine (89). To a solution of LiCl (69 mg, 1.62 mmol) in DMA (1 mL) at 0 °C was added **87** (0.100 g, 0.40 mmol), followed by isoamyl nitrite (123 μL, 0.61 mmol) and SOCl₂ (32 μL, 0.45 mmol). The reaction

mixture was stirred at room temperature for 18 h, cooled, and EtOAc (45 mL) was added. The resultant solution was washed sequentially with aqueous saturated NaHCO₃ solution (45 mL) and brine (2 × 45 mL), dried (Na₂SO₄), and the volatiles were evaporated under reduced pressure. The product was purified by chromatography (silica; 5% MeOH:DCM) to give **89** as a yellow solid (65 mg, 60%); *R*_f 0.32 (5% MeOH:DCM), mp 152–153 °C; UV λ_{max} (EtOH) 252 nm. IR (cm⁻¹) 3148, 2922, 2851, 1677, 1579, 1443. ¹H NMR (500 MHz, CDCl₃) δ 0.96–1.95 (11H, m, cyclohexyl), 4.39 (2H, d, *J* = 6.0 Hz, OCH₂), 8.09 (1H, s, CH), 11.19 (1H, br s, NH). ¹³C NMR (125 MHz, DMSO-*d*₆) δ 25.2 (CH₂), 26.0 (CH₂), 29.1 (CH₂), 36.7 (CH), 70.5 (CH₂O), 95.4 (Ar-C), 131.6 (C¹-H), 158.9 (Ar-C), 162.0 (Ar-C), 163.4 (Ar-C). HRMS calcd for C₁₂H₁₆ClN₄O [M + H]⁺ 267.1007, found 267.1012.

4-((7-(Cyclohexylmethoxy)-3H-[1,2,3]triazolo[4,5-*d*]pyrimidin-5-yl)amino) benzenesulfonamide (90). The title compound was prepared following general procedure A from **88** (0.054 g, 0.20 mmol) and isolated by chromatography (silica; 30% EtOAc:petrol) to give the title compound **90** as a white solid (28 mg, 35%); *R*_f 0.18 (70% EtOAc:petrol); mp 217–219 °C; UV λ_{max} (EtOH) 310 nm. IR (cm⁻¹) 3254, 3121, 2924, 2851, 2363, 2342, 2199, 2140, 1992, 19118, 1700, 1582, 1534, 1492, 1447. ¹H NMR (500 MHz, DMSO-*d*₆) δ 1.04–1.92 (11H, m, cyclohexyl), 4.36 (2H, d, *J* = 6.8 Hz, OCH₂), 7.18 (2H, br s, SO₂NH₂), 7.72 (2H, d, *J* = 8.8 Hz, H-2' and H-6'), 7.99 (2H, d, *J* = 9.0 Hz, H-3' and H-5'), 9.92 (1H, br s, NH) 13.34 (1H, br s, NH). ¹³C NMR (125 MHz, DMSO-*d*₆) δ 25.1 (CH₂), 29.1 (CH₂), 35.9 (CH), 118.0 (Ar-C), 126.4 (Ar-C). HRMS calcd for C₁₇H₂₂N₇O₃S [M + H]⁺ 404.1499, found 404.1500. HPLC: >99.5%, XTerra column, retention time 11.64 min, reverse phase, H₂O/MeCN/HCOOH gradient run, 1 mL min⁻¹.

4-((4-(Cyclohexylmethoxy)-1H-pyrazolo[3,4-*d*]pyrimidin-6-yl)amino) benzenesulfonamide (91). The title compound was prepared by general procedure A from **89** (0.042 g, 0.18 mmol). Chromatography (silica; 70% EtOAc:petrol) afforded **91** as a white solid (20 mg, 28%); *R*_f 0.21 (70% EtOAc:petrol), mp 268–269 °C; UV λ_{max} (EtOH) 304 nm. IR (cm⁻¹) 3367, 3333, 3269, 2922, 2853, 2162, 2012, 1691, 1658, 1624, 1583, 1551, 1524. ¹H NMR (500 MHz, DMSO-*d*₆) δ 1.04–1.92 (11H, m, cyclohexyl), 4.36 (2H, d, *J* = 6.8 Hz, OCH₂), 7.18 (2H, br s, SO₂NH₂), 7.72 (2H, d, *J* = 8.8 Hz, H-2' and H-6'), 7.98 (1H, br s, CH), 7.99 (2H, d, *J* = 9.0 Hz, H-3' and H-5'), 9.92 (1H, br s, NH) 13.34 (1H, br s, NH). ¹³C NMR (125 MHz, DMSO-*d*₆) δ 25.4 (CH₂), 26.5 (CH₂), 29.5 (CH₂), 37.1 (CH), 71.5 (CH₂) 118.6 (Ar-C), 136.6 (Ar-C), 144.0 (Ar-C). HRMS calcd for C₁₈H₂₃N₆O₃S [M + H]⁺ 403.1547, found 403.1547. HPLC: >98.5%, XTerra column, retention time 11.92 min, reverse phase, H₂O/MeCN/HCOOH gradient run, 1 mL min⁻¹.

4-(Cyclohexylmethoxy)-7H-pyrrolo[2,3-*d*]pyrimidin-2-amine (93). Prepared in accordance with general method C from **92** (0.15 g, 0.89 mmol), with heating under microwave irradiation (170 °C, 5 h). The product was isolated by chromatography (silica; 10% MeOH:DCM) to yield **93** as a white crystalline solid (37 mg, 17%); *R*_f 0.55 (10% MeOH:DCM); mp 216.3–217.3 °C; UV λ_{max} (EtOH) 221 nm. IR (cm⁻¹) 3494, 3362, 3117, 2922, 2851, 2164, 2045, 1968, 1617, 1585, 1499, 1478. ¹H NMR (500 MHz, DMSO-*d*₆) δ 0.98–1.83 (11H, m, cyclohexyl), 4.15 (2H, d, *J* = 6.5, OCH₂), 5.91 (2H, br s, NH₂), 6.17 (1H, dd, *J* = 1.9 and 3.5, CH), 6.79 (1H, dd, *J* = 2.2 and 3.5, HNCH), 10.97 (1H, br s, NH). ¹³C NMR (125 MHz, DMSO-*d*₆) δ 25.2 (CH₂), 26.0 (CH₂), 29.3 (CH₂), 36.8 (CH), 69.9 (CH₂O), 96.9 (C¹-H), 97.9 (Ar-C), 119.1 (C²-H), 155.0 (Ar-C), 159.3 (Ar-C), 162.7 (Ar-C). HRMS calcd for C₁₃H₁₉N₄O [M + H]⁺ 247.1553, found 247.1556.

4-Chloro-7-((2-(trimethylsilyl)ethoxy)methyl)-7H-pyrrolo[2,3-*d*]pyrimidin-2-amine (94). NaH (60% in mineral oil, 26 mg, 0.65 mmol) was added dropwise to a solution of **92** (0.10 g, 0.59 mmol) in anhydrous MeCN (3 mL) at –30 °C, and the reaction mixture was stirred for 45 min. 2-(Trimethylsilyl)ethoxymethyl chloride (90%, 122 μL, 0.62 mmol) was added dropwise, and the reaction mixture was stirred for a further 3 h at room temperature. After cooling and addition of water (20 mL), the resultant solution was extracted with EtOAc (20 mL), and the combined organic fractions were washed with brine (3 × 20 mL), dried (Na₂SO₄), and the solvent was removed in vacuo. Purification by chromatography (silica; 15% EtOAc:petrol)

gave **94** as a yellow solid (130 mg, 74%); *R*_f 0.41 (15% EtOAc:petrol); mp 63–64 °C; UV λ_{max} (EtOH) 317, 234 nm. IR (cm⁻¹) 3423, 3318, 3212, 3096, 2952, 1631, 1608, 1546, 1493. ¹H NMR (500 MHz, CDCl₃) δ –0.04 (9H, s, (CH₃)₃), 0.91 (2H, t, *J* = 8.0 Hz, OCH₂CH₂Si), 3.81 (2H, t, *J* = 8.3 Hz, OCH₂CH₂Si), 4.96 (2H, br s, NH₂), 5.44 (2H, s, CH₂), 6.43 (1H, d, *J* = 3.7 Hz, CH), 6.99 (1H, d, *J* = 3.7 Hz, NCH). ¹³C NMR (125 MHz, DMSO-*d*₆) δ 0.0 (Si(CH₃)₃), 19.2 (CH₂CH₂Si), 67.7 (OCH₂CH₂), 74.3 (NCH₂O), 102.2 (C⁵H), 112.2 (Ar-C), 127.0 (C⁶H), 154.2 (Ar-C), 155.6 (Ar-C), 160.2 (Ar-C). HRMS calcd for C₁₂H₂₀ClN₄OSi [M + H]⁺ 299.1089, found 299.1090.

4-(Cyclohexylmethoxy)-7-((2-(trimethylsilyl)ethoxy)methyl)-7H-pyrrolo[2,3-*d*]pyrimidin-2-amine (95). Synthesized following general procedure C from **94** (0.30 g, 1.01 mmol), except using THF as solvent. The title compound was isolated by chromatography (silica; 15% EtOAc:petrol) to furnish **95** as a yellow solid (296 mg, 78%); *R*_f 0.17 (15% EtOAc:petrol); mp 88–90 °C; UV λ_{max} (EtOH) 286, 260, 223 nm. IR (cm⁻¹) 3346, 3222, 2926, 2852, 1647, 1602. ¹H NMR (500 MHz, CDCl₃) δ –0.05 (9H, s, (CH₃)₃), 0.91 (2H, t, *J* = 8.3 Hz, OCH₂CH₂Si), 1.01–1.91 (11H, m cyclohexyl), 3.51 (2H, t, *J* = 8.3 Hz, OCH₂CH₂Si), 4.21 (2H, d, *J* = 6.5 Hz, OCH₂), 4.68 (2H, br s, NH₂), 5.41 (2H, s, CH₂), 6.40 (1H, d, *J* = 3.7 Hz, CH), 6.82 (1H, d, *J* = 3.7 Hz, NCH). ¹³C NMR (125 MHz, DMSO-*d*₆) δ 0.0 (Si(CH₃)₃), 19.2 (CH₂CH₂Si), 27.2 (CH₂), 28.0 (CH₂), 31.3 (CH₂), 38.8 (CH), 67.4 (OCH₂CH₂), 72.6 (CH₂), 74.1 (NCH₂O), 101.2 (C⁵H), 123.6 (C⁶H), 160.6 (Ar-C), 165.4 (Ar-C). HRMS calcd for C₁₉H₃₃N₄O₂Si [M + H]⁺ 377.2367, found 377.2367.

2-Chloro-4-(cyclohexylmethoxy)-7-((2-(trimethylsilyl)ethoxy)methyl)-7H-pyrrolo[2,3-*d*]pyrimidine (96). To a solution of LiCl in THF (0.5 M, 2.13 mL, 1.1 mmol) was added **95** (0.10 g, 0.27 mmol), and the reaction mixture was cooled to 0 °C. Isoamyl nitrite (81 μL, 0.40 mmol) and SOCl₂ (21.3 μL, 0.29 mmol) were added, and the solution was stirred at room temperature for 5 h. The reaction mixture was added to a solution of aqueous NH₄Cl (2 M, 30 mL), extracted with EtOAc (2 × 30 mL), and the combined organic layers were washed with water (3 × 30 mL), dried (Na₂SO₄), and evaporated in vacuo. Purification by chromatography (silica; 15% EtOAc:petrol) gave **96** as a clear oil (60 mg, 56%); *R*_f 0.45 (15% EtOAc:petrol); UV λ_{max} (EtOH) 270 nm. IR (cm⁻¹) 2924, 2852, 1750, 1661, 1588, 1558, 1506. ¹H NMR (500 MHz, DMSO-*d*₆) δ –0.09 (9H, s, (CH₃)₃), 0.82 (2H, t, *J* = 8.1 Hz, OCH₂CH₂Si), 0.95–1.89 (11H, m cyclohexyl), 3.50 (2H, t, *J* = 8.1 Hz, OCH₂CH₂Si), 4.28 (2H, d, *J* = 6.0 Hz, OCH₂), 5.52 (2H, s, CH₂), 6.61 (1H, d, *J* = 3.6 Hz, CH), 7.55 (1H, d, *J* = 3.5 Hz, NCH). ¹³C NMR (125 MHz, DMSO-*d*₆) δ 0.0 (Si(CH₃)₃), 18.5 (CH₂CH₂Si), 26.6 (CH₂), 30.5 (CH₂), 31.8 (CH₂), 38.8 (CH), 67.1 (OCH₂CH₂), 73.3 (CH₂), 74.2 (NCH₂O), 100.4 (C⁵H), 126.4 (C⁶H), 129.7 (Ar-C). HRMS calcd for C₁₉H₃₁ClN₄O₂Si [M + H]⁺ 396.1869, found 396.1872.

4-((4-(Cyclohexylmethoxy)-7-((2-(trimethylsilyl)ethoxy)methyl)-7H-pyrrolo[2,3-*d*]pyrimidin-2-yl)amino) benzenesulfonamide (97). To a degassed solution of **96** (0.050 g, 0.13 mmol) in MeCN (2 mL) was added sulfanilamide (24 mg, 0.14 mmol), anhydrous K₂CO₃ (35 mg, 0.25 mmol), Pd₂(dba)₃ (3 mg, 5.0 × 10⁻³ mmol), and XPhos (2.4 mg, 5.0 × 10⁻³ mmol), and the reaction mixture was stirred under N₂ at 80 °C for 2 h. The cooled mixture was filtered (Celite), diluted with EtOAc (20 mL), and washed sequentially with aqueous HCl solution (1 M, 20 mL), saturated aqueous NaHCO₃ solution (20 mL), and brine (30 mL). After drying (Na₂SO₄), volatiles were removed under reduced pressure and the product was isolated by chromatography (silica; 50% EtOAc:petrol) to give **97** as a yellow oil (33 mg, 49%); *R*_f 0.63 (50% EtOAc:petrol); UV λ_{max} (EtOH) 269 nm. IR (cm⁻¹) 3644, 3342, 2924, 2853, 1586, 1528, 1488. ¹H NMR (500 MHz, CDCl₃) δ –0.08 (9H, s, Si(CH₃)₃), 0.88–0.95 (2H, t, *J* = 8.36 Hz, OCH₂CH₂Si), 1.04–1.93 (11H, m, cyclohexyl), 3.56 (2H, t, *J* = 3.4 Hz, OCH₂CH₂Si), 4.29 (2H, d, *J* = 6.5 Hz, OCH₂), 4.72 (2H, br s, NH₂), 5.52 (2H, s, CH₂), 6.49 (1H, d, *J* = 3.5 Hz, CH), 6.96 (1H, d, *J* = 3.6 Hz, NCH), 7.20 (1H, br s, NH), 7.87 (4H, ap s, H-2', H-3', H-5' and H-6'). ¹³C NMR (125 MHz, CDCl₃) δ 0.0 (Si(CH₃)₃), 15.6 (CH₂Si), 27.2 (CH₂), 27.9 (CH₂), 31.3 (CH₂), 38.8 (CH), 67.8 (CH₂), 73.2 (CH₂), 74.5 (CH₂), 101.5 (Ar-C), 101.9 (Ar-C), 118.8 (Ar-C), 125.0 (Ar-C), 129.3 (Ar-C), 146.2 (Ar-C). HRMS calcd for

C₂₅H₃₈N₅O₄SSi [M + H]⁺ 532.2408, found 532.2406. Note: Not all carbons are visible.

4-((4-(Cyclohexylmethoxy)-7H-pyrrolo[2,3-d]pyrimidin-2-yl)-amino)benzenesulfonamide (98). A solution of **97** (0.040 g, 7.5 × 10⁻² mmol) in TFA (1.5 mL) was heated under reflux for 10 min, and the cooled solution was neutralized with saturated aqueous NaHCO₃ solution. The aqueous solution was extracted with EtOAc (2 × 30 mL), and the solvent was evaporated under reduced pressure. The residual yellow solid was redissolved in MeCN:H₂O (1:1, 8 mL), the pH of the solution was adjusted to 10 with dilute aqueous ammonia (28%), and the solution was stirred for 1 h at room temperature and evaporated in vacuo. Isolation by chromatography (silica; 70% EtOAc:petrol) to give **98** as a white solid (12 mg, 40%); R_f 0.39 (70% EtOAc:petrol); mp 251.2–252.0 °C; UV λ_{max} (EtOH) 312 nm. IR (cm⁻¹) 3374, 3322, 2058, 2924, 2856, 2012, 1615, 1595, 1579. ¹H NMR (500 MHz, DMSO-*d*₆) δ 1.07–1.95 (11H, m, cyclohexyl), 4.31 (2H, d, *J* = 6.3 Hz, OCH₂), 6.34 (1H, dd, *J* = 2.1 and 2.1 Hz, CH), 7.05 (1H, dd, *J* = 2.2 and 2.2 Hz, HNCH), 7.13 (2H, br s, SO₂NH₂), 7.68 (2H, d, *J* = 8.9 Hz, H-2' and H-6') 7.99 (2H, d, *J* = 8.8 Hz, H-3' and H-5'), 9.57 (1H, br s, NH), 11.54 (1H, br s, NH). ¹³C NMR (125 MHz, DMSO-*d*₆) δ 25.2 (CH₂), 26.0 (CH₂), 29.3 (CH₂), 36.8 (CH), 70.6 (CH₂), 98.2 (C⁵H), 116.9 (Ar-C), 126.4 (C⁶H), 134.9 (Ar-C), 144.6 (Ar-C), 146.4 (Ar-C), 153.6 (Ar-C), 154.4 (Ar-C), 162.4 (Ar-C). HRMS calcd for C₁₉H₂₄N₅O₃S [M + H]⁺ 402.1594, found 402.1596. HPLC: >99.5%, XSELECT column, retention time 8.45 min, reverse phase, H₂O/MeCN/HCOOH gradient run, 1 mL min⁻¹.

7-Chloro-5-cyclohexylmethoxyimidazo[1,2-*a*]pyrimidine (100). Synthesized in accordance with general procedure C from **99** (1.0 g, 5.4 mmol). The product was purified by precipitation with glacial acetic acid, and repeated washing with water, to afford **100** as a pale-orange powder (1.16 g, 81%); R_f 0.23 (60% EtOAc:petrol). ¹H NMR (300 MHz, DMSO-*d*₆) δ 1.05–1.95 (11H, m, cyclohexyl), 4.30 (2H, d, *J* = 6.1 Hz, OCH₂), 6.63 (1H, s, H-6), 7.63 (2H, d, *J* = 1.2 Hz, H-3), 7.81 (2H, d, *J* = 1.3 Hz, H-2). ¹³C NMR (75 MHz, DMSO-*d*₆) δ 25.4, 26.2, 29.0, 36.9, 76.3, 88.8, 107.6, 134.2, 148.1, 152.1, 155.6. LRMS (ES⁺) *m/z* 266.0 [M + H]⁺.

5-Cyclohexylmethoxy-7-anilinoimidazo[1,2-*a*]pyrimidine (101). The title compound was synthesized according to general procedure A from **100** (0.075 mg, 0.28 mmol). Purification by chromatography (silica; EtOAc), followed by crystallization from MeOH, gave **101** (31 mg, 34%); R_f 0.11 (100% EtOAc); mp 252–253 °C; UV λ_{max} (EtOH) 305, 257, 212 nm. IR (cm⁻¹) 3153, 2926, 2847, 2161, 1627, 1595, 1568, 1531, 1489, 1291, 1259, 1122. ¹H NMR (300 MHz, DMSO-*d*₆) δ 1.05–1.85 (11H, m, cyclohexyl), 4.35 (2H, d, *J* = 6.2 Hz, OCH₂), 5.98 (1H, s, H-6), 6.99 (1H, t, *J* = 7.3 Hz, PhH), 7.26 (1H, d, *J* = 1.6 Hz, H-3), 7.33 (2H, dd, *J* = 7.6, 8.2 Hz, 2 × PhH), 7.44 (1H, d, *J* = 1.6 Hz, H-2), 7.83 (2H, d, *J* = 7.7 Hz, 2 × PhH), 9.45 (1H, s, NH). LRMS (ES⁺) *m/z* 323.0 [M + H]⁺. Anal. Found: C, 70.41; H, 6.59; N, 17.40. C₁₉H₂₂N₄O requires C, 70.78; H, 6.88; N, 17.38%.

5-Cyclohexylmethoxy-7-(4'-sulfamoylanilino)imidazo[1,2-*a*]pyrimidine (102). Prepared in accordance with general procedure A from **100** (0.25 g, 0.94 mmol), and purified by chromatography (silica; 5% MeOH:EtOAc) to give **102** as a colorless solid (28 mg, 7%); R_f 0.47 (10% MeOH:EtOAc); mp 228–229 °C. IR (cm⁻¹) 3348, 3292, 2926, 2847, 1653, 1617, 1591, 1536, 1429, 1310, 1155. UV λ_{max} (EtOH) 315, 266, 216 nm. ¹H NMR (300 MHz, DMSO-*d*₆) δ 1.15–1.90 (11H, m, cyclohexyl), 4.16 (2H, d, *J* = 6.1 Hz, OCH₂), 6.03 (1H, s, H-6), 7.24 (2H, s, SO₂NH₂), 7.32 (1H, d, *J* = 1.6 Hz, H-3), 7.51 (1H, d, *J* = 1.6 Hz, H-2), 7.78 (2H, d, *J* = 8.9 Hz, H-2' and H-6'), 7.99 (2H, d, *J* = 8.9 Hz, H-3' and H-5'), 9.85 (1H, s, NH). ¹³C NMR (75 MHz, DMSO-*d*₆) δ 25.45, 26.19, 29.16, 36.85, 75.00, 78.91, 105.83, 118.42, 127.05, 131.95, 137.03, 144.01, 154.73, 155.68. LRMS (ES⁺) *m/z* 402.0 [M + H]⁺; Anal. Found: C, 57.06; H, 5.53; N, 17.09. C₁₉H₂₃N₅O₃S requires: C, 56.84; H, 5.77; N, 17.44%.

Protein Kinase Assays. Human CDK1/B was purchased from New England Biolabs or prepared as described,⁴⁶ CDK2/A2 was prepared as described,⁴⁶ CDK4/D was provided by AstraZeneca, and CDK7/H and CDK9/T were purchased from Upstate. IC₅₀ values for CDK1/B, CDK2/A, and CDK4/D were determined as described.²³ IC₅₀ values for CDK7/H and CDK9/H were determined according to

the supplier's instructions (Upstate). CDK1/B, CDK2/A, and CDK4/D were assayed at an ATP concentration of 12.5 μM, CDK7/H and CDK9/T at an ATP concentration of 100 μM. Published K_m(ATP) values for CDK1-cyclin B, CDK2-cyclin A, CDK4-cyclin D2, CDK7-cyclin H-MAT, and CDK9-cyclin T are 0.8, 0.58, 3.8, 4.1, and 0.7 μM, respectively (www.proqinase.com).

Growth Inhibition Assays. The ability of selected compounds to inhibit cell growth was assessed in a panel of human cancer cell lines using an SRB assay.⁴⁵ Briefly, cells in 96-well plates were exposed to inhibitor (0.1–30 μM, three replicates per drug concentration) or 0.5% (v/v) DMSO (for compound **73** treatments) or 0.1% (v/v) DMSO (for all other inhibitors) for 120 h, stained with sulforhodamine B and the absorbance at 570 nm measured (SpectraMax250; Molecular Devices, Wokingham, UK). At least three independent experiments were performed. Growth inhibitory GI₅₀ values were calculated using GraphPad Prism software (GraphPad Software, Inc., San Diego, CA).

Crystallization and Structure Determination. Crystals of CDK2-cyclin A bound to **73** and CDK1-cyclin B-CKS2 bound to **3** were grown as previously described.⁴⁶ Briefly, the inhibitors were added in 2-fold molar excess (which was also a 10-fold higher concentration than the inhibitor IC₅₀) to a low concentration solution of the appropriate protein complex and then the samples were concentrated to 10–12 mg mL⁻¹ for crystallization. CDK2-cyclin A–**73** crystals were grown from a well solution containing 0.6–0.8 M KCl, 0.9–1.2 M (NH₄)₂SO₄, and 100 mM HEPES (pH 7.0). CDK1-cyclin B-CKS2 was cocrystallized with **3** from a screen covering the conditions 0.1 M MES/imidazole buffer (pH 6.7), 6.5% MPD, 5% PEG4K, 10% PEG1K. Before data collection, crystals were cryoprotected by brief immersion in either 8 M sodium formate (CDK2-cyclin A) or 25% ethylene glycol (CDK1-cyclin B-CKS2).

Data processing was carried out using XDS, MOSFLM, POINTLESS/AIMLESS,⁵² and other programs of the CCP4 suite,⁵³ run through the CCP4i2 GUI.⁵⁴ The structures of the different complexes were solved by molecular replacement using Phaser,⁵⁵ and search models drawn from PDB entries 1QMZ (cyclin A-bound CDK2) and 4Y72 (CDK1-cyclin B-CKS2). Models of **73** and **3** and ligand restraints were generated using ACEDRG within the CCP4i2 suite,⁵⁴ and built into the electron density using Coot.⁵⁶ Structures were refined using REFMAC,⁵⁷ interspersed with manual rebuilding in Coot, including TLS refinement.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jmedchem.6b01254.

Structure of compound **102** and associated crystal data and refinement statistics; ¹H NMR spectra for compounds **70**, **73**, **83**, **90**, **91**, and **98** (PDF)
Molecular formula strings (CSV)

Accession Codes

CDK2-cyclin A in complex with compound **73** (SLQE), and CDK1-cyclin B-CKS2 in complex with compound **3** (SLQF) have been deposited with the PDB and the atomic coordinates and experimental data will be released upon article publication.

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Author Contributions

C.R.C. and E.A. contributed equally to the work described in this paper. C.R.C., synthesis and characterization of compounds; E.A., protein expression, crystallization, and structure determination; S.J.H., synthesis and characterization of compounds and manuscript preparation; M.P.M., protein structure determination and interpretation; Benoit Carbain, synthesis and characterization of compounds; I.R.H., experimental design and interpretation; L.K.H., synthesis and characterization of compounds; S.K., protein expression and crystallization; C.J.M., synthesis and characterization of compounds; M.E.M.N., protein structure determination, data analysis and interpretation, and manuscript preparation; D.R.N., project conception, experimental design and interpretation; D.M.T., synthesis and characterization of compounds; M.S., synthesis and characterization of compounds; S.J.T., design, conduct, and interpretation of the cellular studies; L.Z.W., design, conduct, and interpretation of CDK2 inhibition experiments; C.W., synthesis and characterization of compounds; S.R.W., experimental design and interpretation and manuscript preparation; B.T.G., project conception, experimental design and interpretation, data analysis, and manuscript preparation; R.J.G., project conception, experimental design and interpretation, data analysis, and manuscript preparation; J.A.E., experimental design and interpretation and manuscript preparation; C.C., project conception, experimental design and interpretation, data analysis, and manuscript preparation.

Notes

The authors declare no competing financial interest.

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ABBREVIATIONS USED

CDK, cyclin-dependent kinase; CKS, cyclin-dependent kinase subunit; GI50, half-maximum growth inhibition; HEPES, 2-[4-(2-hydroxyethyl)piperazin-1-yl]ethanesulfonic acid; MES, 2-morpholin-4-ylethanesulfonic acid; MPD, 2-methyl-2,4-pentanediol; SRB, sulforhodamine B; TFE, 2,2,2-trifluoroethanol; TLS, translation, libration, and screw-rotation

REFERENCES

(1) Lim, S.; Kaldis, P. Cdks, Cyclins and CKIs: Roles Beyond Cell Cycle Regulation. *Development* **2013**, *140*, 3079–3093.

(2) Malumbres, M. Cyclin-Dependent Kinases. *Genome Biol.* **2014**, *15*, 122–132.

(3) Malumbres, M.; Barbacid, M. Cell Cycle, CDKs and Cancer: A Changing Paradigm. *Nat. Rev. Cancer* **2009**, *9*, 153–166.

(4) Casimiro, M. C.; Velasco-Velazquez, M.; Aguirre-Alvarado, C.; Pestell, R. G. Overview of Cyclins D1 Function in Cancer and the CDK Inhibitor Landscape: Past and Present. *Expert Opin. Invest. Drugs* **2014**, *23*, 295–304.

(5) Sherr, C. J.; Beach, D.; Shapiro, G. I. Targeting CDK4 and CDK6: From Discovery to Therapy. *Cancer Discovery* **2016**, *6*, 353–367.

(6) Bruyere, C.; Meijer, L. Targeting Cyclin-Dependent Kinases in Anti-Neoplastic Therapy. *Curr. Opin. Cell Biol.* **2013**, *25*, 772–779.

(7) Asghar, U.; Witkiewicz, A. K.; Turner, N. C.; Knudsen, E. S. The History and Future of Targeting Cyclin-Dependent Kinases in Cancer Therapy. *Nat. Rev. Drug Discovery* **2015**, *14*, 130–146.

(8) Bose, P.; Simmons, G. L.; Grant, S. Cyclin-Dependent Kinase Inhibitor Therapy for Hematologic Malignancies. *Expert Opin. Invest. Drugs* **2013**, *22*, 723–738.

(9) Aleem, E.; Arceci, R. J. Targeting Cell Cycle Regulators in Hematologic Malignancies. *Front. Cell Dev. Biol.* **2015**, *3*, 16–37.

(10) O'Leary, B.; Finn, R. S.; Turner, N. C. Treating Cancer with Selective CDK4/6 Inhibitors. *Nat. Rev. Clin. Oncol.* **2016**, *13*, 417–430.

(11) Sánchez-Martínez, C.; Gelbert, L. M.; Lallena, M. J.; de Dios, A. Cyclin Dependent Kinase (CDK) Inhibitors as Anticancer Drugs. *Bioorg. Med. Chem. Lett.* **2015**, *25*, 3420–3435.

(12) Tetsu, O.; McCormick, F. Proliferation of Cancer Cells Despite CDK2 Inhibition. *Cancer Cell* **2003**, *3*, 233–245.

(13) Ortega, S.; Prieto, I.; Odajima, J.; Martin, A.; Dubus, P.; Sotillo, R.; Barbero, J. L.; Malumbres, M.; Barbacid, M. Cyclin-Dependent Kinase 2 is Essential for Meiosis but not for Mitotic Cell Division in Mice. *Nat. Genet.* **2003**, *35*, 25–31.

(14) Berthet, C.; Aleem, E.; Coppola, V.; Tessarollo, L.; Kaldis, P. Cdk2 Knockout Mice are Viable. *Curr. Biol.* **2003**, *13*, 1775–1785.

(15) Horiuchi, D.; Huskey, N. E.; Kusdra, L.; Wohlbold, L.; Merrick, K. A.; Zhang, C.; Creasman, K. J.; Shokat, K. M.; Fisher, R. P.; Goga, A. Chemical-Genetic Analysis of Cyclin Dependent Kinase 2 Function Reveals an Important Role in Cellular Transformation by Multiple Oncogenic Pathways. *Proc. Natl. Acad. Sci. U. S. A.* **2012**, *109*, E1019–E1027.

(16) Cheng, C. K.; Gustafson, W. C.; Charron, E.; Houseman, B. T.; Zunder, E.; Goga, A.; Gray, N. S.; Pollok, B.; Oakes, S. A.; James, C. D.; Shokat, K. M.; Weiss, W. A.; Fan, Q. W. Dual Blockade of Lipid and Cyclin-Dependent Kinases Induces Synthetic Lethality in Malignant Glioma. *Proc. Natl. Acad. Sci. U. S. A.* **2012**, *109*, 12722–12727.

(17) Deans, A. J.; Khanna, K. K.; McNeese, C. J.; Mercurio, C.; Heierhorst, J.; McArthur, G. A. Cyclin-Dependent Kinase 2 Functions in Normal DNA Repair and is a Therapeutic Target in BRCA1-Deficient Cancers. *Cancer Res.* **2006**, *66*, 8219–8226.

(18) Molenaar, J. J.; Ebus, M. E.; Geerts, D.; Koster, J.; Lamers, F.; Valentijn, L. J.; Westerhout, E. M.; Versteeg, R.; Caron, H. N. Inactivation of CDK2 is Synthetically Lethal to MYCN Over-Expressing Cancer Cells. *Proc. Natl. Acad. Sci. U. S. A.* **2009**, *106*, 12968–12973.

(19) Yang, L.; Fang, D.; Chen, H.; Lu, Y.; Dong, Z.; Ding, H. F.; Jing, Q.; Su, S. B.; Huang, S. Cyclin-Dependent Kinase 2 is an Ideal Target for Ovary Tumors with Elevated Cyclin E1 Expression. *Oncotarget* **2015**, *6*, 20801–20812.

(20) Guha, M. Cyclin-Dependent Kinase Inhibitors Move into Phase III. *Nat. Rev. Drug Discovery* **2012**, *11*, 892–894.

(21) Galons, H.; Oumata, N.; Gloulou, O.; Meijer, L. Cyclin-Dependent Kinase Inhibitors Closer to Market Launch? *Expert Opin. Ther. Pat.* **2013**, *23*, 945–963.

(22) Santo, L.; Siu, K. T.; Raje, N. Targeting Cyclin-Dependent Kinases and Cell Cycle Progression in Human Cancers. *Semin. Oncol.* **2015**, *42*, 788–800.

- (23) Arris, C. E.; Boyle, F. T.; Calvert, A. H.; Curtin, N. J.; Endicott, J. A.; Garman, E. F.; Gibson, A. E.; Golding, B. T.; Grant, S.; Griffin, R. J.; Jewsbury, P.; Johnson, L. N.; Lawrie, A. M.; Newell, D. R.; Noble, M. E.; Sausville, E. A.; Schultz, R.; Yu, W. Identification of Novel Purine and Pyrimidine Cyclin-Dependent Kinase Inhibitors with Distinct Molecular Interactions and Tumor Cell Growth Inhibition Profiles. *J. Med. Chem.* **2000**, *43*, 2797–2804.
- (24) Davies, T.-G.; Bentley, J.; Arris, C.-E.; Boyle, F. T.; Curtin, N.-J.; Endicott, J.-A.; Gibson, A.-E.; Golding, B.-T.; Griffin, R.-J.; Hardcastle, I.-R.; Jewsbury, P.; Johnson, L.-N.; Mesguiche, V.; Newell, D.-R.; Noble, M.-E. M.; Tucker, J.-A.; Wang, L.; Whitfield, H.-J. Structure-Based Design of a Potent Purine-Based Cyclin-Dependent Kinase Inhibitor. *Nat. Struct. Biol.* **2002**, *9*, 745–749.
- (25) Hardcastle, I. R.; Arris, C. E.; Bentley, J.; Boyle, F. T.; Chen, Y.; Curtin, N. J.; Endicott, J. A.; Gibson, A. E.; Golding, B. T.; Griffin, R. J.; Jewsbury, P.; Menyerol, J.; Mesguiche, V.; Newell, D. R.; Noble, M. E.; Pratt, D. J.; Wang, L. Z.; Whitfield, H. J. N2-Substituted O6-Cyclohexylmethylguanine Derivatives: Potent Inhibitors of Cyclin-Dependent Kinases 1 and 2. *J. Med. Chem.* **2004**, *47*, 3710–3722.
- (26) Marchetti, F.; Cano, C.; Curtin, N. J.; Golding, B. T.; Griffin, R. J.; Haggerty, K.; Newell, D. R.; Parsons, R. J.; Payne, S. L.; Wang, L. Z.; Hardcastle, I. R. Synthesis and Biological Evaluation of 5-Substituted O4-Alkylpyrimidines as CDK2 Inhibitors. *Org. Biomol. Chem.* **2010**, *8*, 2397–2407.
- (27) Jonkers, I.; Lis, J. T. Getting up to Speed with Transcription Elongation by RNA Polymerase II. *Nat. Rev. Mol. Cell Biol.* **2015**, *16*, 167–177.
- (28) Fischer, P. M.; Gianella-Borradori, A. Recent Progress in the Discovery and Development of Cyclin-Dependent Kinase Inhibitors. *Expert Opin. Invest. Drugs* **2005**, *14*, 457–477.
- (29) Parry, D.; Guzi, T.; Shanahan, F.; Davis, N.; Prabhavalkar, D.; Wiswell, D.; Seghezzi, W.; Paruch, K.; Dwyer, M. P.; Doll, R.; Nomeir, A.; Windsor, W.; Fischmann, T.; Wang, Y.; Oft, M.; Chen, T.; Kirschmeier, P.; Lees, E. M. Dinaciclib (SCH 727965), A Novel and Potent Cyclin-Dependent Kinase Inhibitor. *Mol. Cancer Ther.* **2010**, *9*, 2344–2353.
- (30) Pratt, D. J.; Bentley, J.; Jewsbury, P.; Boyle, F. T.; Endicott, J. A.; Noble, M. E. Dissecting the Determinants of Cyclin-Dependent Kinase 2 and Cyclin-Dependent Kinase 4 Inhibitor Selectivity. *J. Med. Chem.* **2006**, *49*, 5470–5477.
- (31) Gibson, A. E.; Arris, C. E.; Bentley, J.; Boyle, F. T.; Curtin, N. J.; Davies, T. G.; Endicott, J. A.; Golding, B. T.; Grant, S.; Griffin, R. J.; Jewsbury, P.; Johnson, L. N.; Mesguiche, V.; Newell, D. R.; Noble, M. E.; Tucker, J. A.; Whitfield, H. J. Probing the ATP Ribose-Binding Domain of Cyclin-Dependent Kinases 1 and 2 with O(6)-Substituted Guanine Derivatives. *J. Med. Chem.* **2002**, *45*, 3381–3393.
- (32) Montgomery, J. A.; Hewson, K. Synthesis of Potential Anticancer Agents XX. 2-Fluoropurines. *J. Am. Chem. Soc.* **1960**, *82*, 463–468.
- (33) Carbain, B.; Coxon, C. R.; Lebraud, H.; Elliott, K. J.; Matheson, C. J.; Meschini, E.; Roberts, A. R.; Turner, D. M.; Wong, C.; Cano, C.; Griffin, R. J.; Hardcastle, I. R.; Golding, B. T. Trifluoroacetic Acid in 2,2,2-Trifluoroethanol Facilitates S_NAr Reactions of Heterocycles with Arylamines. *Chem. - Eur. J.* **2014**, *20*, 2311–2317.
- (34) Lembicz, N. K.; Grant, S.; Clegg, W.; Griffin, R. J.; Heath, S. L.; Golding, B. T. Facilitation of Displacements at the 6-Position of Purines by the Use of 1,4-Diazabicyclo[2.2.2]octane as Leaving Group. *J. Chem. Soc., Perkin Trans. 1* **1997**, 185–186.
- (35) Kato, N.; Sakata, T.; Breton, G.; Le Roch, K. G.; Nagle, A.; Andersen, C.; Bursulaya, B.; Henson, K.; Johnson, J.; Kumar, K. A.; Marr, F.; Mason, D.; McNamara, C.; Plouffe, D.; Ramachandran, V.; Spooner, M.; Tuntland, T.; Zhou, Y.; Peters, E. C.; Chatterjee, A.; Schultz, P. G.; Ward, G. E.; Gray, N.; Harper, J.; Winzler, E. A. Gene Expression Signatures and Small-Molecule Compounds Link a Protein Kinase to Plasmodium Falciparum Motility. *Nat. Chem. Biol.* **2008**, *4*, 347–356.
- (36) Molander, G. A.; Gormisky, P. E. Cross-Coupling of Cyclopropyl- and Cyclobutyltrifluoroborates with Aryl and Heteroaryl Chlorides. *J. Org. Chem.* **2008**, *73*, 7481–7485.
- (37) Temple, C.; Smith, B. H.; Montgomery, J. A. Preparation of 2,5-Diamino-4,6-dichloropyrimidine. *J. Org. Chem.* **1975**, *40*, 3141–3142.
- (38) Shealy, Y. F.; Clayton, J. D.; O'Dell, C. A.; Montgomery, J. A. v-Triazolo[4,5-d]pyrimidines. II. O-Substituted Derivatives of 8-Azaguanine and 8-Azahypoxanthine. *J. Org. Chem.* **1962**, *27*, 4518–4523.
- (39) Seela, F.; Steker, H. Synthesis of 2'-Deoxyribofuranosides of 8-Aza-7-deazaguanine and Related Pyrazolo[3,4-d]pyrimidines. *Helv. Chim. Acta* **1986**, *69*, 1602–1613.
- (40) Brückl, T.; Thoma, I.; Wagner, A. J.; Knochel, P.; Carell, T. Efficient Synthesis of Deazaguanosine-Derived tRNA Nucleosides PreQ0, PreQ1, and Archaeosine Using the Turbo-Grignard Method. *Eur. J. Org. Chem.* **2010**, *2010*, 6517–6519.
- (41) Revankar, G. R.; Matthews, T. R.; Robins, R. K. Synthesis and Antimicrobial Activity of Certain Imidazo[1,2-a]pyrimidines. *J. Med. Chem.* **1975**, *18*, 1253–1255.
- (42) Harada, H.; Asano, O.; Hoshino, Y.; Yoshikawa, S.; Matsukura, M.; Kabasawa, Y.; Nijima, J.; Kotake, Y.; Watanabe, N.; Kawata, T.; Inoue, T.; Horioze, T.; Yasuda, N.; Minami, H.; Nagata, K.; Murakami, M.; Nagaoka, J.; Kobayashi, S.; Tanaka, I.; Abe, S. 2-Alkynyl-8-aryl-9-methyladenines as Novel Adenosine Receptor Antagonists: Their Synthesis and Structure-Activity Relationships Toward Hepatic Glucose Production Induced Via Agonism of the A(2B) Receptor. *J. Med. Chem.* **2001**, *44*, 170–179.
- (43) Rao, R. P.; Robins, R. K.; O'Brien, D. E. The Synthesis of Certain 5,7-Dihydroxyimidazo[1,2- α] pyrimidines. *J. Heterocycl. Chem.* **1973**, *10*, 1021–1023.
- (44) Revankar, R. R.; Robins, R. K. Synthesis and Biological Activity of some Nucleosides Resembling Guanosine: Imidazo[1,2-a]-pyrimidine Nucleosides. *Ann. N. Y. Acad. Sci.* **1975**, *255*, 166–172.
- (45) Thomas, H. D.; Wang, L.-Z.; Roche, C.; Bentley, J.; Cheng, Y.; Hardcastle, I. R.; Golding, B. T.; Griffin, R. J.; Curtin, N. J.; Newell, D. R. Preclinical In Vitro and In Vivo Evaluation of the Potent and Specific Cyclin-Dependent Kinase 2 Inhibitor NU6102 and a Water Soluble Prodrug NU6301. *Eur. J. Cancer* **2011**, *47*, 2052–2059.
- (46) Brown, N. R.; Korolchuk, S.; Martin, M. P.; Stanley, W. A.; Moukhametzianov, R.; Noble, M. E. M.; Endicott, J. A. CDK1 Structures Reveal Conserved and Unique Features of the Essential Cell Cycle CDK. *Nat. Commun.* **2015**, *6*, 6769–6780.
- (47) Beach, J. W.; Kim, H. O.; Jeong, L. S.; Nampalli, S.; Islam, Q.; Ahn, S. K.; Babu, J. R.; Chu, C. K. A Highly Stereoselective Synthesis of Anti-HIV 2',3'-Dideoxy- and 2',3'-Didehydro-2',3'-dideoxynucleosides. *J. Org. Chem.* **1992**, *57*, 3887–3894.
- (48) Baumgarten, H. E.; Petersen, J. M. Reactions of Amines. V. Synthesis of α -Amino Ketones. *J. Am. Chem. Soc.* **1960**, *82*, 459–463.
- (49) Kos, N. J.; Jongejan, H.; Van Der Plas, H. C. Deamination, Involving Ring Opening, in Reactions of 1-Aminopurinium Mesitylenesulfonates with Methanolic Ammonia. *Tetrahedron* **1987**, *43*, 4841–4848.
- (50) Balsiger, R. W.; Montgomery, J. A. Synthesis of Potential Anticancer Agents. XXV. Preparation of 6-Alkoxy-2-aminopurines. *J. Org. Chem.* **1960**, *25*, 1573–1575.
- (51) Griffin, R. J.; Arris, C. E.; Bleasdale, C.; Boyle, F. T.; Calvert, A. H.; Curtin, N. J.; Dalby, C.; Kanugula, S.; Lembicz, N. K.; Newell, D. R.; Pegg, A. E.; Golding, B. T. Resistance-Modifying Agents. 8. Inhibition of O(6)-Alkylguanine-DNA Alkyltransferase by O(6)-Alkenyl-, O(6)-Cycloalkenyl-, and O(6)-(2-Oxoalkyl)guanines and Potentiation of Temozolomide Cytotoxicity In Vitro by O(6)-(1-Cyclopentenylmethyl)guanine. *J. Med. Chem.* **2000**, *43*, 4071–4083.
- (52) Evans, P. R. An Introduction to Data Reduction: Space-Group Determination, Scaling and Intensity Statistics. *Acta Crystallogr., Sect. D: Biol. Crystallogr.* **2011**, *D67*, 282–292.
- (53) Winn, M. D.; Ballard, C. C.; Cowtan, K. D.; Dodson, E. J.; Emsley, P.; Evans, P. R.; Keegan, R. M.; Krissinel, E. B.; Leslie, A. G.; McCoy, A. J.; McNicholas, S. J.; Murshudov, G. N.; Pannu, N. S.; Potterton, E. A.; Powell, H. R.; Read, R. J.; Vagin, A. A.; Wilson, K. S. Overview of the CCP4 Suite and Current Developments. *Acta Crystallogr., Sect. D: Biol. Crystallogr.* **2011**, *D67*, 235–242.

- (54) Potterton, E.; Briggs, P.; Turkenburg, M.; Dodson, E. A Graphical User Interface to the CCP4 Program Suite. *Acta Crystallogr., Sect. D: Biol. Crystallogr.* **2003**, *D59*, 1131–1137.
- (55) McCoy, A. J.; Grosse-Kunstleve, R. W.; Adams, P. D.; Winn, M. D.; Storoni, L. C.; Read, R. J. Phaser Crystallographic Software. *J. Appl. Crystallogr.* **2007**, *40*, 658–674.
- (56) Emsley, P.; Lohkamp, B.; Scott, W. G.; Cowtan, K. Features and Development of Coot. *Acta Crystallogr., Sect. D: Biol. Crystallogr.* **2010**, *D66*, 486–501.
- (57) Murshudov, G. N.; Vagin, A. A.; Dodson, E. J. Refinement of Macromolecular Structures by the Maximum-Likelihood Method. *Acta Crystallogr., Sect. D: Biol. Crystallogr.* **1997**, *D53*, 240–255.